Lecture notes for Nonlinear Wave Equations (NWI-WM153)

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1 Introduction

Wave equations are partial differential equations (pde) that play an important role in describing a variety of phenomena in physics, like the propagation of oscillations in electric and magnetic fields at the speed of light, the squeezing and stretching of space and time in Einstein's theory of general relativity, the propagation of density fluctuations in compressible fluids and matter displacements in elastic bodies. Many of these phenomena have an inherently nonlinear character.

In this course, we will be mainly concerned with equations of the following form:

$$\Box_g \phi = F(\phi, D\phi), \tag{1.1}$$

with $\phi : I \times \mathbb{R}^n \to \mathbb{R}^m$, $n, m \in \mathbb{N}_+$, $I \subseteq \mathbb{R}$ an interval, $D\phi : I \times \mathbb{R}^n \to \mathbb{R}^{m(n+1)}$ the corresponding derivative function, and where $\Box_g$ denotes the *Laplace–Beltrami differential operator*, which is defined as follows:

$$\Box_g(\cdot) := \sum_{0 \leq \alpha, \beta \leq n} \frac{1}{\sqrt{-\det g(\phi, D\phi)}} \partial_\alpha \left(\sqrt{-\det g(\phi, D\phi)} (g^{-1})^{\alpha\beta}(\phi, D\phi) \partial_\beta (\cdot) \right),$$
$$g : I \times \mathbb{R}^n \times U \times V \to \mathbb{R}^{(n+1)^2},$$

with

- $U \subseteq \mathbb{R}^m$ and $V \subseteq \mathbb{R}^{m(n+1)}$ open subsets,
- g a $(n+1) \times (n+1)$ symmetric matrix-valued function with eigenvalues $\lambda_0 < 0 < \lambda_1 \leq \ldots \leq \lambda_n$ (depending on $(x, \phi, D\phi) \in (I \times \mathbb{R}^n) \times U \times V$) and coefficients $g_{\alpha\beta}$, where $\alpha, \beta \in \{0, \ldots n\}$,
- g^{-1} the corresponding inverse matrix with coefficients $(g^{-1})^{\alpha\beta}$,[1]
- $F : I \times \mathbb{R}^n \times U \times V \to \mathbb{R}^m$.

In analysing (1.1), we will develop the tools needed to understand the physically motivated examples mentioned above.

We say (1.1) is a:

1. **linear** wave equation, if $F(x, \cdot, \cdot)$ is linear for fixed $x \in I \times \mathbb{R}^n$ and $g(x, \cdot, \cdot)$ is constant for all fixed $x \in I \times \mathbb{R}^n$ (independent of ϕ and $D\phi$),
2. **semilinear** wave equation, if F is allowed to be nonlinear, but $g(x, \cdot, \cdot)$ is constant for all fixed $x \in I \times \mathbb{R}^n$,
3. **quasilinear** wave equation, if F is allowed to be nonlinear and $g(x, \cdot, \cdot)$ is allowed to be non-constant for some fixed $x \in I \times \mathbb{R}^n$ (i.e. it can depend on ϕ or $D\phi$).

We will refer to semilinear and quasilinear wave equations as *nonlinear wave equations.*[2]

We will be considering solutions to (1.1) arising from a *Cauchy problem* or *initial value* problem. It will be useful to introduce the following decomposition of points $x \in I \times \mathbb{R}^n$: $t = x^0 \in I$ and $\mathbf{x} \in \mathbb{R}^n$, with $\mathbf{x}^i = x^i$, for $i \in 1, \ldots n$. Assume that $(g_{ij})_{1 \leq i,j \leq n}$ is a strictly positive definite $n \times n$ matrix-valued function. We can interpret the restriction

$$(\phi, \partial_t \phi)|_{t=0}$$

as initial data on $\{t = 0\}$. That is to say, given a pair $(\Phi, \Phi') : \mathbb{R}^n \to \mathbb{R}^m \times \mathbb{R}^m$ in a suitable function space, we may ask whether there *exists* a solution ϕ to (1.1) such that $(\phi, \partial_t \phi)|_{t=0} = (\Phi, \Phi')$, and if so, whether that solution is *unique*. We say the Cauchy problem for (1.1) is **well-posed** if the following criteria are satisfied:

(i) there **exists** a solution ϕ corresponding to prescribed data (in a suitable function space),

[1] Exercise: Why is the inverse g^{-1} well-defined at each point in $I \times \mathbb{R}^n$?

[2] One could also consider *fully nonlinear* pde, by allowing g and F to depend also on $D^2\phi$. That is to say, by allowing the pde to depend nonlinearly on the highest-order derivatives. These type of pde lie outside the scope of this course.

(ii) this solution ϕ is **unique** (i.e. any other solution corresponding to the same initial data must be equal to ϕ),

(iii) at each time $t \geq 0$, the solution ϕ **depends continuously** on the data. That is to say, given a sequence (Φ_i, Φ_i') of data, such that $(\Phi_i, \Phi_i') \to (\Phi, \Phi)$ as $i \to \infty$ with respect to a choice of norm for functions on $\mathbb{R}^n$, we must have that the corresponding solutions satisfy $\phi_i(t, \cdot) \to \phi(t, \cdot)$, with respect to a choice of norm for functions on $\mathbb{R}^n$ (which does not need to agree with the initial data norm).

The above criteria are known as *Hadamard's criteria for well-posedness*.

We say the Cauchy problem for (1.1) is **locally well-posed** if there exists a $T_0 > 0$ such that (i)–(iii) hold in the region $[0, T_0] \times \mathbb{R}^n \subset \mathbb{R}^{n+1}$. We say the Cauchy problem for (1.1) is **globally well-posed** if T_0 can be taken arbitrarily large.

We will moreover investigate the **dispersive properties** of $\phi(t, \mathbf{x})$ as $t \to \infty$. That is to say, we will study how ϕ decays in time. As we will see, the question of global well-posedness is intimately connected to the dispersive properties of solutions. That is to say, showing that global well-posedness holds requires sufficiently fast decay in time.

The route towards addressing both the local and global behaviour of solutions to *nonlinear* equations starts with developing a **sufficiently robust theory of estimates** for *linear* wave equations. A significant part of the course will therefore be concerned with linear equations.

The discussion in §3–§5 follows closely the lecture notes

Luk, J., *Introduction to Nonlinear Wave Equations*, https://web.stanford.edu/ jluk/NWnotes.pdf

as well as the book:

Sogge, C.D., 1995. *Lectures on non-linear wave equations* (Vol. 2). Boston, MA: International Press.

The content of the remaining sections will deviate from the above sources.

Throughout the notes there will be EXERCISES, which are meant to encourage the reader to absorb the material actively rather than passively and often involve side-computations that are not carried out in full detail in the lectures. At the end of each section, there will also be **Problems**, which are typically more self-contained questions.

If you spot any errors or typos in these lecture notes, please send me an email!

1.1 Some examples of (non)linear wave equations

Here, we present some important examples of wave equations.

Example 1.1 (Standard wave equation, D'ALEMBERT 1746, EULER 1766)**.** *Let $\phi : I \times \mathbb{R}^n \to \mathbb{R}$. We refer to the following equation as the "standard wave equation":*

$$0 = -\partial_t^2 \phi + \sum_{1 \leq i \leq n} \partial_i^2 \phi, \tag{1.2}$$

with $\partial_\alpha := \partial_{x^\alpha}$.

The above equation is of the form (1.1) with $x^0 = t$ and $g = m = \mathrm{diag}(-1, 1, \ldots)$.[3] *We denote $\Box := \Box_m$.*

Example 1.2 (Maxwell's equations and electromagnetic waves, MAXWELL 1861–1862)**.** *Let*

$$E : I \times \mathbb{R}^3 \to \mathbb{R}^3 \quad \text{and} \quad B : I \times \mathbb{R}^3 \to \mathbb{R}^3.$$

Then the Maxwell equations in vacuum *constitute the following first-order system of pde:*

$$\partial_t E = \underline{\nabla} \times B (= \mathrm{curl}\, B),$$
$$\partial_t B = -\underline{\nabla} \times E (= -\mathrm{curl}\, E),$$

[3] When interpreted as a "metric tensor field" on $\mathbb{R}^{n+1}$, m is known as the *Minkowski metric*.

$$\underline{\nabla} \cdot E(= \operatorname{div} E) = 0,$$
$$\underline{\nabla} \cdot B(= \operatorname{div} B) = 0,$$

with $\underline{\nabla}$ *denoting the gradient with respect to* $\mathbb{R}^3$*. The Maxwell equations imply*[4]*:*

$$\Box E = 0,$$
$$\Box B = 0.$$

We obtain a well-posed initial value problem by prescribing as initial data:

$$(E, \partial_t E)|_{t=0} = (E_0, \underline{\nabla} \times B_0) \in \mathbb{R}^3 \times \mathbb{R}^3, \quad \text{with}$$
$$\underline{\nabla} \cdot E_0 = \underline{\nabla} \cdot B_0 = 0, \quad \text{or}$$
$$(B, \partial_t B)|_{t=0} = (B_0, -\underline{\nabla} \times E_0) \in \mathbb{R}^3 \times \mathbb{R}^3, \quad \text{with}$$
$$\underline{\nabla} \cdot E_0 = \underline{\nabla} \cdot B_0 = 0.$$

The divergence-free conditions on E_0 *and* B_0 *are initial data constraints.*[5]

In the example that follows, we will apply the *Einstein summation convention*, where we omit the "$\Sigma_{\alpha,\beta=0}^{n}$" symbol when summing over repeated indices α, β. **This notation will be used throughout the rest of the lecture notes!**

Example 1.3 (The Einstein vacuum equations and gravitational waves, Einstein 1915, Lanczos 1922)**.** *The Einstein vacuum equations take the form:*

$$\operatorname{Ric}_{\mu\nu}(g) = 0 \quad \mu, \nu = 0, \dots, n,$$

where $g : U \to \mathbb{R}^{(n+1)^2}$ *is a symmetrix-matrix-valued function on a open subset* $U \subset \mathbb{R}^{n+1}$ *with one negative eigenvalue and* n *positive eigenvalues and* $\operatorname{Ric}(\cdot)$ *its associated "Ricci tensor". The Ricci tensor* $\operatorname{Ric}(g)$ *is defined as the matrix with components:*

$$\operatorname{Ric}_{\mu\nu}(g) = -\frac{1}{2}(g^{-1})^{\alpha\beta}\left[\partial_\alpha\partial_\beta g_{\mu\nu} + \partial_\mu\partial_\nu g_{\alpha\beta} - \partial_\alpha\partial_\nu g_{\beta\mu} - \partial_\alpha\partial_\mu g_{\beta\nu}\right] + F_{\mu\nu}(g, Dg), \tag{1.3}$$

with $F_{\mu\nu}(g, Dg)$ *denoting matrix coefficients that are nonlinear in* $g_{\alpha\beta}$ *and* $\partial_\gamma g_{\alpha\beta}$ *but do not depend on second-order derivatives.*

The Einstein vacuum equations do not take the form of (1.1) *for general matrices* g *satisfying the assumptions above. However, it turns out (see* ***Problem*** *1.2) that if* $\Box_g x^\alpha = 0$ *for all* $\alpha = 0, \dots n$ *("harmonic coordinate condition"), then*

$$\operatorname{Ric}_{\mu\nu}(g) = -\frac{1}{2}(g^{-1})^{\alpha\beta}\partial_\alpha\partial_\beta g_{\mu\nu} + \widetilde{F}_{\mu\nu}(g, Dg), \tag{1.4}$$

where $\widetilde{F}_{\mu\nu}$ *groups together terms that are independent of second-order derivatives of* g*.*[6] *It can be shown that the Einstein vacuum equations satisfy a well-posed initial value problem.*[7]

Example 1.4 (Irrotational, compressible Euler equations and sound waves, Euler 1752)**.** *Let* $u : I \times \mathbb{R}^3 \to \mathbb{R}^3$ *denote the fluid velocity vector field and* $h : I \times \mathbb{R}^3 \to \mathbb{R}_{\geq 0}$ *the enthalpy. We define the pressure* $p : \mathbb{R}_{\geq 0} \supseteq h(I \times \mathbb{R}^3) \to \mathbb{R}_{\geq 0}$ *as a function of* h*, such that* $\rho := \frac{dp}{dh} > 0$*, with* ρ *the density of the fluid, and* $\eta^2 := \rho(\frac{d^2p}{dh^2})^{-1} > 0$*, with* η *the speed of sound.*

[4] Exercise: Show this.

[5] Exercise: Starting from the above wave equations with initial data, show conversely that (E, B) solve the vacuum Maxwell equations.

[6] As defined here, g is a matrix and therefore a coordinate-dependent object. More generally, g can be defined as a coordinate-independent object called a "Lorentzian metric tensor field" which can be associated to a matrix after making a choice of coordinates on U. The matrix will change in a particular way after making a change of coordinates. The condition $\Box_g x^\alpha = 0$ can therefore be interpreted as a condition on the coordinates, rather than on the Lorentzian metric tensor field g. Furthermore, the subset U can also be replaced by an $n+1$-dimensional "manifold".

[7] This was proved by Choquet–Bruhat in 1952.

The compressible Euler equations then take the form:

$$\partial_t u + u^i \partial_i u = -\underline{\nabla} h, \tag{1.5}$$

$$\partial_t \rho + \underline{\nabla} \cdot (\rho u) = 0. \tag{1.6}$$

To obtain a second-order equation, we assume the fluid is ***irrotational****, i.e.* $\nabla \times u = 0$*. Then there exists a* $\phi : I \times \mathbb{R}^3 \to \mathbb{R}$*, such that* $u = -\underline{\nabla}\phi$*, with* ϕ *unique up to a function of* $t \in I$*. Then* (1.5) *becomes:*

$$\underline{\nabla}\left(\partial_t \phi + \frac{1}{2}|\underline{\nabla}\phi|^2 - h\right) = 0,$$

so $\partial_t \phi + \frac{1}{2}|\nabla\phi|^2 - h$ *is a function only of time. Using the freedom of adding a function of time to* ϕ*, we can choose* ϕ *such that*

$$h = \partial_t \phi + \frac{1}{2}|\underline{\nabla}\phi|^2.$$

Applying now (1.6)*, we can obtain*[8]*:*

$$0 = -\eta^{-2}\partial_t^2 \phi + (\delta^{ij} - \eta^{-2}\partial_i\phi\partial_j\phi)\partial_i\partial_j\phi - 2\eta^{-2}\partial_i\phi\partial_t\partial_i\phi, \tag{1.7}$$

where δ^{ij} *denotes the Kronecker delta, i.e.* $\delta^{ij} = 1$ *if* $i = j$ *and* $\delta^{ij} = 0$ *if* $i \neq j$ *.*

1.2 A remark on hyperbolicity

Nonlinear wave equations, such as (1.1), are examples of *hyperbolic equations*. The notion of hyperbolicity for a pde of of order m (i.e. involving derivatives up to order m) is synonymous with having a locally well-posed initial value problem with the first $m-1$ derivatives prescribed along an initial hypersurface ("at time zero").

This nomenclature is motivated by the following second-order constant coefficient equation on $\mathbb{R}^2$:

$$\sum_{0\leq\alpha,\beta\leq 1} a^{\alpha\beta}\partial_\alpha\partial_\beta\phi + \sum_{0\leq\alpha\leq 1} b^\alpha \partial_\alpha u = 0.$$

If we denote $a^{00} = A$, $a^{01} = a^{10} = B$ and $a^{11} = C$, it can be shown that the above equation is equivalent to the standard wave equation with extra first-order terms in a different set of coordinates (and hence, has in particular a locally well-posed initial value problem), provided

$$B^2 - AC > 0.$$[9]

In the case of the planar curves, defined by $Ax^2 + 2Bxy + Cy^2 = 0$, the above condition implies that the curve is a hyperbola.[10]

For general second-order non-constant coefficient or nonlinear pde, it is harder to deduce local well-posedness directly from algebraic properties of $a^{\alpha\beta}$, which is also called the *principal symbol* of the equation. There are various notions of "hyperbolicity" which give *sufficient* conditions for establishing local well-posedness of the initial value problem. For example: *symmetric hyperbolicity* and *regular hyperbolicity* (equation (1.1) satisfies both these notions of hyperbolicity). What these notions have in common is that they provide some algebraic constraints on the principal symbol, which allow one to derive *energy estimates*.

As we will see later in this course, *energy estimates* form a key property for establishing local well-posedness of solutions to (1.1). Our strategy will be to derive *energy estimates* directly for (1.1), rather than showing it satisfies one of the above mentioned notions of hyperbolicity.

[8] Exercise: Show this and convince yourself that (1.7) is included in the class of equations (1.1).

[9] Exercise: Show this.

[10] The names of other classifications of pde can similarly be viewed in terms of planar curves describing conic sections in this simple setting: the pde is elliptic if $B^2 - AC < 0$ and parabolic if $B^2 - AC = 0$.

1.3 Problems

Problem 1.1.

a) Write down the linearization *of the Euler equations from Example 1.4 around the zero solution* $(u, h) \equiv (0, 0)$ *by omitting all nonlinear terms in* u, h *and their derivatives.*

b) Assuming existence and uniqueness of smooth solutions to the initial value problem for the standard wave equation with smooth initial data, argue why the linearization of the irrotational Euler equations around the zero solution with smooth initial data $(u|_{t=0}, h|_{t=0})$ *must also have existence and uniqueness of solutions. [Hint: determine and use the linearization of (1.7).]*

Problem 1.2.

a) Show that the Cartesian coordinates $(x^0, x^1, \dots, x^n)$ *on* $\mathbb{R}^{n+1}$ *satisfy* $\Box x^\alpha = \Box_m x^\alpha = 0$.

b) Show that the Einstein vacuum equations (1.3) reduce to (1.4) if $\Box_g x^\alpha = 0$ *for all* $\alpha = 0, \dots, n$. *[Hint: derive and use that* $\partial_\mu \det g = \det g (g^{-1})^{\alpha\beta} \partial_\mu g_{\alpha\beta}$.*]*

1.4 Acknowledgements

The author thanks Luuk de Ridder, Quinten Rutgers, Dave Verweg and Jorn van Voorthuizen, for spotting errors in earlier versions of these lecture notes and Gustav Holzegel for sharing his lecture notes for a similar course on nonlinear wave equations.

2 The standard wave equation

In this section, we will focus in more detail on the standard wave equation (Example 1.1). We will assume throughout this section the existence and uniqueness of smooth solutions, given smooth initial data. This fact is typically proved in undergraduate pde courses, but it will also be shown for much more general linear wave equations in the next section.

2.1 The Huygens principles and dispersion for the standard wave equation

We consider the initial problem for the standard wave equation in $\mathbb{R}^{n+1}$:

$$\begin{aligned} \Box\phi = -\partial_t^2\phi + \sum_{1\le i\le n} \partial_i^2\phi = 0 \quad \text{in } \mathbb{R}^{n+1}, \\ (\phi|_{t=0}, \partial_t\phi|_{t=0}) = (\Phi, \Phi') \quad \text{on } \mathbb{R}^n. \end{aligned} \tag{2.1}$$

In order to investigate the dispersive properties of solutions to (2.1), it is useful to define *ingoing and outgoing lightcones*.

Definition 2.1. *Let* $r = \sqrt{\sum_{1\le i\le n} (x^i)^2}$ *denote the radius function on* $\mathbb{R}^n$. *We define the* ***outgoing lightcone*** *emanating from the spacetime point* $p \in \mathbb{R}^{n+1}$ *as the subset:*

$$\mathcal{C}_p^+ = p + \left\{ x \in \mathbb{R}^{n+1} \,\middle|\, t - r = 0 \right\} \subset \mathbb{R}^{n+1}.$$

We define the ***ingoing lightcone*** *emanating from the spacetime point* $p \in \mathbb{R}^{n+1}$ *as the subset:*

$$\mathcal{C}_p^- = p + \left\{ x \in \mathbb{R}^{n+1} \,\middle|\, t + r = 0 \right\} \subset \mathbb{R}^{n+1}.$$

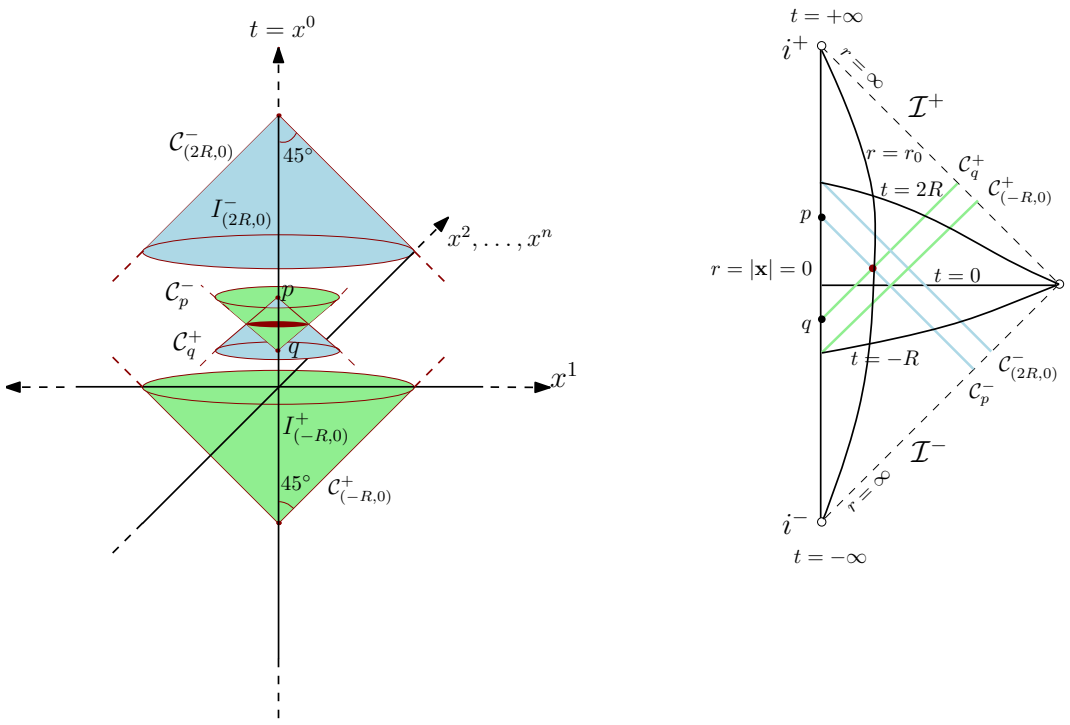


Figure 1: Each point in $\mathbb{R}^{n+1}$ lies in the intersection of ingoing and outgoing lightcones emanating from the the time axis $r = |\mathbf{x}| = 0$. The figure on the right depicts all these lightcones in a compact region of $\mathbb{R}^2$. The lightcones are depicted as diagonal lines at 45° with the time axis and these intersect in points with represent $n-1$-spheres.

We define the ***interior of the outgoing lightcone*** *emanating from the spacetime point $p \in \mathbb{R}^{n+1}$ as the subset:*

$$I^+_p = p + \left\{ x \in \mathbb{R}^{n+1} \,\middle|\, t - r > 0 \right\} \subset \mathbb{R}^{n+1}.$$

We define the ***interior of the ingoing lightcone*** *emanating from the spacetime point $p \in \mathbb{R}^{n+1}$ as the subset:*

$$I^-_p = \left\{ x \in \mathbb{R}^{n+1} \,\middle|\, t + r < 0 \right\} \subset \mathbb{R}^{n+1}.$$

It will be useful to introduce the following schematic two-dimensional representation of $\mathbb{R}^{n+1}$, which is called a *Penrose diagram*; see Figure 1 below.

In this picture, ingoing and outgoing lightcones emanating from points on the t-axis are represented by diagonal lines at 45° angle with the t-axis. Furthermore, the limits of the points $(\pm T, 0, \ldots, 0) = (\pm T, 0)$ on the t-axis as $T \to \infty$ are represented at a finite distance as the points $i^\pm$ (future/past timelike infinity) and the limiting cones $C^-_{(T,0)}$ and $C^+_{(-T,0)}$ as $T \to \infty$ are represented by the dashed lines $\mathcal{I}^+$ ("future null infinity") and $\mathcal{I}^-$ ("past null infinity"), respectively. These pictures can be made mathematically precise as representing conformal compactifications of $\mathbb{R}^{n+1}$. In this course, however, they will serve merely as useful schematic representations of $\mathbb{R}^{n+1}$.

For the standard wave equation in odd spatial dimensions, the main dispersive properties are captured via the *Huygens principles*.

Theorem 2.1. *Let $n \in \mathbb{N}_{\geq 3}$ be odd. Suppose $(\Phi, \Phi') \in C^\infty_c(\mathbb{R}^n) \times C^\infty_c(\mathbb{R}^n)$. Let $R > 0$ be such that $\operatorname{supp}(\Phi) \cup \operatorname{supp}(\Phi') \subseteq B_R(0)$. Then, a corresponding smooth solution ϕ to (2.1) necessarily:*

(i) *vanishes in $\mathbb{R}^{n+1} \setminus (I^+_{(-R,0)} \cup I^-_{(+R,0)})$ (***weak Huygens principle***) .*

(ii) *vanishes in* $\mathbb{R}^{n+1} \setminus (I^+_{(-R,0)} \cup I^-_{(+R,0)})$ *and* $I^+_{(+R,0)} \cup I^-_{(-R,0)}$ *(***strong Huygens principle***).*

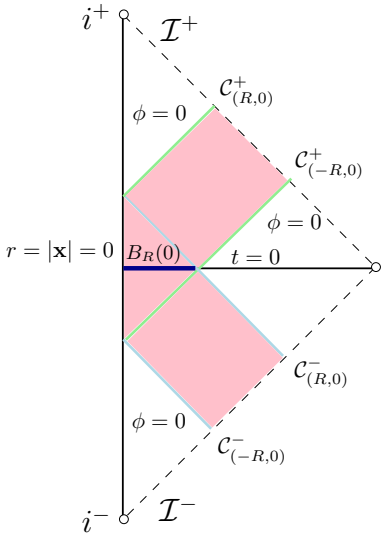


Figure 2: The strong Huygens principle.

Proof of Theorem 2.1 for $n = 3$. Let $n = 3$ and consider standard spherical coordinates (r, θ, φ) with $r \in \mathbb{R}_+$, $\theta \in (0, \pi)$ and $\varphi \in (0, 2\pi)$ defined as follows:

$$\begin{aligned} x^1 &= r \sin\theta \cos\varphi, \\ x^2 &= r \sin\theta \sin\varphi, \\ x^3 &= r \cos\theta. \end{aligned}$$

It follows then that (2.1) is equivalent to[11]

$$-\partial_t^2 \phi + r^{-2} \partial_r (r^2 \partial_r \phi) + r^{-2} \not{\Delta}_{\mathbb{S}^2} \phi = 0, \tag{2.2}$$

$$\not{\Delta}_{\mathbb{S}^2}(\cdot) := \frac{1}{\sin\theta} \partial_\theta (\sin\theta \partial_\theta (\cdot)) + \frac{1}{\sin^2\theta} \partial_\varphi^2 (\cdot).$$

Introduce now the rescaled variable $\psi = r\phi$ and the *null coordinates*[12]

$$\begin{aligned} u &= \frac{t - r}{2}, \\ v &= \frac{t + r}{2}. \end{aligned}$$

Then (2.2) is equivalent to[13]

$$\partial_u \partial_v \psi = r^{-2} \not{\Delta}_{\mathbb{S}^2} \psi. \tag{2.3}$$

We prove both (i) and (ii) by decomposing ψ into *spherical harmonic modes*. Let $Y_{\ell m}(\theta, \varphi)$ denote the standard spherical harmonics, satisfying $\not{\Delta}_{\mathbb{S}^2} Y_{\ell m} = -\ell(\ell+1) Y_{\ell m}$. Then, for suitably regular functions on $\mathbb{R}^{n+1}$, we can write[14]

$$f(t, r, \theta, \varphi) = \sum_{\ell \in \mathbb{N}_0} f_\ell(t, r, \theta, \varphi), \quad \text{with}$$

[11] Exercise: Show this.
[12] The coordinates u and v are sometimes also called the *retarded* and *advanced* time coordinates, respectively.
[13] Exercise: Show this.
[14] This follows from the fact that $\{Y_{\ell m}\}$ form an orthonormal basis for $L^2(\mathbb{S}^2)$; see Appendix A for a discussion on L^2-spaces.

$$f_\ell(r,\theta,\varphi) = \sum_{-\ell\leq m\leq \ell} f_{\ell m}(t,r) Y_{\ell m}(\theta,\varphi).$$

In particular, $\not{\Delta}_{\mathbb{S}^2} f_\ell = -\ell(\ell+1) f_\ell$.

Step 1: $\ell = 0$ modes: Let us consider first the $\ell = 0$ mode ψ_0, or the spherically symmetric part of ψ. Then

$$\partial_u \partial_v \psi_0 = 0,$$

which we can interpret as a conservation law in u or v: since $\partial_u \psi_0|_{t=0} = 0$ and $\partial_v \psi_0|_{t=0} = 0$ in $\{t=0\}\setminus\overline{B_R(0)}$ and by the above equation $\partial_u \psi_0$ is conserved in v and $\partial_v \psi_0$ is conserved in u, we can integrate from $t = 0$ to conclude that in fact $\partial_u \psi_0 = 0$ and $\partial_v \psi_0 = 0$ in $\mathbb{R}^{n+1} \setminus (I^+_{(-R,0)} \cup I^-_{(+R,0)})$, thus proving (i).

Without loss of generality (WLOG), we will show (ii) holds in $\{t \geq 0\}$ (the $t < 0$ case follows simply from exchanging u and v). By (i), we have in particular that $\partial_v \psi_0 = 0$ along

$$\mathcal{C}^+_{(-R,0)} \cap \{t \geq 0\} = \left\{ u = -\frac{R}{2} \right\} \cap \{t \geq 0\}.$$

Hence, we can integrate $\partial_u \partial_v \psi_0$ in u to conclude that $\partial_v \psi_0 = 0$ in

$$I^+_{(R,0)} = \left\{ u \geq \frac{R}{2} \right\}.$$

Finally, using that $\psi = r\phi = 0$ along $r = 0$, we can integrate $\partial_v \psi_0$ in v from $r = 0$ to conclude that $\psi = 0$ in $I^+_{(R,0)}$, thereby concluding (ii) for the $\ell = 0$ modes.

Step 2: general ℓ modes:

We will obtain (i) and (ii) for $\ell \geq 1$ by using that solutions ψ_ℓ to (2.3) satisfy the following higher order conservation laws[15]:

$$\partial_u [r^{-2\ell} \partial_v ((r^2\partial_v)^\ell \psi_\ell)] = 0,$$
$$\partial_v [r^{-2\ell} \partial_u ((r^2\partial_u)^\ell \psi_\ell)] = 0.$$

We then proceed as in the $\ell = 0$ case, with the role of ψ_0 taken on by $(r^2\partial_v)^\ell \psi_\ell$: we integrate the above conservation laws from $\{t = 0\} \setminus \overline{B_R(0)}$ to conclude that $\partial_v((r^2\partial_v)^\ell \psi_\ell) = 0$ and $\partial_u((r^2\partial_u)^\ell \psi_\ell) = 0$ in $\mathbb{R}^{n+1} \setminus (I^+_{(-R,0)} \cup I^-_{(+R,0)})$. By integrating multiple times in v or u we can then conclude that in fact $(r^2\partial_v)^k \psi_\ell = 0$ and $(r^2\partial_u)^k \psi_\ell = 0$ for all $0 \leq k \leq \ell$ in $\mathbb{R}^{n+1} \setminus (I^+_{(-R,0)} \cup I^-_{(+R,0)})$. So (i) holds for each $\ell \in \mathbb{N}_0$, which means (i) holds for the full solution ψ.

To obtain (ii), we restrict again to $\{t \geq 0\}$ WLOG. By (i) we have in particular that $\partial_v (r^2\partial_v)^\ell \psi_\ell = 0$ along $\{u = -\frac{1}{2}R\} \cap \{t \geq 0\}$, so we can integrate the conservation law in u to conclude that $\partial_v (r^2\partial_v)^\ell \psi_\ell = 0$ in $\{u \geq \frac{1}{2}R\}$. Then (ii) follows by integrating $\partial_v (r^2\partial_v)^\ell \psi_\ell$ a number $\ell + 1$ times starting from $r = 0$ and using that $(r^2\partial_v)^k \psi_\ell = 0$ at $r = 0$ for all $0 \leq k \leq \ell$.

The general n case is proved in **Problem** 2.1. □

Remark 2.1. *The weak Huygens principle is also known as the domain of dependence property of the standard wave equation. It is in fact valid also for even n, as will be shown in the next section using energy estimates. An analogue of this property holds moreover for much more general wave equations. The strong Huygens principle on the other hand is a very special property of the standard wave equation (2.1) in odd space dimensions.*

Remark 2.2. *Alternatively, one can read off the weak and strong Huygens principles from* Kirchoff's formula*, typically introduced in undergraduate pde courses, which provides an explicit expression of ϕ in terms of initial data (Φ, Φ'). However, such a formula is not available when considering more general (linear) wave equations.*

[15] Exercise: Prove this by deriving first the equation satisfied by $(r^2\partial_v)^k \psi$ for general $k \in \mathbb{N}_0$ via a proof by induction.

2.2 Energy estimates

In this section, we will derive estimates for *energies* which are non-negative definite integrals defined as follows:

$$E[\phi](t) := \frac{1}{2}\int_{\mathbb{R}^n} (\partial_t\phi)^2 + \sum_{i=1}^n (\partial_i\phi)^2\, d\mathbf{x}.$$

We will show that $E[\phi](t)$ naturally comes up when integrating $\partial_t\phi\Box\phi = 0$ in spacetime $\mathbb{R}^{n+1}$ and that it is *conserved in time* as a consequence of the t-invariance of the coefficients of the standard wave equation. We refer to ∂_t as a *vector field multiplier*, because we multiply $\partial_t\phi$ with $\Box\phi$.

It will moreover be useful to introduce the following notation for energies restricted to balls of radius $R > 0$:

$$E[\phi](t; R) := \frac{1}{2}\int_{B_R(0)} (\partial_t\phi)^2 + \sum_{i=1}^n (\partial_i\phi)^2\, d\mathbf{x}.$$

Proposition 2.2. *Let ϕ be a suitably regular solution of (2.1). For all $0 \le t_1 < t_2$ and $R > t_2$:*

$$E[\phi](t_2; R - t_2) \le E[\phi](t_1; R - t_1). \tag{2.4}$$

Proof. We multiply $-\partial_t\phi\Box\psi$ and apply the Leibniz rule to obtain:

$$\begin{aligned}
-\partial_t\phi\Box\phi &= \partial_t\phi\partial_t^2\phi - \sum_{1\le i\le n} \partial_t\phi\partial_i^2\phi \\
&= \frac{1}{2}\partial_t\left((\partial_t\phi)^2\right) - \sum_i \partial_i(\partial_t\phi\partial_i\phi) + \sum_i \partial_i\partial_t\phi\partial_i\phi \\
&= \frac{1}{2}\partial_t\left((\partial_t\phi)^2 + \sum_i (\partial_i\phi)^2\right) - \sum_i \partial_i(\partial_t\phi\partial_i\phi).
\end{aligned}$$

Let $0 \le t_1 < t_2$ and $R > t_2$. We will integrate the above identity over the region $I^-_{(R,0)} \cap \{t_1 \le t \le t_2\}$. Note that

$$\begin{aligned}
\int_{I^-_{(R,0)}\cap\{t_1\le t\le t_2\}} \frac{1}{2}\partial_t\left((\partial_t\phi)^2 + \sum_i (\partial_i\phi)^2\right) d\mathbf{x}dt &= \frac{1}{2}\int_{\{t=t_2\}\cap\{r\le R-t_2\}} (\partial_t\phi)^2 + \sum_i (\partial_i\phi)^2\, d\mathbf{x} \\
&\quad - \frac{1}{2}\int_{\{t=t_1\}\cap\{r\le R-t_1\}} (\partial_t\phi)^2 + \sum_i (\partial_i\phi)^2\, d\mathbf{x} \\
&\quad + \frac{1}{2}\int_{\mathcal{C}^-_{(R,0)}\cap\{t_1\le t\le t_2\}} (\partial_t\phi)^2 + \sum_i (\partial_i\phi)^2\, d\mathbf{x}.
\end{aligned}$$

Furthermore,

$$-\int_{I^-_{(R,0)}\cap\{t_1\le t\le t_2\}} \sum_i \partial_i(\partial_t\phi\partial_i\phi)\, d\mathbf{x}dt = \sum_i \int_{\mathcal{C}^-_{(R,0)}\cap\{t_1\le t\le t_2\}} \partial_t\phi\partial_i\phi\, \frac{x^i}{r} d\mathbf{x},$$

where the final inequality follows from a change of variables from t to x^i along $\mathcal{C}^-_{(R,0)}$; see **Problem** 2.4. Combining the above, we therefore obtain:

$$E[\phi](t; R - t_2) + \frac{1}{2}\int_{\mathcal{C}^-_{(R,0)}\cap\{t_1\le t\le t_2\}} (\partial_t\phi)^2 + \sum_i (\partial_i\phi)^2 + 2\sum_i \frac{x^i}{r}\partial_t\phi\partial_i\phi\, d\mathbf{x} = E[\phi](t; R - t_1).$$

By applying Young's inequality $2ab \le a^2 + b^2$, it follows that

$$\left|2\sum_i \frac{x^i}{r}\partial_i\phi\partial_t\phi\right| \le \sum_i (\partial_i\phi)^2 + \sum_i r^{-2}(x^i)^2(\partial_t\phi)^2 = \sum_i (\partial_i\phi)^2 + (\partial_t\phi)^2,$$

so the integral over $\mathcal{C}^-_{(R,0)}$ is non-negative definite and we conclude that (2.4) holds. □

Corollary 2.3 (Domain of dependence property)**.** *Let ϕ be a suitably regular solution of (2.1).*

(i) *Let $R > 0$ and suppose $(\phi, \partial_t\phi)|_{t=0} \equiv 0$ on $B_R((0, \mathbf{x}))$. Then $\phi \equiv 0$ in $I^-_{(R,\mathbf{x})} \cap I^+_{(-R,\mathbf{x})}$.*

(ii) *If $\mathrm{supp}(\phi, \partial_t\phi)|_{t=0} \subset B_R((0, \mathbf{x}))$ for some $R > 0$, then $\phi \equiv 0$ in $\mathbb{R}^{n+1} \setminus (I^+_{(-R,\mathbf{x})} \cup I^-_{(+R,\mathbf{x})})$.*

Proof. WLOG, we can translate out coordinates on $\mathbb{R}^n$ so that $\mathbf{x} = 0$, and we can moreover restrict to $t \geq 0$. Then (i) follows by applying (2.4), using that the RHS vanishes. Exercise: prove (ii) □

Corollary 2.4 (Energy conservation)**.** *Let ϕ be a suitably regular solution of (2.1) with $(\phi, \partial_t\phi)|_{t=0} \in C^\infty_c(\mathbb{R}^n) \times C^\infty_c(\mathbb{R}^n)$. For all $0 \leq t_1 < t_2$:*

$$E[\phi](t_2) = E[\phi](t_1). \tag{2.5}$$

Proof. We apply Corollary 2.3 (ii) to conclude that $\phi \equiv 0$ on $\mathbb{R}^{n+1} \setminus (I^+_{(-R,0)} \cup I^-_{(+R,0)})$ for some suitably large $R > 0$. Then for fixed $0 \leq t_1 \leq t_2$, there exists a $\widetilde{R} > 0$ suitably large such that moreover $\phi \equiv 0$ on $\mathcal{C}^-_{(\widetilde{R},0)} \cap \{t_1 \leq t \leq t_2\}$, so we can apply the proof of Proposition 2.2 to conclude (2.5). □

Remark 2.3. *Energy conservation results from time-translation invariance (a generalization of this to other types of symmetry is known as "Noether's (first) theorem").*

2.3 Problems

Problem 2.1 (The strong Huygens principle for general odd space dimensions)**.** *In this problem we will take $n \geq 3$ and we will work with spherical coordinates $(r, \theta_1, \ldots, \theta_{n-2}, \varphi)$ on $\mathbb{R}^n$, which are defined as follows:*

$$\begin{aligned}
x^1 &= r \sin\theta_1 \cdot \sin\theta_2 \ldots \sin\theta_{n-2} \sin\varphi,\\
x^2 &= r \sin\theta_1 \cdot \sin\theta_2 \ldots \sin\theta_{n-2} \cos\varphi,\\
x^3 &= r \sin\theta_1 \cdot \sin\theta_2 \ldots \sin\theta_{n-3} \cdot \cos\theta_{n-2},\\
x^4 &= r \sin\theta_1 \cdot \sin\theta_2 \ldots \sin\theta_{n-4} \cdot \cos\theta_{n-3},\\
&\vdots\\
x^{n-1} &= r \sin\theta_1 \cos\theta_2,\\
x^n &= r \cos\theta_1,
\end{aligned}$$

with $r = \sqrt{\sum_{i=1}^n (x^i)^2} \in \mathbb{R}_+$, $\theta_i \in (0, \pi)$ and $\varphi \in (0, 2\pi)$.

a) Show that for general $n \in \mathbb{N}_{\geq 2}$, (2.1) is equivalent to:

$$-\partial_t^2\phi + r^{-(n-1)}\partial_r(r^{n-1}\partial_r\phi) + r^{-2}\not{\Delta}_{\mathbb{S}^{n-1}}\phi = 0,$$

with $\not{\Delta}_{\mathbb{S}^{n-1}}$ a differential operator involving only angular derivatives and determine $\not{\Delta}_{\mathbb{S}^{n-1}}$ precisely.

b) Let $Y_{\ell m}(\theta_1, \ldots, \theta_{n-2}, \varphi)$, with $\ell \in \mathbb{N}_0$ and $m \in \{-\ell, -\ell+1, \ldots, \ell\}$ denote the spherical harmonics *on $\mathbb{S}^{n-1} := \{r = 1\} \subset \mathbb{R}^n$. You may assume that*

$$\not{\Delta}_{\mathbb{S}^{n-1}} Y_{\ell m} = -\ell(\ell + n - 2) Y_{\ell m}$$

and that we can uniquely decompose

$$\phi = \sum_{\ell \in \mathbb{N}_0} \phi_\ell, \quad \text{with}$$

$$\phi_\ell(t, r, \theta_1, \ldots, \theta_{n-2}, \varphi) := \sum_{m=-\ell}^{\ell} \phi_{\ell m}(t, r) Y_{\ell m}(\theta_1, \ldots, \theta_{n-2}, \varphi).$$

Prove the $n \geq 5$ case in Theorem 2.1 by generalizing the arguments in the proof of Theorem 2.1 to the setting of $n \geq 3$. [Hint: it is convenient to write odd $n \geq 3$ as $n = 2k + 3$, with $k \in \mathbb{N}_0$ and to work with an equation for the rescaled variable $\psi = r^p \phi$, given an appropriate choice of p.]

c) Which of the Huygens principles hold in the case $n = 1$?

Problem 2.2. *Consider the linearization of Euler's equations, introduced in **Problem** 1.1. By taking compactly supported initial data and applying the Huygens principles, explain why the coefficient η can be interpreted as the speed of sound in the linearized setting.*

Problem 2.3 (Duhamel's principle)**.** *Consider the standard wave equation with an inhomogeneity that is independent of ϕ:*

$$\Box \phi = F,$$

with $F : \mathbb{R}^{n+1} \to \mathbb{R}$ a smooth function and initial data $(\phi|_{t=0}, \partial_t \phi|_{t=0})$ a smooth pair of functions. Show that the solution ϕ can be expressed in terms of solutions ϕ_s to the initial value problems

$$\begin{aligned} \Box \phi_s &= 0, \\ (\phi_s|_{t=s}, \partial_t \phi|_{t=s})(\mathbf{x}) &= (0, F(s, \mathbf{x})). \end{aligned}$$

You may assume $s \mapsto \phi_s(t, \mathbf{x})$ is continuous for all $(t, \mathbf{x}) \in \mathbb{R}^{n+1}$. [Hint: integrate in ϕ_s in s.]

Problem 2.4. *Show that for any integrable function $f : \mathbb{R}^{n+1} \supset I^-_{(R,0)} \to \mathbb{R}$ and all $t_1 < t_2$:*

$$\int_{I^-_{(R,0)} \cap \{t_1 \leq t \leq t_2\}} \sum_{i=1}^{n} \partial_i f(t, \mathbf{x}) \, d\mathbf{x} dt = - \sum_{i=1}^{n} \int_{\mathcal{C}^-_{(R,0)} \cap \{t_1 \leq t \leq t_2\}} f(t, \mathbf{x}) \frac{x^i}{r} d\mathbf{x}.$$

3 Global well-posedness for general linear wave equations

In this section we consider the initial value problem for the inhomogeneous linear wave equation:

$$\begin{aligned} L\phi := a^{\alpha\beta} \partial_\alpha \partial_\beta \phi + b^\alpha \partial_\alpha \phi + d \cdot \phi = F \quad &\text{in } [0, \infty) \times \mathbb{R}^n, \\ (\phi, \partial_t \phi)|_{t=0} = (\Phi, \Phi') \quad &\text{on } \mathbb{R}^n, \end{aligned} \tag{3.1}$$

where $a^{\alpha\beta}$ denote the coefficients of a symmetric $(n+1) \times (n+1)$-dimensional matrix-valued map on $[0, \infty) \times \mathbb{R}^n$, satisfying the condition: there exists an $\epsilon_0 > 0$, such that for all $(t, x) \in [0, \infty) \times \mathbb{R}^n$:

$$|a^{00} - m^{00}|(t, \mathbf{x}) \leq \epsilon_0, \tag{3.2}$$

$$\sqrt{\sum_{i,j=1}^{n} |a^{ij} - m^{ij}|^2(t, \mathbf{x})} \leq \epsilon_0, \tag{3.3}$$

where $\epsilon_0 > 0$ will be chosen suitably small later on. We moreover assume $\partial_\mu a^{\alpha\beta}, b^\beta, d$ are bounded functions on $[0, \infty) \times \mathbb{R}^n$ for all $\alpha, \beta, \mu \in \{0, 1, 2, 3\}$. Furthermore, $F : [0, \infty) \times \mathbb{R}^n \to \mathbb{R}$ denotes an inhomogeneity that is independent of ϕ and its derivatives.

In order to make sense of the initial value problem, we will later specify further the appropriate function spaces and regularity properties of the maps: a, b, d, Φ, Φ' and F.

Throughout the lecture notes we will make the simplifying assumption

$$d \equiv 0,$$

unless explicitly stated otherwise.

The case of non-zero d is covered in **Problem** 3.3.

3.1 Energy estimates

At the core of well-posedness theory for general linear wave equations are energy estimates, analogous to those derived in § 2.2. In order to work with energy quantities, it will be useful to introduce the following notations:

$$\begin{aligned}|\partial^k \phi|^2 &:= \sum_{|I|=k} (\partial^I \phi)^2,\\ |\partial^k b|^2 &:= \sum_{|I|=k} \sup_{\alpha} |\partial^I b^\alpha|^2,\\ |\partial^k a|^2 &:= \sum_{|I|=k} \sup_{\alpha,\beta} |\partial^I a^{\alpha\beta}|^2,\end{aligned}$$

for $k \in \mathbb{N}_0$, $I = (I_0, \dots, I_n)$ a multindex, $|I| = \sum_{\alpha=0}^n I_\alpha$ and $\partial^I = \partial_0^{I_0}\partial_1^{I_1}\dots\partial_n^{I_n}$. We will also use the notation ∂_x^k when restricting to derivatives in $\mathbb{R}^n$.

To prevent introducing more notation, we will make use of the following abuse of notation: we will take the positive constant "C" in any inequality to satisfy the "algebra of constants":

$$C \cdot C = C + C = C.$$

Theorem 3.1. *Let ϕ be a smooth solution to (3.1), such that $|\partial\phi|(t,\mathbf{x}) \to 0$ as $|\mathbf{x}| \to \infty$. Then, for $\epsilon_0 < 1$:*

(i) *There exists a constant $C = C(\epsilon_0, n) > 0$, such that for all $0 \le t_1 < t_2$*

$$\sup_{t\in[t_1,t_2]} ||\partial\phi||^2_{L^2(\mathbb{R}^n)}(t) \le C\left[||\partial\phi||^2_{L^2(\mathbb{R}^n)}(t_1) + \left(\int_{t_1}^{t_2} ||F||_{L^2(\mathbb{R}^n)}(t)\,dt\right)^2\right] e^{C\int_{t_1}^{t_2} ||\partial a||_{L^\infty(\mathbb{R}^n)}(t) + ||b||_{L^\infty(\mathbb{R}^n)}(t)\,dt}. \tag{3.4}$$

(ii) *Furthermore, for all $T > 0$:*

$$\begin{aligned}\sup_{t\in[0,T]} ||(\phi, \partial_t\phi)||_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}(t) \le C(1+T)&\left[||(\Phi,\Phi')||_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)} + \int_0^T ||F||_{L^2(\mathbb{R}^n)}(t)\,dt\right]\\ &\times e^{C\int_0^T ||\partial a||_{L^\infty(\mathbb{R}^n)}(t) + ||b||_{L^\infty(\mathbb{R}^n)}(t)}.\end{aligned} \tag{3.5}$$

Proof. We apply the Leibniz rule to write

$$\begin{aligned}0 =& -\partial_t\phi(a^{\alpha\beta}\partial_\alpha\partial_\beta\phi + b^\alpha\partial_\alpha\phi - F)\\ =& -a^{00}\partial_t\phi\partial_t^2\phi - 2a^{0i}\partial_t\phi\partial_t\partial_i\phi - a^{ij}\partial_t\phi\partial_i\partial_j\phi - b^\alpha\partial_t\phi\partial_\alpha\phi + \partial_t\phi F\\ =& \partial_t\left(-a^{00}\frac{1}{2}(\partial_t\phi)^2\right) - \partial_i\left(a^{0i}(\partial_t\phi)^2 + a^{ij}\partial_t\phi\partial_j\phi\right) + a^{ij}\partial_i\partial_t\phi\partial_j\phi\\ &+ \frac{1}{2}\partial_t a^{00}(\partial_t\phi)^2 + (\partial_t\phi)((\partial_i a^{\alpha i})\partial_\alpha\phi - b^\alpha\partial_\alpha\phi) + \partial_t\phi F\\ =& \partial_t\left(-\frac{1}{2}a^{00}(\partial_t\phi)^2 + \frac{1}{2}a^{ij}\partial_i\phi\partial_j\phi\right) + \partial_i\left(-a^{0i}(\partial_t\phi)^2 - a^{ij}\partial_t\phi\partial_j\phi\right)\\ &+ \frac{1}{2}(\partial_t a^{00})(\partial_t\phi)^2 - \frac{1}{2}\partial_t a^{ij}\partial_i\phi\partial_j\phi + (\partial_t\phi)((\partial_i a^{\alpha i})\partial_\alpha\phi - b^\alpha\partial_\alpha\phi) - \frac{1}{2}(\partial_t a^{ij})\partial_i\phi\partial_j\phi + \partial_t\phi F,\end{aligned} \tag{3.6}$$

where we used symmetry of a to arrive at the last equality.

We now integrate the terms above in the region $I^-_{(R,0)} \cap \{t_1 \le t \le t_2\}$. First of all the total derivative

terms contribute as boundary terms in the integral:

$$\begin{aligned}\int_{I^-_{(R,0)}\cap\{t_1\le t\le t_2\}} &\partial_t\left(-\frac{1}{2}a^{00}(\partial_t\phi)^2+\frac{1}{2}a^{ij}\partial_i\phi\partial_j\phi\right)+\partial_i\left(-a^{0i}(\partial_t\phi)^2-a^{ij}\partial_t\phi\partial_j\phi\right)d\mathbf{x}dt\\ &=\frac{1}{2}\int_{\{t=t_2\}\cap I^-_{(R,0)}}-a^{00}(\partial_t\phi)^2+a^{ij}\partial_i\phi\partial_j\phi\,d\mathbf{x}-\frac{1}{2}\int_{\{t=t_1\}\cap I^-_{(R,0)}}-a^{00}(\partial_t\phi)^2+a^{ij}\partial_i\phi\partial_j\phi\,d\mathbf{x}\\ &\quad+\frac{1}{2}\int_{\mathcal{C}^-_{(R,0)}\cap\{t_1\le t\le t_2\}}-a^{00}(\partial_t\phi)^2+a^{ij}\partial_i\phi\partial_j\phi+\frac{2x^i}{r}a^{0i}(\partial_t\phi)^2+\frac{2x^i}{r}a^{ij}\partial_t\phi\partial_j\phi\,d\mathbf{x},\end{aligned}$$

where we apply **Problem** 2.4 to obtain the integral over $\mathcal{C}^-_{(R,0)}$ with respect to the volume form $d\mathbf{x}$. Now we take $R\to\infty$ use that $|\partial\phi|(t,x)\to 0$ as $x\to\infty$ to conclude that the integral over $\mathcal{C}^-_{(R,0)}$ vanishes. We apply Cauchy–Schwarz two times to estimate

$$\begin{aligned}\sum_{i,j}a^{ij}\partial_i\phi\partial_j\phi&=\sum_{i,j}\delta^{ij}\partial_i\phi\partial_j\phi+\sum_{i,j}(a^{ij}-\delta^{ij})\partial_i\phi\partial_j\phi\\ &\ge\sum_i|\partial_i\phi|^2-\sum_i|\partial_i\phi|\sqrt{\sum_j|\partial_j\phi|^2}\sqrt{\sum_j|a^{ij}-\delta^{ij}|^2}\\ &\ge\sum_i|\partial_i\phi|^2-\sqrt{\sum_j|\partial_j\phi|^2}\sqrt{\sum_i|\partial_j\phi|^2}\sqrt{\sum_{i,j}|a^{ij}-\delta^{ij}|^2}\\ &\ge\left(1-\sqrt{\sum_{i,j}|a^{ij}-\delta^{ij}|^2}\right)\sum_i|\partial_i\phi|^2.\end{aligned}$$

Hence, assuming (3.2) and (3.3) with $\epsilon_0<1$, there exists a constant $C=C(\epsilon_0)>0$, such that

$$-a^{00}(\partial_t\phi)^2+\sum_{i,j}a^{ij}\partial_i\phi\partial_j\phi\ge\frac{1}{C}|\partial\phi|^2.$$

Applying the identity (3.6) and integrating over $[t_1,t_2]\times\mathbb{R}^n$, we then obtain:

$$||\partial\phi||^2_{L^2(\mathbb{R}^n)}(t_2)\le C||\partial\phi||^2_{L^2(\mathbb{R}^n)}(t_1)+C\int_{t_1}^{t_2}\int_{\mathbb{R}^n}(\sup_{\alpha,\beta,\gamma}|\partial_\alpha a^{\beta\gamma}|+\sup_\alpha|b^\alpha|)|\partial\phi|^2(t,\mathbf{x})\,d\mathbf{x}dt+C\int_{t_1}^{t_2}\int_{\mathbb{R}^n}|\partial_t\phi||F|(t,x)\,d\mathbf{x}dt$$

By Cauchy–Schwarz in $\mathbb{R}^n$, we obtain for all $0\le t_1\le t_2$:

$$\begin{aligned}||\partial\phi||^2_{L^2(\mathbb{R}^n)}(t_2)\le C||\partial\phi||^2_{L^2(\mathbb{R}^n)}(t_1)&+C\int_{t_1}^{t_2}(||\partial a||_{L^\infty(\mathbb{R}^n)}(t)+||b||_{L^\infty(\mathbb{R}^n)}(t))||\partial\phi||^2_{L^2(\mathbb{R}^n)}(t)dt\\ &+C\int_{t_1}^{t_2}||\partial_t\phi||_{L^2(\mathbb{R}^n)}(t)||F||_{L^2(\mathbb{R}^n)}(t)dt.\end{aligned}$$

Note that since the above inequality holds for all $t_2\ge t_1$, it remains valid if we take the supremum over $t\in[t_1,t_2]$ on the LHS.

By Young's inequality with ϵ:

$$AB\le\frac{\epsilon}{2}A^2+\frac{1}{2\epsilon}B^2,$$

with $A=\sup_{t\in[t_1,t_2]}||\partial_t\phi||_{L^2(\mathbb{R}^n)}(t)$, $B=C\int_{t_1}^{t_2}||F||_{L^2(\mathbb{R}^n)}(t)\,dt$ and $\epsilon=1$, we obtain

$$C\int_{t_1}^{t_2}||\partial_t\phi||_{L^2(\mathbb{R}^n)}(t)||F||_{L^2(\mathbb{R}^n)}(t)dt\le\frac{1}{2}\sup_{t\in[t_1,t_2]}||\partial_t\phi||^2_{L^2(\mathbb{R}^n)}(t)+\frac{C^2}{2}\left(\int_{t_1}^{t_2}||F||_{L^2(\mathbb{R}^n)}(t)dt\right)^2$$

and hence,

$$\sup_{t\in[t_1,t_2]} ||\partial\phi||^2_{L^2(\mathbb{R}^n)}(t) \leq C||\partial\phi||^2_{L^2(\mathbb{R}^n)}(t_1) + C\int_{t_1}^{t_2} (||\partial a||_{L^\infty(\mathbb{R}^n)}(t) + ||b||_{L^\infty(\mathbb{R}^n)}(t))||\partial\phi||^2_{L^2(\mathbb{R}^n)}(t)dt$$
$$+ C^2\left(\int_{t_1}^{t_2} ||F||_{L^2(\mathbb{R}^n)}(t)\,dt\right)^2.$$

To conclude (3.4), we will apply Grönwall inequality, which is proved below:

Lemma 3.2 (Grönwall inequality)**.** *Let $f \in C^0([0,T];\mathbb{R})$ be a non-negative function and let $g : [0,T] \to \mathbb{R}$ be Lebesgue/Riemann integrable and non-negative. Assume that there exists a constant $D > 0$, such that*

$$f(t) \leq D + \int_0^t f(s)g(s)\,ds \quad \forall t \in [0,T]. \tag{3.7}$$

Then

$$f(t) \leq De^{\int_0^t g(s)\,ds} \quad \forall t \in [0,T].$$

To obtain (3.4), we take $t_1 = 0$ and $t_2 = T$ and we control $||\phi|||_{L^2(\mathbb{R}^n)}(t)$ as follows: by the Fundamental Theorem of Calculus, followed by two applications of the Minkowski inequality[16]

$$\begin{aligned}||\phi||_{L^2(\mathbb{R}^n)}(t) &= \left|\left|\phi(0,\cdot) + \int_0^t \partial_t\phi(t,\cdot)\,dt'\right|\right|_{L^2(\mathbb{R}^n)}(t)\\ &\leq ||\phi||_{L^2(\mathbb{R}^n)}(0) + \left|\left|\int_0^t \partial_t\phi(t,\cdot)\,dt'\right|\right|_{L^2(\mathbb{R}^n)}(t)\\ &\leq ||\phi||_{L^2(\mathbb{R}^n)}(0) + \int_0^t ||\partial_t\phi(t,\cdot)||_{L^2(\mathbb{R}^n)}(t')\,dt'.\end{aligned}$$

□

Proof of Lemma 3.2. In anticipation of later analysis for nonlinear wave equations, we will prove the Grönwall inequality using a *continuity argument*, which is also known as the *bootstrap method*. You may think of this as an induction argument on $\mathbb{R}$, rather than $\mathbb{N}$.

We start with a *bootstrap assumption*: there exists an $\epsilon > 0$, such that

$$f(t) \leq (1+\epsilon)De^{(1+\epsilon)\int_0^t g(s)\,ds} \quad \forall t \in [0,T]. \tag{3.8}$$

Now define the subset $S \subset [0,T]$ as follows:

$$S = \{\tau \in [0,T] \,|\, (3.8) \text{ holds for every } t \in [0,\tau]\}.$$

The aim will be to show that $S = [0,T]$. Since $[0,T]$ is connected, this follows if:

- S is non-empty,
- S is closed,
- S is open.

[16] Discrete Minkowski inequality: if $f, g \in L^p(\mathbb{R}^n)$, then $||f+g||_{L^p(\mathbb{R}^n)} \leq ||f||_{L^p(\mathbb{R}^n)} + ||g||_{L^p(\mathbb{R}^n)}$. Integral Minkowski inequality: for $f \in L^1([0,T];L^p(\mathbb{R}^n))$, $||\int_0^T f(t)\,dt||_{L^p(\mathbb{R}^n)} \leq \int_0^t ||f(t)||_{L^p(\mathbb{R}^n)}\,dt$.

Non-emptyness of S follows from the fact that $0 \in S$, by the assumption (3.7) evaluated at $t = 0$.

Closedness of S follows from continuity of f. Indeed, consider a sequence $t_i \in S$, such that $t_i \to t_* \in [0, T]$. Then we must have that $[0, t_*) \subset S$, since for every $t \in [0, t_*)$, we can find $i \in \mathbb{N}$ such that $t_i > t$, so $[0, t] \in S$. Since (3.8) holds on $[0, t_*)$, we can appeal to continuity of $f(t) - (1+\epsilon)De^{(1+\epsilon)\int_0^t g(s)\,ds}$ in t to conclude that $f(t_*) - (1+\epsilon)De^{(1+\epsilon)\int_0^{t_*} g(s)\,ds} \leq 0$ so $t_* \in S$.

Openness of S is the most difficult part. For this, we will show that we can *improve* the bootstrap assumption (3.8), by proving that for all $t \in S$, $t \neq T$:

$$f(t) \leq De^{(1+\epsilon)\int_0^t g(s)\,ds}. \tag{3.9}$$

Indeed, we can then appeal to continuity of f to conclude that there exists $\delta > 0$ such that $t + \delta \in S$. To derive (3.9), we apply both (3.7) and (3.8), as follows:

$$\begin{aligned} f(t) &\leq D + \int_0^t f(s)g(s)\,ds \leq D + (1+\epsilon)D\int_0^t g(s)e^{(1+\epsilon)\int_0^s g(r)\,dr}\,ds, \\ &= D + D\int_0^t \frac{d}{ds}e^{(1+\epsilon)\int_0^s g(r)\,dr}\,ds \\ &= De^{(1+\epsilon)\int_0^s g(r)\,dr}\,ds. \end{aligned}$$

Hence, we can conclude that $S = [0, T]$. Finally, $\epsilon > 0$ appearing in (3.8) was arbitrary, we can take the limit $\epsilon \downarrow 0$ and conclude that (3.8) also holds for all $t \in [0, T]$ with $\epsilon = 0$. □

Corollary 3.3. *Let $k \in \mathbb{N}_{\geq 1}$ and let ϕ be a smooth solution to (3.1) such that $|\partial\partial_x^I\phi|(t, \mathbf{x}) \to 0$ for all $|I| \leq k-1$ as $|\mathbf{x}| \to \infty$. Then, for all $T > 0$, there exists a $C = C(\epsilon_0, n, k) > 0$, such that*

$$\begin{aligned} &\sup_{t\in[0,T]} ||(\phi, \partial_t\phi)||_{H^k(\mathbb{R}^n)\times H^{k-1}(\mathbb{R}^n)}(t) \\ &\quad\leq C(1+T)\Bigg[||(\Phi, \Phi')||_{H^k(\mathbb{R}^n)\times H^{k-1}(\mathbb{R}^n)} + \int_0^T ||F||_{H^{k-1}(\mathbb{R}^n)}(t)\,dt \\ &\qquad + \sum_{1\leq|I|+|J|\leq k-2}\int_0^T ||\partial_x^I\partial_x a\cdot\partial^2\partial_x^J\phi||_{L^2(\mathbb{R}^n)}(t)\,dt + \sum_{1\leq|I|+|J|\leq k-2}\int_0^T ||\partial_x^I\partial_x b\cdot\partial\partial_x^J\phi||_{L^2(\mathbb{R}^n)}(t)\,dt\Bigg] \\ &\quad\times e^{C\int_0^T ||\partial a||_{L^\infty(\mathbb{R}^n)}(t)+||b||_{L^\infty(\mathbb{R}^n)}(t)\,dt}, \end{aligned} \tag{3.10}$$

where we used the following schematic notation: for scalar, vector or matrix quantities A, B

$$\partial^p A\cdot\partial^q B$$

groups together all terms which are products of coefficients of p-th order derivatives of A with coefficients of q-th order derivatives of B. Furthermore, we denote $\partial A = \partial^1 A$.

Proof. See **Problem** 3.2. □

Equipped with a higher-order energy estimate, we will be able to prove global existence by applying standard results from functional analysis. First, we need to extend the energy estimates to the dual spaces:

$$H^{-m}(\mathbb{R}^n) = (H^m(\mathbb{R}^n))^* = \{f : H^m(\mathbb{R}^n) \to \mathbb{R} \text{ bounded and linear}\},$$

with

$$||f||_{H^{-m}(\mathbb{R}^n)} = \sup_{v\in H^m(\mathbb{R}^n)\setminus\{0\}} \frac{|f(v)|}{||v||_{H^m(\mathbb{R}^n)}}.$$

In the proposition below, we will extend the energy estimates to the spaces $H^{-m}(\mathbb{R}^n)$. For this we will use that, given $f \in \bigcap_{k\in\mathbb{N}} H^k(\mathbb{R}^n)$, there exists a unique solution $u \in C^\infty(\mathbb{R}^n) \cap \bigcap_{k\in\mathbb{N}} H^k(\mathbb{R}^n)$ to

$(I-\Delta)u = f$. This is proved in Theorem B.1. More generally, we can define $(I-\Delta)^{-k}f$ to be the solution u to $(1-\Delta)^k u = f$. We can then express for all $m \in \mathbb{N}_0$:

$$||(1-\Delta)^{-m}f||_{H^m(\mathbb{R}^n)} \leq ||f||_{H^{-m}(\mathbb{R}^n)} \leq C(m)||(1-\Delta)^{-m}f||_{H^m(\mathbb{R}^n)} \quad \text{if } m \geq 1, \tag{3.11}$$

$$c(m)||(1-\Delta)^{-1}f||_{H^{-m}(\mathbb{R}^n)} \leq ||f||_{H^{-m-2}(\mathbb{R}^n)} \leq C(m)||(1-\Delta)^{-1}f||_{H^{-m}(\mathbb{R}^n)} \tag{3.12}$$

where $c(m), C(m) > 0$ are positive constants, and we interpret $||f||_{H^{-m}(\mathbb{R}^n)} = ||\langle f, \cdot\rangle_{L^2(\mathbb{R}^n)}||_{H^{-m}(\mathbb{R}^n)}$, with $\langle f, \cdot\rangle_{L^2(\mathbb{R}^n)} \in L^2(\mathbb{R}^n)^* \subset H^{-m}(\mathbb{R}^n)$ the canonical dual element associated to $f \in L^2(\mathbb{R}^n)$ via the Riesz representation theorem.[17] The identities (3.11) – (3.12) are proved in **Problem** 3.1.

Proposition 3.4. *Let $\phi \in C^\infty([0,T]; \bigcap_{k\in\mathbb{N}_0} H^k(\mathbb{R}^n))$ and consider the differential operator*

$$L = a^{\alpha\beta}\partial_\alpha\partial_\beta(\cdot) + b^\alpha\partial_\alpha(\cdot).$$

Then, for all $m \in \mathbb{Z}$, there exists a constant $C_{T,a,b} = C(a,b,n,m,T) > 0$, such that:

$$\begin{aligned} &\sup_{t\in[0,T]} ||(\phi, \partial_t\phi)||_{H^m(\mathbb{R}^n)\times H^{m-1}(\mathbb{R}^n)}(t) \\ &\quad \leq C_{T,a,b}\left[||(\phi,\partial_t\phi)||_{H^m(\mathbb{R}^n)\times H^{m-1}(\mathbb{R}^n)}(0) + \int_0^T ||L\phi||_{H^{m-1}(\mathbb{R}^n)}(t)\,dt\right]. \end{aligned} \tag{3.13}$$

Proof. We first consider the case $m \geq 1$. Applying (3.10), together with L^∞ boundedness of arbitrarily many derivatives of $a^{\alpha\beta}$ and b^α and Grönwall to absorb the terms involving ϕ on the RHS into the LHS, where we moreover need to apply

$$-a^{00}\partial_t^2\phi = 2a^{0i}\partial_i\partial_t\phi + a^{ij}\partial_i\partial_j\phi + b^\alpha\partial_\alpha\phi - L\phi$$

to absorb terms of the form $\partial_x^J\partial_t^2\phi$.[18]

In order to extend to non-positive m, we do a proof by induction. Assume as our induction hypothesis that (3.13) holds for some value of $m \in \mathbb{Z}$. Let $\widetilde{\phi}(t,\cdot)$ be the unique solution to $(I-\Delta)\widetilde{\phi}(t,\cdot) = \phi(t,\cdot)$, defined in Theorem B.1. Applying Theorem B.1 to $\partial_t^k\phi(t,\cdot)$, we can conclude that $\widetilde{\phi} \in C^\infty([0,T]; \bigcap_{k\in\mathbb{N}_0} H^k(\mathbb{R}^n))$. Then by the induction hypothesis, (3.11)–(3.12) and (B.2):

$$\begin{aligned} \sup_{t\in[0,T]} ||(\phi,\partial_t\phi)||_{H^{m-2}(\mathbb{R}^n)\times H^{m-3}(\mathbb{R}^n)}(t) &\leq \sup_{t\in[0,T]} ||(\widetilde{\phi},\partial_t\widetilde{\phi})||_{H^m(\mathbb{R}^n)\times H^{m-1}(\mathbb{R}^n)}(t) \\ &\leq C_{T,a,b}\left[||(\widetilde{\phi},\partial_t\widetilde{\phi})||_{H^m(\mathbb{R}^n)\times H^{m-1}(\mathbb{R}^n)}(0) + \int_0^T ||L\widetilde{\phi}||_{H^{m-1}(\mathbb{R}^n)}(t)\,dt\right] \\ &\leq C_{T,a,b}\left[C(m)||(\phi,\partial_t\phi)||_{H^{m-2}(\mathbb{R}^n)\times H^{m-3}(\mathbb{R}^n)}(0) + \int_0^T ||L\widetilde{\phi}||_{H^{m-1}(\mathbb{R}^n)}(t)\,dt\right]. \end{aligned}$$

Note that

$$|L\phi - (I-\Delta)L\widetilde{\phi}| = |[L,(I-\Delta)]\widetilde{\phi}| \leq C_{a,b}\sum_{1\leq J\leq 3}|\partial^J\widetilde{\phi}|,$$

so, by (3.11)–(3.12) and (B.2),

$$\int_0^T ||L\widetilde{\phi}||_{H^{m-1}(\mathbb{R}^n)}(t)\,dt \leq C(m)\int_0^T ||(I-\Delta)L\widetilde{\phi}||_{H^{m-3}(\mathbb{R}^n)}(t)\,dt \leq C\int_0^T ||L\phi||_{H^{m-3}(\mathbb{R}^n)}(t)\,dt + C\int_0^T ||\widetilde{\phi}||_{H^m(\mathbb{R}^n)}(t)\,dt$$

and we have that

$$\begin{aligned} &\sup_{t\in[0,T]} ||(\phi,\partial_t\phi)||_{H^{m-2}(\mathbb{R}^n)\times H^{m-3}(\mathbb{R}^n)}(t) \\ &\quad \leq C_{T,a,b}\left[||(\phi,\partial_t\phi)||_{H^{m-2}(\mathbb{R}^n)\times H^{m-3}(\mathbb{R}^n)}(0) + \int_0^T ||L\phi||_{H^{m-3}(\mathbb{R}^n)}(t)\,dt + \int_0^T ||\phi||_{H^{m-2}(\mathbb{R}^n)}(t)\,dt\right]. \end{aligned}$$

[17] Riesz Representation theorem: Let H be a Hilbert space. Then for all $f \in H^*$, there exists a unique $u \in H$ such that $f(v) = \langle u, v\rangle_H$ for all $v \in H$.

[18] Exercise: Convince yourself that the condition $|\partial\partial_x^J\phi|(t,\mathbf{x}) \to 0$ as $|\mathbf{x}| \to \infty$ for all $|J| \leq m-1$ follows from the assumption that $\phi \in C^\infty([0,T]; \bigcap_{k\in\mathbb{N}_0} H^k(\mathbb{R}^n))$.

Now we apply Grönwall to get rid on the last integral on the RHS above and conclude (3.13) with m replaced by $m-2$ (and a different constant $C>0$). Hence starting with $m=1$ or $m=2$ and applying the above induction argument gives (3.13) for an appropriate choice of constant $C_{T,a,b}$ (depending also on m) and for all $m \in \mathbb{Z}$. □

3.2 Global well-posedness

In this section, we will apply the energy estimates from the previous section to deduce global well-posedness for general linear wave equations. There are multiple ways of obtaining global well-posedness, but what they all have in common is that they rely on energy estimates. The method we will apply to go from energy estimates to global well-posedness relies on results from introductory courses in functional analysis, which will be reviewed.

To simplify the exposition, we restrict to L of the form

$$L = \partial_\alpha(a^{\alpha\beta}\partial_\beta(\cdot))$$

and we define $L^* = L$. The more general case will be considered in **Problem** 3.5.

Definition 3.1. *Let $k \in \mathbb{N}$, $F \in L^1([0,T);H^{k-1}(\mathbb{R}^n))$ and $(\Phi,\Phi') \in H^k(\mathbb{R}^n)\times H^{k-1}(\mathbb{R}^n)$. We say $\phi \in L^\infty([0,T);H^k(\mathbb{R}^n))$ is a **weak solution to** (3.1) if it satisfies the equation:*

$$\begin{aligned}\int_0^T\int_{\mathbb{R}^n} \phi\cdot L^*v\,d\mathbf{x}dt &= \int_0^T\int_{\mathbb{R}^n} F\cdot v\,d\mathbf{x}dt\\ &+\int_{\mathbb{R}^n} a^{00}(0,\mathbf{x})(\Phi'(\mathbf{x})v(0,\mathbf{x})-\Phi(\mathbf{x})\partial_t v(0,\mathbf{x})) - 2a^{0i}(0,\mathbf{x})\Phi(\mathbf{x})\partial_i v(0,\mathbf{x}) - \partial_i a^{0i}(0,\mathbf{x})\Phi(\mathbf{x})v(0,\mathbf{x})\,d\mathbf{x}\end{aligned} \tag{3.14}$$

for all $v \in C_c^\infty((-\infty,T)\times\mathbb{R}^n)$.

We say $\phi \in C^2([0,T)\times\mathbb{R}^n)$ is a classical solution *to (3.1) it satisfies the equation (3.1) pointwise.*

Exercise: Verify that classical solutions to (3.1) are also weak solutions to (3.1).

Theorem 3.5. *Let $k \in \mathbb{N}$, $T>0$, $F \in L^1([0,T);H^{k-1}(\mathbb{R}^n))$ and $(\Phi,\Phi') \in H^k(\mathbb{R}^n)\times H^{k-1}(\mathbb{R}^n)$. Then exists a unique weak solution $\phi \in C^0([0,T);H^k(\mathbb{R}^n))\cap C^1([0,T);H^{k-1}(\mathbb{R}^n))$ to (3.1) satisfying*

$$\sup_{t\in[0,T)} ||(\phi,\partial_t\phi)||_{H^k(\mathbb{R}^n)\times H^{k-1}(\mathbb{R}^n)}(t) \leq C_{T,a,b}\left[||(\Phi,\Phi')||_{H^k(\mathbb{R}^n)\times H^{k-1}(\mathbb{R}^n)} + \int_0^T ||F||_{H^{k-1}(\mathbb{R}^n)}(t)\,dt\right]. \tag{3.15}$$

For $k > 2+\frac{n}{2}$, $\phi \in C^2([0,T)\times\mathbb{R}^n)$ is a classical solution to (3.1).

Proof. The proof can be divided into four steps.

Step 1: uniqueness Given existence and (3.15), uniqueness follows immediately from linearity. Indeed, suppose ϕ_1 and ϕ_2 are two weak solutions to (3.1) satisfying (3.15). Then $\phi_1-\phi_2$ satisfies (3.1) with zero initial data and vanishing F, so by (3.15), $\phi_1-\phi_2 \equiv 0$.

Step 2: existence of weak solutions with trivial data To prove existence, we first consider the case where $(\Phi,\Phi')=0$. Consider the map

$$L^* : C_c^\infty((-\infty,T)\times\mathbb{R}^n) \to C_c^\infty((-\infty,T)\times\mathbb{R}^n).$$

The above map is injective by (3.13) and $L=L^*$, so given $f \in L^*(C_c^\infty((-\infty,T)\times\mathbb{R}^n))$ there exists a unique $v \in C_c^\infty((-\infty,T)\times\mathbb{R}^n)$ such that $L^*v=f$. The map

$$A : L^*v \mapsto \int_{-\infty}^T\int_{\mathbb{R}^n} v\cdot F\,d\mathbf{x}dt$$

is therefore well-defined and also linear.

By trivially extending F to $(-\infty, 0)$ as $F(t,\cdot) = 0$ for $t < 0$, we can estimate

$$\begin{aligned}|A(L^*v)| &= \left|\int_{-\infty}^{T}\int_{\mathbb{R}^n} v \cdot F \, d\mathbf{x}dt\right| \leq \int_{-\infty}^{T} ||v||_{H^{-(k-1)}(\mathbb{R}^n)}(t)||F||_{H^{k-1}(\mathbb{R}^n)}(t)\, dt \\ &\leq \sup_{t\in(-\infty,T)} ||v||_{H^{-k+1}(\mathbb{R}^n)}(t) \int_0^T ||F||_{H^{k-1}(\mathbb{R}^n)}(t)\, dt \\ &\overset{(3.13)}{\leq} C\int_0^T ||F||_{H^{k-1}(\mathbb{R}^n)}(t)\, dt \int_{-\infty}^{T} ||L^*v||_{H^{-k}(\mathbb{R}^n)}\, dt,\end{aligned}$$

where in the first inequality above, we identified $v(t,\cdot)$ with the bounded linear operator $\int_{\mathbb{R}^n} v(t,\cdot)\cdot(\cdot)\, d\mathbf{x} \in H^{(-k-1)}(\mathbb{R}^n)$.

Hence, we have that A is a bounded linear functional on the subspace $L^*(C_c^\infty((-\infty,T)\times\mathbb{R}^n)) \subset L^1((-\infty,T); H^{-k}(\mathbb{R}^n))$. We can then apply the Hahn–Banach theorem to extend A as a bounded linear functional on the larger space $L^1((-\infty,T); H^{-k}(\mathbb{R}^n))$:

Theorem 3.6 (Hahn–Banach)**.** *Let $(X, ||\cdot||_X)$ a normed vector space and $Y \subsetneq X$ a subspace with the subspace norm. Suppose $A \in Y^*$ (a bounded linear functional). Then there exists $\widetilde{A} \in X^*$ such that $\widetilde{A}|_Y = A$ and $||\widetilde{A}||_{X^*} = ||A||_{Y^*}$.*[19]

We therefore conclude that there exists a $\phi \in (L^1((-\infty,T); H^{-k}(\mathbb{R}^n)))^*$ such that $\phi|_{L^*(C_c^\infty((-\infty,T)\times\mathbb{R}^n))} = A$ and

$$||\phi||_{(L^1(-\infty,T);H^{-k}(\mathbb{R}^n)))^*} \leq C\int_0^T ||F||_{H^{k-1}(\mathbb{R}^n)}(t)\, dt.$$

Now we appeal to the following duality property (see (iii) of Theorem A.5):

$$L^1((-\infty,T); H^{-k}(\mathbb{R}^n))^* = L^\infty((-\infty,T); H^k(\mathbb{R}^n))$$

and we assume $\phi \equiv 0$ in $t \leq 0$ to obtain $\phi \in L^\infty([0,T); H^k(\mathbb{R}^n))$, together with the estimate:

$$||\phi||_{L^\infty([0,T);H^k(\mathbb{R}^n))} \leq C\int_0^T ||F||_{H^{k-1}(\mathbb{R}^n)}(t)\, dt.$$

By definition of the linear operator A:

$$\int_0^T\int_{\mathbb{R}^n} \phi\cdot L^*v\, d\mathbf{x}dt = \int_0^T\int_{\mathbb{R}^n} v\cdot F\, d\mathbf{x}dt.$$

for all $v \in C_c^\infty((-\infty,T)\times\mathbb{R}^n)$. Hence, ϕ is a weak solution to (3.1) arising from trivial initial data.

Step 3: improved regularity Consider a sequence $F_m \in C_c^\infty([0,T)\times\mathbb{R}^n)$ such that $F_m \to F$ in $L^1([0,T); H^{k-1}(\mathbb{R}^n))$. Then, by Step 2, the corresponding weak solutions ϕ_m are elements of $L^\infty([0,T); \bigcap_{l\in\mathbb{N}} H^l(\mathbb{R}^n))$. Let $v \in C_c^\infty([0,T)\times\mathbb{R}^{n+1})$, then

$$\partial_t(a^{00}\partial_t v) + a^{0i}\partial_i\partial_t v + \partial_i(a^{0i}\partial_t v) = -\partial_i(a^{ij}\partial_j v) - (\partial_t a^{0i})\partial_i v + L^*v. \tag{3.16}$$

By using (3.17) and the fact that $\phi_m \in L^\infty([0,T); \bigcap_{l\in\mathbb{N}} H^l(\mathbb{R}^n))$, we can integrate by parts on $\mathbb{R}^n$ and rearrange the equation:

$$\int_0^T\int_{\mathbb{R}^n} \phi_m\cdot L^*v\, d\mathbf{x}dt = \int_0^T\int_{\mathbb{R}^n} v\cdot F_m\, d\mathbf{x}dt$$

[19] In the generality stated here, the Hahn–Banach Theorem requires Zorn's lemma, which is equivalent to the axiom of choice. However, when applied to separable Hilbert spaces, the axiom of countable choice is sufficient.

as follows:

$$\int_0^T \int_{\mathbb{R}^n} \phi_m \cdot [\partial_t(a^{00}\partial_t v) + a^{0i}\partial_i\partial_t v + \partial_i(a^{0i}\partial_t v)]\, d\mathbf{x}dt = \int_0^T \int_{\mathbb{R}^n} v \cdot (F_m + \partial_i(\partial_t a^{0i}\phi_m) - \partial_i(a^{ij}\partial_j\phi_m))\, d\mathbf{x}dt. \tag{3.17}$$

Let $X = a^{00}\partial_t + 2a^{0i}\partial_i$, and let $\gamma(t)$ denote the integrals curves of X parametrized by t, which are smooth by the smoothness assumption on the coefficients $a^{\alpha\beta}$. We can then write the LHS of (3.17) as follows:

$$\int_0^T \int_{\mathbb{R}^n} e^{-\int_0^t (\partial_t a^{00} + \partial_i a^{0i})|_{\gamma(t')}\, dt'} \phi_m \cdot X(e^{\int_0^t (\partial_t a^{00} + \partial_i a^{0i})|_{\gamma(t')}\, dt'} \partial_t v)\, d\mathbf{x}dt.$$

Hence, after integrating by parts in X and ∂_t, we must have that the following weak derivative is well-defined:

$$\begin{aligned}\partial_t(e^{\int_0^t (\partial_t a^{00} + \partial_i a^{0i})|_{\gamma(t')}dt'} X(e^{-\int_0^t (\partial_t a^{00} + \partial_i a^{0i})|_{\gamma(t')}dt'}\phi_m)) &= F_m + \partial_i(\partial_t a^{0i}\phi_m) - \partial_i(a^{ij}\partial_j\phi_m) \\ &\in L^\infty([0,T); \bigcap_{l\in\mathbb{N}} H^l(\mathbb{R}^n)) \subset L^1([0,T); \bigcap_{l\in\mathbb{N}} H^l(\mathbb{R}^n)).\end{aligned} \tag{3.18}$$

It will be convenient to use the shorthand notation $e(x) = e^{\int_0^t (\partial_t a^{00} + \partial_i a^{0i})|_{\gamma(t')}dt'}$.

By the Fundamental Theorem of Calculus in t, using that $\phi_m(t,\cdot) \equiv 0$ for $t \leq 0$ by construction, we have that $X(e(x)^{-1}\phi_m) \in C^0([0,T); \bigcap_{l\in\mathbb{N}} H^l(\mathbb{R}^n))$. After integrating along γ and by applying the Fundamental Theorem of Calculus once more, but now in the X direction, we moreover obtain $\phi_m \in C^0([0,T); \bigcap_{l\in\mathbb{N}} H^l(\mathbb{R}^n))$, from which it moreover automatically follows that $\partial_i\phi_m \in C^0([0,T); \bigcap_{l\in\mathbb{N}} H^l(\mathbb{R}^n))$ for all $1 \leq i \leq n$.

By definition of X, it then follows that

$$\partial_t\phi_m = \frac{1}{a^{00}}\left(X\phi_m - 2a^{0i}\partial_i\phi\right) \in C^0([0,T); \bigcap_{l\in\mathbb{N}} H^l(\mathbb{R}^n)).$$

Now, using the fact that also $\partial_i\partial_t\phi_m, \partial_i\partial_j\phi_m \in C^0([0,T); \bigcap_{l\in\mathbb{N}} H^l(\mathbb{R}^n))$ for all $1 \leq i,j \leq n$, the right-hand side of (3.18) is an element of $C^0([0,T); \bigcap_{l\in\mathbb{N}} H^l(\mathbb{R}^n))$, so after applying (3.18) we conclude that:

$$\partial_t^2\phi_m \in C^0([0,T); \bigcap_{l\in\mathbb{N}} H^l(\mathbb{R}^n)).$$

Hence, ϕ_m is a classical solution to $L\phi_m = F_m$.

Applying (3.17) with v replaced by $\partial_t v$ and integrating by parts in t repeating the above, we can in fact conclude that: $\partial_t^3\phi_m \in C^0([0,T); \bigcap_{l\in\mathbb{N}} H^l(\mathbb{R}^n))$. Repeatedly acting with ∂_t on v and integrating by parts allows us to conclude $\phi_m \in C^\infty([0,T); \bigcap_{l\in\mathbb{N}} H^l(\mathbb{R}^n))$.

We are therefore allowed to apply (3.13) to obtain

$$||(\phi_m, \partial_t\phi_m)||_{C^0([0,T);H^k(\mathbb{R}^n))\times C^0([0,T);H^{k-1}(\mathbb{R}^n))} \leq C\int_0^T ||F_m||_{H^{k-1}(\mathbb{R}^n)}(t)\, dt.$$

Using that the constant C above is uniform in m together with the assumed convergence of $F_m \to F$, we can conclude that the sequence (ϕ_m) is Cauchy in $C^0([0,T); H^k(\mathbb{R}^n))$ and $C^1([0,T); H^{k-1}(\mathbb{R}^n))$ and converges to ϕ, satisfying

$$||(\phi, \partial_t\phi)||_{C^0([0,T);H^k(\mathbb{R}^n))\times C^0([0,T);H^{k-1}(\mathbb{R}^n))} \leq C\int_0^T ||F||_{H^{k-1}(\mathbb{R}^n)}(t)\, dt$$

so in particular

$$\phi \in C^0([0,T); H^k(\mathbb{R}^n)) \cap C^1([0,T); H^{k-1}(\mathbb{R}^n)).$$

Step 4: non-trivial data Now consider non-trivial initial data (Φ, Φ') and consider a sequence $(\Phi_m, \Phi'_m) \in C_c^\infty(\mathbb{R}^n) \times C_c^\infty(\mathbb{R}^n)$ converging to (Φ, Φ') with respect to $H^k(\mathbb{R}^n) \times H^{k-1}(\mathbb{R}^n)$ and F_m as in Step 2 above. Let $u_m = \Phi_m + t\Phi'_m$. Then $Lu_m \in C^\infty([0,T); \cap_{k\in\mathbb{N}} H^k(\mathbb{R}^n))$. We then apply the above argument with F_m replaced by $F_m - Lu_m$, to conclude that there exists a unique $\hat{\phi}_m \in C^\infty([0,T); \bigcap_{k\in\mathbb{N}} H^k(\mathbb{R}^n))$ such that

$$\int_0^T \int_{\mathbb{R}^n} \hat{\phi}_m \cdot L^* v \, d\mathbf{x}dt = \int_0^T \int_{\mathbb{R}^n} v \cdot (F_m - Lu_m) \, d\mathbf{x}dt$$

and by smoothness of u_m and $\hat{\phi}_m$, we can conclude from the above equation that $L\phi_m = F_m$ for $\phi_m := \hat{\phi}_m + u_m$ and $(\hat{\phi}_m, \partial_t \hat{\phi}_m) = (0,0)$. Now, we can apply (3.13) to conclude that

$$||(\phi_m, \partial_t \phi_m)||_{C^0([0,T);H^k(\mathbb{R}^n))\times C^1([0,T);H^{k-1}(\mathbb{R}^n))} \leq C(||(\Phi_m, \Phi'_m)||_{H^k(\mathbb{R}^n)\times H^{k-1}(\mathbb{R}^n)} + C\int_0^T ||F_m||_{H^{k-1}(\mathbb{R}^n)}(t)\, dt.$$

Since the above estimate is uniform in m, we can conclude that $(\phi_m, \partial_t \phi_m)$ is Cauchy in $C^0([0,T); H^k(\mathbb{R}^n) \times H^{k-1}(\mathbb{R}^n))$ so it converges to a limit ϕ which is moreover a weak solution to (3.1) data (Φ, Φ').

EXERCISE: Show that for $k > 2 + \frac{n}{2}$, $\phi \in C^2([0,T) \times \mathbb{R}^n)$ is a classical solution to (3.1), using Theorem A.2. □

3.3 Problems

Problem 3.1. *Let $f \in \bigcap_{m\in\mathbb{N}} H^m(\mathbb{R}^n)$. By Theorem B.1, we know that for all $k \in \mathbb{N}$ there exists a unique solution $u_k \in \bigcap_{m\in\mathbb{N}} H^m(\mathbb{R}^n)$ to the equation $(1-\Delta)^k u = f$. We denote $(1-\Delta)^{-k} f = u_k$. Show that*

$$\begin{aligned} ||(1-\Delta)^{-m} f||_{H^m(\mathbb{R}^n)} \leq ||f||_{H^{-m}(\mathbb{R}^n)} \leq C(m)||(1-\Delta)^{-m} f||_{H^m(\mathbb{R}^n)} \quad \text{if } m \geq 1, \\ c(m)||(1-\Delta)^{-1} f||_{H^{-m}(\mathbb{R}^n)} \leq ||f||_{H^{-m-2}(\mathbb{R}^n)} \leq C(m)||(1-\Delta)^{-1} f||_{H^{-m}(\mathbb{R}^n)}, \end{aligned}$$

where $C(m)$ and $c(m)$ are positive constants and we interpret $||f||_{H^{-m}(\mathbb{R}^n)} = ||\langle f, \cdot\rangle_{L^2(\mathbb{R}^n)}||_{H^{-m}(\mathbb{R}^n)}$, with $\langle f, \cdot\rangle_{L^2(\mathbb{R}^n)} \in L^2(\mathbb{R}^n)^* \subset H^{-m}(\mathbb{R}^n)$ the canonical dual element associated to $f \in L^2(\mathbb{R}^n)$ via the Riesz representation theorem.

Problem 3.2. *Let ϕ be a smooth solution to (3.1) (with $d \equiv 0$) such that $|\partial \partial_x^I \phi|(t, \mathbf{x}) \to 0$ as $|\mathbf{x}| \to \infty$ with $|I| \leq k-1 \in \mathbb{N}$.*

(a) Consider $\partial_x^I \phi$ with $|I| = k-1$ and show that it satisfies an equation of the form

$$a^{\alpha\beta}\partial_\alpha \partial_\beta \partial_x^I \phi + b \cdot \partial \partial_x^I \phi = \partial_x^I F + \sum_{|J|+|M|=k-2} \partial_x^J \partial_x a \cdot \partial^2 \partial_x^M \phi + \sum_{|J|+|M|=k-2} \partial_x^J \partial_x b \cdot \partial \partial_x^M \phi. \tag{3.19}$$

Recall the following schematic notation: for scalar, vector or matrix quantities A, B

$$\partial^p A \cdot \partial^q B$$

groups together all terms which are products of coefficients of p-th order derivatives of A with coefficients of q-th order derivatives of B.

(b) Apply (3.19) to prove Corollary 3.3.

Problem 3.3. *Formulate and prove an analogue of Theorem 3.1 for equations (3.1) with d non-vanishing.*

Problem 3.4. *Explain precisely in what sense Theorem 3.5 implies linear wave equations of the form $\partial_\alpha(a^{\alpha\beta}\partial_\beta \phi) = F$ satisfy Hadamard's three criteria for well-posedness. Is the well-posedness global for these equations?*

Problem 3.5. *Let $L = \partial_\alpha(a^{\alpha\beta}\partial_\beta(\cdot)) + b^\alpha \partial_\alpha(\cdot) + d \cdot (\cdot)$.*

a) Write down the corresponding differential operator L^* that ensures that C^2 solutions to $Lu = F$ with initial data (Φ, Φ') satisfy (3.14).

b) Given ***Problem*** *3.3, prove the analogue of Proposition 3.4 for the above, more general, form of L.*

c) Explain how the proof of Theorem 3.5 changes when replacing $L = \partial_\alpha(a^{\alpha\beta}\partial_\beta(\cdot))$ by the above more general form of L.

Problem 3.6. *Consider $a^{\alpha\beta}$ with $a^{00} = -1$, $a^{0i} = 0$ and $a^{ij} = f(|\mathbf{x}|)^2\delta^{ij}$, with $f : [0,\infty) \to \mathbb{R}_+$ smooth and δ^{ij} the Kronecker delta.*

a) Let $\gamma : \mathbb{R} \to \mathbb{R}^{n+1}$ such that $\gamma(0) = 0$ and the tangent $\dot\gamma(s) = \frac{d\gamma}{ds}(s)$ satisfies

$$a^{\alpha\beta}\dot\gamma^\alpha\dot\gamma^\beta = 0.$$

Consider the following parametrization $s = r = \sqrt{\sum_{i=1}^n (x^i)^2}$ and express $\gamma(r)$ in terms of f.

b) Consider the wave equation:

$$\begin{aligned}&\partial_\alpha(a^{\alpha\beta}\partial_\beta\phi) = 0 \quad \text{in } \mathbb{R}^{n+1},\\ &(\phi, \partial_t\phi)|_{t=0} = (\Phi, \Phi') \in C_c^\infty(\mathbb{R}^n) \times C_c^\infty(\mathbb{R}^n).\end{aligned}$$

Suppose ϕ is a smooth solution to the above initial value problem, arising from initial data $(\Phi, \Phi') = (0,0)$ in $B_R(0)$. Give a description of the subset of $\mathbb{R}^{n+1}$ in which ϕ must vanish. Draw this subset for the special case $n = 2$ and $f(r) = \sqrt{1+r}$.[Hint: integrate by parts $\partial_t\phi\partial_\alpha(a^{\alpha\beta}\partial_\beta\phi) = 0$, as in the proof of Proposition 2.2, but replace the ingoing cone $\mathcal{C}^-_{(R,0)}$ by a different, appropriate surface of revolution emanating from the time axis $|\mathbf{x}| = 0$.]

c) Prove the analogue of (ii) *of Corollary 2.3 (the weak Huygens principle).*

4 Local well-posedness for nonlinear wave equations

We consider the initial value problem for the following quasilinear equation:

$$\begin{aligned}&a^{\alpha\beta}(\phi, D\phi)\partial_\alpha\partial_\beta\phi = F(\phi, D\phi) \quad \text{in } [0,T) \times \mathbb{R}^n,\\ &(\phi|_{t=0}, \partial_t\phi|_{t=0}) = (\Phi, \Phi') \in H^k(\mathbb{R}^n) \times H^{k-1}(\mathbb{R}^n),\end{aligned} \tag{4.1}$$

with $k \in \mathbb{N}$ to be specified later and $a^{\alpha\beta}, F : \mathbb{R} \times \mathbb{R}^{n+1} \to \mathbb{R}$ smooth functions satisfying:

$$F(0,0) = 0.^{20}$$

Analogously to the linear setting in § 3, we also assume that there exists an $\epsilon_0 < 1$ such that:

$$\begin{aligned}&|a^{00} - m^{00}|(v,w) \le \epsilon_0,\\ &\sqrt{\sum_{i,j=1}^n |a^{ij} - m^{ij}|^2(v,w)} \le \epsilon_0,\end{aligned}$$

for all $v \in \mathbb{R}$ and $w \in \mathbb{R}^{n+1}$.

We can appeal to the notion of a classical solution of (4.1) as in Definition 3.1.

We will now prove the first theorem involving *nonlinear* wave equations.

Theorem 4.1 (Local well-posedness for quasilinear wave equations)**.** *Let $n \in \mathbb{N}_{\ge 1}$ and $(\Phi, \Phi') \in H^{n+2}(\mathbb{R}^n) \times H^{n+1}(\mathbb{R}^n)$. Then we can find a $T = T(a, F, n, ||\Phi||_{H^{n+2}(\mathbb{R}^n)}, ||\Phi'||_{H^{n+1}(\mathbb{R}^n)}) > 0$ such that:*

(i) *There exists a unique classical solution $\phi \in C^0([0,T]; H^{n+2}(\mathbb{R}^n)) \cap C^1([0,T]; H^{n+1}(\mathbb{R}^n)) \cap C^2([0,T] \times \mathbb{R}^n)$ to (4.1).*

[20] See **Problem** 4.3 for more general a, F.

(ii) *If* $(\Phi_{(i)}, \Phi'_{(i)}) \in H^{n+2}(\mathbb{R}^n) \times H^{n+1}(\mathbb{R}^n)$ *are a sequence of initial data converging to* (Φ, Φ') *in* $H^{n+2}(\mathbb{R}^n) \times H^{n+1}(\mathbb{R}^n)$*, then the corresponding classical solutions* ϕ_i *satisfy for all* $1 \leq s < n+1$*:*

$$\sup_{t\in[0,T]} ||\phi_{(i)} - \phi||_{H^{s+1}(\mathbb{R}^n)}(t) + ||\partial_t \phi_{(i)} - \partial_t \phi||_{H^s(\mathbb{R}^n)}(t) \to 0 \quad (i \to \infty).$$

Remark 4.1. *Without changing the proof, the statement of Theorem 4.1 holds for* $H^{n+2}(\mathbb{R}^n) \times H^{n+1}(\mathbb{R}^n)$ *replaced with* $H^{m+1}(\mathbb{R}^n) \times H^m(\mathbb{R}^n)$ *with* $m \geq n+1$ *and with* T, K *depending on the choice of* m*.*

Furthermore, it is possible the lower the regularity assumptions, by taking $m > \frac{n}{2} + 1$*, using more refined* interpolation estimates*. We will not pursue such a regularity improvement here, but refer the interested reader instead to Theorem 6.4.11 in:*

Hörmander, L., 1997. *Lectures on nonlinear hyperbolic differential equations* (Vol. 26). Springer Science & Business Media.

Proof of Theorem 4.1. **Strategy of the proof**:

"(i)": We will only prove "existence", since "uniqueness" follows from the estimates leading to (ii). We consider a sequence (ϕ_i), $i \in \mathbb{N}_0$ of classical solutions to linear problems, defined iteratively as follows:[21]

$$\begin{aligned} a^{\alpha\beta}(\phi_{i-1}, D\phi_{i-1})\partial_\alpha\partial_\beta\phi_i &= F(\phi_{i-1}, D\phi_{i-1}) \quad \text{in } [0,T]\times\mathbb{R}^n, \\ (\phi_i|_{t=0}, \partial_t\phi_i|_{t=0}) &= (\Phi, \Phi') \in H^{m+1}(\mathbb{R}^n) \times H^m(\mathbb{R}^n) \quad (i \geq 1), \\ \phi_{i=0} &\equiv 0. \end{aligned} \tag{4.2}$$

The goal is to show that, for $T > 0$ suitably small depending only on $||\Phi||_{H^{m+1}(\mathbb{R}^n)}$ and $||\Phi'||_{H^m(\mathbb{R}^n)}$ (and the coefficients of the equation), the sequence (ϕ_i) is:

(A) uniformly bounded in $C^0([0,T]; H^{m+1}(\mathbb{R}^n)) \cap C^1([0,T]; H^m(\mathbb{R}^n))$ if $m \geq n+1$,

(B) Cauchy in the larger space $C^0([0,T]; H^1(\mathbb{R}^n)) \cap C^1([0,T]; L^2(\mathbb{R}^n))$,

which may be thought of as an analogue of *Picard iteration* from ode theory.

Since is a Cauchy sequence in a Banach space, it converges to a limit $\phi \in C^0([0,T]; H^1(\mathbb{R}^n)) \cap C^1([0,T]; L^2(\mathbb{R}^n))$. In **Problems** 4.1 and 4.5, it is shown that in fact $\phi \in C^0([0,T]; H^{m+1}(\mathbb{R}^n)) \cap C^1([0,T]; H^m(\mathbb{R}^n))$ and, after applying the Sobolev embedding theorem, we obtain that $\phi \in C^2([0,T]\times\mathbb{R}^n)$ is a classical solution to (4.1).[22]

"(ii)": We consider the difference $\phi - \phi_{(i)}$ and apply energy estimates. Since (4.1) is nonlinear and $\phi - \phi_{(i)}$ therefore need not be a solution to (4.1), the relevant estimates for $\phi - \phi_{(i)}$ do <u>not</u> follow immediately.

We will first establish (A). We will show that for $T > 0$ suitably small, there exists a constant $K > 1$ such that for all $t \leq T$:

$$||(\phi_j, \partial_t\phi_j)||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t) \leq K \quad \forall j \leq i-1 \text{ implies } \quad ||(\phi_i, \partial_t\phi_i)||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t) \leq K.$$

From this, it follows moreover that $||(\phi, \partial_t\phi)||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t) \leq K$.

We will make the following inductive assumption:

$$||(\phi_{i-1}, \partial_t\phi_{i-1})||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t) \leq K. \tag{4.3}$$

[21] Note that the regularity in the Sobolev spaces is sufficiently high to conclude that weak solutions are also classical solutions.

[22] Exercise: Convince yourself that $\phi \in C^2([0,T]\times\mathbb{R}^n)$ and ϕ is a classical solution to (4.1).

By (3.10), we have that

$$\begin{aligned}
&\sup_{t\in[0,T]} ||(\phi_i, \partial_t\phi_i)||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t) \\
&\quad\leq C(1+T)\Bigg[||(\Phi,\Phi')||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)} + \sum_{|I|\leq m}\int_0^T \underbrace{||\partial_x^I F(\phi_{i-1}, D\phi_{i-1})||_{L^2(\mathbb{R}^n)}(t)}_{=:\mathcal{T}_1}\, dt \\
&\qquad + \sum_{|I|+|J|\leq m-1}\int_0^T \underbrace{||\partial_x^I\partial_x a(\phi_{i-1}, D\phi_{i-1})\cdot\partial^2\partial_x^J\phi_i||_{L^2(\mathbb{R}^n)}(t)}_{=:\mathcal{T}_2}\, dt\Bigg] \\
&\qquad \times \exp\left(C\int_0^T \underbrace{||\partial a(\phi_{i-1}, D\phi_{i-1})||_{L^\infty(\mathbb{R}^n)}(t)}_{=:\mathcal{T}_3}\, dt\right).
\end{aligned} \tag{4.4}$$

By Theorem A.2, we can estimate for $m > \frac{n}{2}$:

$$\begin{aligned}
\sum_{|I|<m+1-\frac{n}{2}} ||\partial_x^I\phi_{i-1}||_{L^\infty(\mathbb{R}^n)}(t) + \sum_{|I|<m-\frac{n}{2}} ||\partial_x^I\partial_t\phi_{i-1}||_{L^\infty(\mathbb{R}^n)}(t) &\leq C||(\phi_{i-1},\partial_t\phi_{i-1})||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t) \\
&\leq CK,
\end{aligned} \tag{4.5}$$

with $C>0$ a constant depending only on n. Hence, the arguments ϕ_{i-1} and $\partial\phi_{i-1}$ appearing in ∂_x-derivatives of $F(\phi_{i-1}, D\phi_{i-1})$ and $a(\phi_{i-1}, D\phi_{i-1})$, after applying the chain rule, lie within a ball of radius CK. So by smoothness of F and a, together with the assumption $F(0,0)=0$: $||\partial_x^I F(\phi_{i-1}, D\phi_{i-1})||_{L^\infty(\mathbb{R}^n)}(t) \leq C_{|I|,K,F}$ $|I| < m-\frac{n}{2}$ and $||\partial_x^I a(\phi_{i-1}, D\phi_{i-1})||_{L^\infty(\mathbb{R}^n)}(t)\leq C_{|I|,K,a}$ for $1\leq|I|<m-\frac{n}{2}$, provided we take $m>\frac{n}{2}+1$.

We first consider $\mathcal{T}_3$. We note that by applying (4.2), we can also obtain L^∞ estimates for terms containing two t-derivatives, so (for $m>\frac{n}{2}+1$):

$$\begin{aligned}
\sum_{|I|<m-1-\frac{n}{2}} ||\partial_x^I\partial_t^2\phi_{i-1})||_{L^\infty(\mathbb{R}^n)}(t) &\leq \sum_{|I|<m-1-\frac{n}{2}} ||\partial_x^I((a^{tt})^{-1}a(\phi_{i-2}, D\phi_{i-2})\cdot\partial_x^2\phi_{i-1})||_{L^\infty(\mathbb{R}^n)}(t) \\
&\quad + ||\partial_x^I((a^{tt})^{-1}a(\phi_{i-2}, D\phi_{i-2})\cdot\partial_x\partial_t\phi_{i-1})||_{L^\infty(\mathbb{R}^n)}(t) + ||\partial_x^I((a^{tt})^{-1}F(\phi_{i-2}, D\phi_{i-2})||_{L^\infty(\mathbb{R}^n)}(t) \\
&\leq C_{m,K,F,a}
\end{aligned}$$

for some constant $C_{m,K,F,a}>0$. Taking $|I|=1$ above then implies $|\mathcal{T}_3|\leq C_{1,K,F,a}$ after applying the chain rule and choosing $C_{1,K,F,a}>0$ suitably large.

Consider now $\mathcal{T}_1$. By the chain rule and the assumption $F(0,0)=0$, $\partial_x^I F(\phi_{i-1}, D\phi_{i-1})$ will be a sum of linear terms in ϕ_{i-1} and $D\phi_{i-1}$, quadratic terms, cubic terms, etc. all the way up to terms of order m. Indeed, we obtain:

$$\begin{aligned}
\sum_{|I|\leq m} ||\partial_x^I F(\phi_{i-1}, D\phi_{i-1})||_{L^2(\mathbb{R}^n)}(t) \leq C_{m,K,F}\Bigg[&\sum_{|I|\leq m}||\partial_x^I\partial\phi_{i-1}||_{L^2(\mathbb{R}^n)} \\
&+ \sum_{|I_1|+|I_2|\leq m}||\partial_x^{I_1}\partial\phi_{i-1}\cdot\partial_x^{I_2}\partial\phi_{i-1}||_{L^2(\mathbb{R}^n)} + \text{cubic terms} + \dots \\
&+ \text{terms with } m \text{ factors in } \partial\phi_{i-1} \text{ and } \phi_{i-1}\Bigg].
\end{aligned}$$

Note that the linear term can immediately be estimated via (4.3). To estimate the quadratic term via (4.3), we will take out the L^∞-norm of one of the factors. Note that in the sum of quadratic terms $|I_1|\leq\frac{m}{2}$ or $|I_2|\leq\frac{m}{2}$. Assume WLOG that $|I_1|\leq\frac{m}{2}$. Then we estimate

$$\begin{aligned}
\sum_{|I_1|+|I_2|\leq m}||\partial_x^{I_1}\partial\phi_{i-1}\cdot\partial_x^{I_2}\partial\phi_{i-1}||_{L^2(\mathbb{R}^n)} &\leq \sum_{|I_1|\leq\frac{m}{2}}||\partial_x^{I_1}\partial\phi_{i-1}||_{L^\infty(\mathbb{R}^n)}\sum_{|I_2|\leq m}||\partial_x^{I_2}\partial\phi_{i-1}||_{L^2(\mathbb{R}^n)} \\
&\leq CK\sum_{|I_1|\leq\frac{m}{2}}||\partial_x^{I_1}\partial\phi_{i-1}||_{L^\infty(\mathbb{R}^n)},
\end{aligned}$$

where we applied (4.3) to estimate the factor involving I_2. Now we apply Theorem A.2 and (4.3) to estimate the term involving I_1, using that

$$\sum_{|I_1|\leq \frac{m}{2}} ||\partial_x^{I_1}\partial\phi_{i-1}||_{L^\infty(\mathbb{R}^n)} \leq C||(\phi_{i-1},\partial_t\phi_{i-1})||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t) \leq CK,$$

provided we require

$$\frac{m}{2}+1+\frac{n}{2} < m+1.$$

It is here that we used that the regularity in our Sobolev spaces is sufficiently high! We thus require $m \geq n+1$. Then we conclude that

$$\sum_{|I_1|+|I_2|\leq m} ||\partial_x^{I_1}\partial\phi_{i-1}\cdot\partial_x^{I_2}\partial\phi_{i-1}||_{L^2(\mathbb{R}^n)} \leq CK^2.$$

For the cubic terms, we similarly observe that when considering sums of the form $|I_1|+|I_2|+|I_3|\leq m$, all but of of the multiindices must satisfy $|I_i|\leq \frac{m}{3}$, so all but one of the factors can be estimated in L^∞. The same argument applies for the higher-order terms. We are finally left with:

$$\sum_{|I|\leq m} ||\partial_x^I F(\phi_{i-1},D\phi_{i-1})||_{L^2(\mathbb{R}^n)}(t) \leq CC_{m,K,F}K^m,$$

with $C>0$ a constant depending only on n.

To estimate the remaining term $\mathcal{T}_2$, we notice two key differences: 1) this term involves not only ϕ_{i-1} but also ϕ_i, 2) we see ∂^2, which includes two ∂_t derivatives. We note that 1) implies that we need to apply the Sobolev embedding (the first inequality of (4.5)) also to ϕ_i and we deal with 2) by applying the equation (4.2) to express two ∂_t derivatives in terms of ∂_x derivatives and terms involving ϕ_{i-1}. We leave the details as an exercise. Exercise: show that

$$\mathcal{T}_2 \leq CC_{m,K,a,F}K^m \sup_{t\in[0,T)} ||(\phi_i,\partial_t\phi_i)||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t).$$

Plugging this into (4.4), we arrive at

$$\begin{aligned}
&\sup_{t\in[0,T]} ||(\phi_i,\partial_t\phi_i)||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t)\\
&\leq C(1+T)\Bigg[||(\Phi,\Phi')||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)} + C_{m,K,F}T\\
&\quad + C_{m,K,a,F}T\sup_{t\in[0,T]} ||(\phi_i,\partial_t\phi_i)||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t)\Bigg]\exp\left(CC_{K,a,F}T\right)
\end{aligned}$$

for all $m\geq n+1$. We now choose $T>0$ suitably small, depending on a,K, such that $C(1+T)\cdot CC_{m,K,a,F}T \leq \frac{1}{4}$ and $\exp\left(CC_{K,a,F}T\right)\leq 2$ to obtain:

$$\sup_{t\in[0,T]} ||(\phi_i,\partial_t\phi_i)||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t) \leq 2C(1+T)||(\Phi,\Phi')||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)} + 4C(1+T)C_{m,K,F}T.$$

We can choose $T>0$ suitably small, depending on F,K, to guarantee moreover that $4C(1+T)C_{m,K,F}T < \frac{1}{2}K$. Finally, choosing K suitably large compared to $||(\Phi,\Phi')||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}$, we can conclude that

$$\sup_{t\in[0,T]} ||(\phi_i,\partial_t\phi_i)||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t) < K,$$

which finishes the proof of (A).

We will now prove (B). Let $i\geq 2$. Then the difference $\phi_i-\phi_{i-1}$ satisfies the equation:

$$\begin{aligned}
a^{\alpha\beta}(\phi_{i-1},D\phi_{i-1})\partial_\alpha\partial_\beta(\phi_i-\phi_{i-1}) = &-\left(a^{\alpha\beta}(\phi_{i-1},D\phi_{i-1})-a^{\alpha\beta}(\phi_{i-2},D\phi_{i-2})\right)\partial_\alpha\partial_\beta\phi_{i-1}\\
&+F(\phi_{i-1},D\phi_{i-1})-F(\phi_{i-2},D\phi_{i-2}),
\end{aligned}$$

with $(\phi_i - \phi_{i-1}|_{t=0}, \partial_t\phi_i - \partial_t\phi_{i-1}|_{t=0}) \equiv 0$, since the data for ϕ_i are independent of i.

We apply the mean-value theorem to estimate:

$$|a^{\alpha\beta}(\phi_{i-1}, D\phi_{i-1}) - a^{\alpha\beta}(\phi_{i-2}, D\phi_{i-2})|(x) \leq \sup_{B_{C\cdot K}(0)_v \times B_{C\cdot K}(0)_w} |\partial_v a(v,w)|\phi_{i-1} - \phi_{i-2}|(x)$$
$$+ \sum_{k=\alpha}^{n} \sup_{B_{C\cdot K}(0)_v \times B_{C\cdot K}(0)_w} |\partial_{w^\alpha} a(v,w)|\partial_\alpha\phi_{i-1} - \partial_\alpha\phi_{i-2}|(x),$$

using that $||\phi_{i-j}||_{L^\infty}(t) + ||D\phi_{i-j}||_{L^\infty}(t) \leq CK$ for $j = 1, 2$ by (4.5). We then obtain:[23]

$$||(a^{\alpha\beta}(\phi_{i-1}, D\phi_{i-1}) - a^{\alpha\beta}(\phi_{i-2}, D\phi_{i-2}))\partial_\alpha\partial_\beta\phi_{i-1}||_{L^2(\mathbb{R}^n)}(t) \leq C_{1,K,a}K(||\phi_{i-1} - \phi_{i-2}||_{H^1(\mathbb{R}^n)} + ||\partial_t\phi_{i-1} - \partial_t\phi_{i-2}||_{L^2(\mathbb{R}^n)}).$$

Similarly, we obtain

$$||F(\phi_{i-1}, D\phi_{i-1}) - F(\phi_{i-2}, D\phi_{i-2})||_{L^2(\mathbb{R}^n)}(t) \leq C_{1,K,F}(||\phi_{i-1} - \phi_{i-2}||_{H^1(\mathbb{R}^n)} + ||\partial_t\phi_{i-1} - \partial_t\phi_{i-2}||_{L^2(\mathbb{R}^n)}).$$

Now we apply (3.5), using that by the arguments in (A):

$$||\partial a(\phi_{i-1}, D\phi_{i-1})||_{L^\infty(\mathbb{R}^n)} \leq C_{1,K,a,F}K,$$

to obtain:

$$\sup_{t\in[0,T]} ||(\phi_i - \phi_{i-1}, \partial_t\phi_i - \partial_t\phi_{i-1})||_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}(t)$$
$$\leq CK(C_{1,K,a,F} + C_{1,K,F})T \sup_{t\in[0,T)} ||(\phi_{i-1} - \phi_{i-2}, \partial_t\phi_{i-1} - \partial_t\phi_{i-2})||_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}(t).$$

Now we can make $T > 0$ suitably small such that $CK(C_{2,K,a} + C_{1,K,F})T < \frac{1}{2}$ and we obtain:

$$\sup_{t\in[0,T]} ||(\phi_i - \phi_{i-1}, \partial_t\phi_i - \partial_t\phi_{i-1})||_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}(t) \leq \frac{1}{2} \sup_{t\in[0,T]} ||(\phi_{i-1} - \phi_{i-2}, \partial_t\phi_{i-1} - \partial_t\phi_{i-2})||_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}(t),$$

so we can conclude that for all $j > i \geq 2$:

$$\sup_{t\in[0,T]} ||(\phi_j - \phi_i, \partial_t(\phi_j - \phi_i))||_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}(t) \leq \sum_{k=i+1}^{j} \sup_{t\in[0,T]} ||(\phi_k - \phi_{k-1}, \partial_t(\phi_k - \phi_{k-1}))||_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}(t)$$
$$\leq CK \sum_{k=i+1}^{\infty} 2^{-k} \leq 2^{-i}CK,$$

from which the Cauchy property follows.

We turn to proving (ii). Consider the sequence of solutions $\phi_{(i)}$, arising from the sequence of given data (do not confuse with sequence ϕ_i in (A) and (B) above!). By the above, we have that there exist $T > 0$ and $B > 0$ such that:

$$\sup_{t\in[0,T]} ||(\phi, \partial_t\phi)||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t) + ||(\phi_{(i)}, \partial_t\phi_{(i)})||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t) \leq B.$$

for all $i \in \mathbb{N}_0$.

The difference $\phi_{(i)} - \phi$ satisfies the equation:

$$a^{\alpha\beta}(\phi_{(i)}, D\phi_{(i)})\partial_\alpha\partial_\beta(\phi_{(i)} - \phi) = -(a^{\alpha\beta}(\phi_{(i)}, D\phi_{(i)}) - a^{\alpha\beta}(\phi, D\phi))\partial_\alpha\partial_\beta\phi + F(\phi_{(i)}, D\phi_{(i)}) - F(\phi, D\phi),$$

[23] Note that in this step we applied a uniform L^∞ bound for $\partial^2\phi_{i-1}$, which does not follow from just $H^1 \times L^2$ boundedness of $(\phi_{i-1}, \partial_t\phi_{i-1})$. For this reason, we cannot bound the $H^{m+1} \times H^m$ norm of the difference $\phi_i - \phi_{i-1}$ directly, as this would require control over the L^∞ norm for $\partial^2\partial_x^m\phi_{i-1}$, which we do not have.

so we can apply the same arguments as in (B), to conclude that

$$\sup_{t\in[0,T]} ||(\phi_{(i)}-\phi,\partial_t\phi_{(i)}-\partial_t\phi)||_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}(t) \leq C\Bigg[||(\Phi_{(i)}-\Phi,\Phi'_{(i)}-\Phi')||_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}$$

$$+K(C_{1,B,F}+C_{1,B,a,F})T\sup_{t\in[0,T]} ||(\phi_{(i)}-\phi,\partial_t\phi_{(i)}-\partial_t\phi)||_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}(t)\Bigg].$$

Hence, for $T>0$ suitably small, we can bound

$$\sup_{t\in[0,T]} ||(\phi_{(i)}-\phi,\partial_t\phi_{(i)}-\partial_t\phi)||_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}(t) \leq C||(\Phi_{(i)}-\Phi,\Phi'_{(i)}-\Phi)||_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}, \tag{4.6}$$

from which it follows that the LHS goes to zero as $i\to\infty$.

To conclude (ii) and finish the proof of the theorem, we will make use of the following *interpolation inequality*:

Lemma 4.2. *Let $0\leq s_1\leq s\leq s_2$, $\theta_1,\theta_2\in\mathbb{R}_+$ such that $\theta_1+\theta_2=1$ and $s_1\theta_1+s_2\theta_2=s$. Assume $f\in H^{s_1}(\mathbb{R}^n)\cap H^{s_2}(\mathbb{R}^n)$. Then there exists $C=C(s_1,s_2,s,n)>0$, such that*

$$||f||_{H^s(\mathbb{R}^n)}\leq C||f||^{\theta_1}_{H^{s_1}(\mathbb{R}^n)}||f||^{\theta_2}_{H^{s_2}(\mathbb{R}^n)}.$$

Proof. Problem 4.2. □

We take $(s_1,s_2,\theta_1,\theta_2)=(1,m+1,\frac{m+1-s}{m},\frac{s-1}{m})$, with $1\leq s<m+1$ and $(s_1,s_2,\theta_1,\theta_2)=(0,m,\frac{m+1-s}{m},\frac{s-1}{m})$ with $0\leq s-1<m$ to obtain

$$\sup_{t\in[0,T]} ||(\phi_{(i)}-\phi,\partial_t\phi_{(i)}-\partial_t\phi)||_{H^s(\mathbb{R}^n)\times H^{s-1}(\mathbb{R}^n)}(t)$$

$$\leq C\sup_{t\in[0,T]}\left(||(\phi_{(i)}-\phi,\partial_t\phi_{(i)}-\partial_t\phi)||_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}(t)\right)^{\frac{m+1-s}{m}}\cdot\left(||(\phi_{(i)}-\phi,\partial_t\phi_{(i)}-\partial_t\phi)||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t)\right)^{\frac{s-1}{m}}$$

$$\to 0$$

as $i\to\infty$, since the first factor goes to 0 by (4.6) and the second factor is bounded by the triangle inequality and the boundedness properties of ϕ and $\phi_{(i)}$. □

In order to construct more regular solutions, we can assume more regularity for the initial data. For example, we could consider smooth initial data. However, *a priori* it could happen that the time of existence T goes to zero as we consider initial data norms with higher and higher regularity. The aim of the theorem below is to show that this does *not* happen.

Theorem 4.3 (Persistence of regularity)**.** *Let $(\Phi,\Phi')\in H^{n+3}(\mathbb{R}^n)\times H^{n+2}(\mathbb{R}^n)$. Define the corresponding* maximal time of existence *as follows:*

$$T_*=\sup\{T>0\,\big|\,\exists!\,\phi\in C^0([0,T];H^{n+3}(\mathbb{R}^n))\cap C^1([0,T];H^{n+2}(\mathbb{R}^n))\text{ solving (4.1) with initial data }(\Phi,\Phi')\}.$$

Then:

(i) *If $m\in\mathbb{N}$ such that $m\geq n+2$ and $(\Phi,\Phi')\in H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)$, then the corresponding solution ϕ satisfies: $C^0([0,T];H^{m+1}(\mathbb{R}^n))\cap C^1([0,T];H^m(\mathbb{R}^n))$ for all $0<T<T_*$.*

(ii) *If $(\Phi,\Phi')\in\cap_{m\in\mathbb{N}}H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)$, then the corresponding solution ϕ is smooth in $[0,T_*)\times\mathbb{R}^n$.*

Proof. By the Sobolev Embedding Theorem, (ii) follows immediately from (i). We will prove (i) by induction. The case $m=n+2$ follows from Theorem 4.1 (see Remark 4.1). As the induction step, we suppose that (i) holds for some $m\geq n+2$ and

$$\sup_{t\in[0,T_*]} ||\phi||_{H^{m+1}(\mathbb{R}^n)}(t)+||\partial_t\phi||_{H^m(\mathbb{R}^n)}(t)\leq D.$$

The goal now is to estimate the $H^{m+2}\times H^{m+1}$ norm of $(\phi,\partial_t\phi)$, using the inductive assumption. Note that we do not know *a priori* that $\phi\in C^0([0,T_*);H^{m+2}(\mathbb{R}^n))\cap C^1([0,T_*);H^{m+1}(\mathbb{R}^n))$, so we cannot apply energy estimates directly in this time interval. By Theorem 4.1 and Remark 4.1, we do know that there exists a $T_m = T_m(a,b,m,||(\Phi,\Phi')||_{H^{m+2}(\mathbb{R}^n)\times H^{m+1}(\mathbb{R}^n)}||)>0$ such that $\phi\in C^0([0,T];H^{m+2}(\mathbb{R}^n))\cap C^1([0,T];H^{m+1}(\mathbb{R}^n))$ for all $T\leq T_m$.

By (3.10),[24] we have that for all $0<T\leq T_m$:

$$\begin{aligned}
&\sup_{t\in[0,T]}||(\phi,\partial_t\phi)||_{H^{m+2}(\mathbb{R}^n)\times H^{m+1}(\mathbb{R}^n)}(t)\\
&\leq C(1+T)\Bigg[||(\Phi,\Phi')||_{H^{m+2}(\mathbb{R}^n)\times H^{m+1}(\mathbb{R}^n)}+\sum_{|I|\leq m+1}\int_0^T||\partial_x^I(F(\phi,D\phi))||_{L^2(\mathbb{R}^n)}(t)\,dt\\
&\quad+\sum_{|I|+|J|\leq m}\int_0^T||(\partial_x^I\partial_x a(\phi,D\phi))\cdot\partial^2\partial_x^J\phi||_{L^2(\mathbb{R}^n)}(t)\,dt\Bigg]\\
&\times e^{C\int_0^T||\partial a(\phi,D\phi)||_{L^\infty(\mathbb{R}^n)}(t)\,dt}.
\end{aligned}\tag{4.7}$$

Observe that $||\partial a(\phi,D\phi)||_{L^\infty(\mathbb{R}^n)}(t)\leq C_{1,a}\cdot D$ by applying the inductive assumption together with the Sobolev Embedding Theorem. It remains to show that the remaining integrals in t above can be bounded linearly in the highest-order derivatives, i.e. linearly in

$$\sup_{t\in[0,T]}||(\phi,\partial_t\phi)||_{H^{m+2}(\mathbb{R}^n)\times H^{m+1}(\mathbb{R}^n)}(t).$$

We proceed as in the proof of (i) of Theorem 4.1 by noting that:

$$\begin{aligned}
\sum_{|I|\leq m+1}||\partial_x^I F(\phi,D\phi)||_{L^2(\mathbb{R}^n)}(t)\leq C_{m,D,F}\Bigg[&\sum_{|I|\leq m+1}||\partial_x^I\partial\phi||_{L^2(\mathbb{R}^n)}\\
&+\sum_{|I_1|+|I_2|\leq m+1}||\partial_x^{I_1}\partial\phi\cdot\partial_x^{I_2}\partial\phi||_{L^2(\mathbb{R}^n)}+\text{cubic}+\dots\\
&+\text{terms with } m+1 \text{ factors in } \partial\phi,\phi\Bigg].
\end{aligned}$$

Restricting to the quadratic term, we can assume WLOG that $|I_1|\leq\frac{m+1}{2}$, so

$$\begin{aligned}
\sum_{|I_1|+|I_2|\leq m+1}||\partial_x^{I_1}\partial\phi\cdot\partial_x^{I_2}\partial\phi||_{L^2(\mathbb{R}^n)}&\leq\sum_{|I_1|\leq\frac{m+1}{2}}||\partial_x^{I_1}\partial\phi||_{L^\infty(\mathbb{R}^n)}\sum_{|I_2|\leq m+1}||\partial_x^{I_2}\partial\phi||_{L^2(\mathbb{R}^n)}\\
&\leq CD||(\phi,\partial_t\phi)||_{H^{m+2}(\mathbb{R}^n)\times H^{m+1}(\mathbb{R}^n)},
\end{aligned}$$

where we used the Sobolev Embedding Theorem to estimate the L^∞ norm by the inductive assumption, using that

$$1+\frac{m+1}{2}+\frac{n}{2}=\frac{m+n+3}{2}<m+1,$$

since we assumed $m>n+1$.

We can estimate the remaining terms of the RHS of (4.7) in a similar way to conclude that for all $T<T_m$:

$$\begin{aligned}
&\sup_{t\in[0,T]}||(\phi,\partial_t\phi)||_{H^{m+2}(\mathbb{R}^n)\times H^{m+1}(\mathbb{R}^n)}(t)\\
&\leq C(1+T)\left[||(\Phi,\Phi')||_{H^{m+2}(\mathbb{R}^n)\times H^{m+1}(\mathbb{R}^n)}+C_{m,D,F,a}D\int_0^T||(\phi,\partial_t\phi)||_{H^{m+2}(\mathbb{R}^n)\times H^{m+1}(\mathbb{R}^n)}\,dt\right]e^{CC_{K,a}TD}.
\end{aligned}$$

[24] Technically, (3.10) applies only to functions in $C^\infty([0,T);\cap_{k\in\mathbb{N}_0}H^k(\mathbb{R}^n))$. However, we can approximate $\phi\in C^0([0,T);H^{m+1}(\mathbb{R}^n))\cap C^1([0,T);H^m(\mathbb{R}^n))$ by a sequence of smooth functions (ϕ_i) such that ϕ_i converges with respect to $C^0([0,T);H^{m+1}(\mathbb{R}^n))$ and $\partial_t\phi_i$ converges with respect to $C^0([0,T);H^m(\mathbb{R}^n))$. Taking the limit $i\to\infty$, both the LHS and the RHS of (3.10) applied to ϕ_i converge.

By Grönwall (Lemma 3.2), we can now conclude that for all $T \leq T_m = T_m(a, b, m, ||(\Phi, \Phi')||_{H^{m+2}(\mathbb{R}^n)\times H^{m+1}(\mathbb{R}^n)}||)$

$$\begin{aligned}&\sup_{t\in[0,T)} ||(\phi, \partial_t\phi)||_{H^{m+2}(\mathbb{R}^n)\times H^{m+1}(\mathbb{R}^n)}(t)\\ &\quad\leq C(1+T_*)(||(\Phi,\Phi')||_{H^{m+2}(\mathbb{R}^n)\times H^{m+1}(\mathbb{R}^n)})e^{C_{m,D,F,a}T_*D},\end{aligned} \tag{4.8}$$

extending thus the inductive assumption to $m+1$ replacing m, for $T \leq T_m$. It remains to show that the above estimate actually holds for all $T \leq T_*$ to conclude the induction argument.

Denote with

$$S = \Big\{ T \in [0, T_*) \,|\, \phi \in C^0([0,T]; H^{m+2}(\mathbb{R}^n)) \cap C^1([0,T]; H^{m+1}(\mathbb{R}^n))$$
$$\text{is a solution to (4.1) with initial data } (\Phi, \Phi') \in H^{m+2}(\mathbb{R}^n) \times H^{m+1}(\mathbb{R}^n) \text{ satisfying (4.8)} \Big\}.$$

Since $\phi \in C^0([0, T_m); H^{m+2}(\mathbb{R}^n)) \cap C^1([0, T_m); H^{m+1}(\mathbb{R}^n))$ satisfies (4.8), S is non-empty. Let $T \in S$. Then, for $\epsilon > 0$ suitably small, we can interpret $(\phi|_{t=T-\epsilon}, \partial_t\phi_{t=T-\epsilon})$ as initial data and apply (4.8) to conclude that $\phi \in C^0((T-\epsilon, T+\epsilon); H^{m+2}(\mathbb{R}^n)) \cap C^1((T-\epsilon, T+\epsilon); H^{m+1}(\mathbb{R}^n))$, so in fact $(T-\epsilon, T+\epsilon) \in S$, so S is open. Closedness of S follows easily after considering a sequence of times $T_i \in S$ converging to $\tilde{T} \in [0, T_*)$ (Exercise: Prove closedness.). Hence, S is a closed and open subset of the connected set $[0, T_*)$, from which it follows that $S = [0, T_*)$. It therefore follows that in fact (4.8) holds for all $T \leq T_*$. □

4.1 Breakdown criteria

In order to pass from local to global well-posedness, it is useful to identify *breakdown criteria* that prevent the maximal time of existence T_* (see Theorem 4.3) from going to infinity.

Theorem 4.4 (Breakdown criteria for well-posedness)**.** *Let $(\Phi, \Phi') \in H^{n+2}(\mathbb{R}^n) \times H^{n+1}(\mathbb{R}^n)$ and denote with ϕ the corresponding solution to (4.1) and T_* the maximal time of existence, as in Theorem 4.3. If $T_* < \infty$, then:*

(i)
$$\liminf_{t\uparrow T_*} ||\phi||_{H^{n+2}(\mathbb{R}^n)}(t) + ||\partial_t\phi||_{H^{n+1}(\mathbb{R}^n)}(t) = \infty,$$

(ii)
$$\limsup_{t\uparrow T_*} \sum_{|I|<\frac{n}{2}+1} ||\partial_x^I\phi||_{L^\infty(\mathbb{R}^n)}(t) + \sum_{|I|<\frac{n}{2}} ||\partial_x^I\partial_t\phi||_{L^\infty(\mathbb{R}^n)}(t) = \infty,$$

(iii)
$$\limsup_{t\uparrow T_*} \sum_{|I|\leq 2} ||\partial_x^I\phi||_{L^\infty(\mathbb{R}^n)}(t) + \sum_{|I|\leq 1} ||\partial_x^I\partial_t\phi||_{L^\infty(\mathbb{R}^n)}(t) = \infty.$$

and if we assume that a is independent of $D\phi$, i.e. $a(\phi, D\phi) = a(\phi)$, then

$$\limsup_{t\uparrow T_*} \sum_{|I|\leq 1} ||\partial_x^I\phi||_{L^\infty(\mathbb{R}^n)}(t) + ||\partial_t\phi||_{L^\infty(\mathbb{R}^n)}(t) = \infty.$$

Proof. It is clear that the first part of (iii)⇒(ii) if $n \geq 2$. However, we prove (ii) independently, as it uses less sophisticated tools.

Suppose (i) does not hold. Then there exists a constant $B > 0$ and a sequence of times (t_i) with $t_i \to T_*$, such that

$$||\phi||_{H^{n+2}(\mathbb{R}^n)}(t_i) + ||\partial_t\phi||_{H^{n+1}(\mathbb{R}^n)}(t_i) \leq B.$$

for all $i \in \mathbb{N}$. By Theorem 4.1, there exists a $T = T(a, F, B) > 0$ such that for all $i \in \mathbb{N}$: $\phi \in C^0([t_i, t_i + T]; H^{n+2}(\mathbb{R}^n)) \cap C^1([t_i, t_i + T]; H^{n+1}(\mathbb{R}^n))$. In particular, by taking $I \in \mathbb{N}$ suffciently large, we have that $t_I + T > T_*$, which contradicts the maximality of T_*. We conclude that (i) must hold.

Now suppose (ii) does not hold. Then there exists a constant $B > 0$ such that

$$\sup_{t\in[0,T_*)} \sum_{|I|<\frac{n}{2}+1} ||\partial_x^I \phi||_{L^\infty(\mathbb{R}^n)}(t) + \sum_{|I|<\frac{n}{2}} ||\partial_x^I \partial_t \phi||_{L^\infty(\mathbb{R}^n)}(t) \leq B.$$

The strategy for reaching a contradiction will be to show that the above assumption actually implies finiteness of the $H^{n+2} \times H^{n+1}$, from which a contradiction follows as in the proof of (i).

The first step is to estimate the $H^{n+1} \times H^n$ norm. By (3.10), we have that for all $T < T_*$

$$\begin{aligned}
&\sup_{t\in[0,T]} ||(\phi, \partial_t\phi)||_{H^{n+1}(\mathbb{R}^n)\times H^n(\mathbb{R}^n)}(t) \\
&\quad \leq C(1+T)\Bigg[||(\Phi,\Phi')||_{H^{n+1}(\mathbb{R}^n)\times H^n(\mathbb{R}^n)} + \sum_{|I|\leq n} \int_0^T ||\partial_x^I (F(\phi, D\phi))||_{L^2(\mathbb{R}^n)}(t)\, dt \\
&\qquad + \sum_{|I|+|J|\leq n-1} \int_0^T ||\partial_x^I \partial_x a(\phi, D\phi)\cdot \partial^2 \partial_x^J \phi||_{L^2(\mathbb{R}^n)}(t)\, dt\Bigg] \\
&\quad \times e^{C\int_0^T ||\partial a(\phi, D\phi)||_{L^\infty(\mathbb{R}^n)}(t)\, dt}.
\end{aligned} \tag{4.9}$$

We proceed as in the proof of (i) of Theorem 4.1 by noting that:

$$\begin{aligned}
\sum_{|I|\leq n} ||\partial_x^I F(\phi, D\phi)||_{L^2(\mathbb{R}^n)}(t) \leq C_{n,D,F}\Bigg[&\sum_{|I|\leq n} ||\partial_x^I \partial\phi||_{L^2(\mathbb{R}^n)} \\
&+ \sum_{|I_1|+|I_2|\leq n} ||\partial_x^{I_1}\partial\phi\cdot\partial_x^{I_2}\partial\phi||_{L^2(\mathbb{R}^n)} + \text{cubic} + \dots \\
&+ \text{terms with } n \text{ factors in } \partial\phi, \phi\Bigg].
\end{aligned}$$

Restricting to the quadratic term, we can assume WLOG that $|I_1| \leq \frac{n}{2}$, so

$$\sum_{|I_1|+|I_2|\leq n} ||\partial_x^{I_1}\partial\phi\cdot\partial_x^{I_2}\partial\phi||_{L^2(\mathbb{R}^n)} \leq \sum_{|I_1|\leq\frac{n}{2}} ||\partial_x^{I_1}\partial\phi||_{L^\infty(\mathbb{R}^n)} \sum_{|I_2|\leq n} ||\partial_x^{I_2}\partial\phi||_{L^2(\mathbb{R}^n)} \leq CB||(\phi,\partial_t\phi)||_{H^{n+1}(\mathbb{R}^n)\times H^n(\mathbb{R}^n)},$$

Note that the first factor in the product on the right-hand side is bounded by B. The remaining terms on the right-hand side of (4.9) can be estimated analogously[25] and we conclude that:

$$\begin{aligned}
&\sup_{t\in[0,T]} ||(\phi, \partial_t\phi)||_{H^{n+1}(\mathbb{R}^n)\times H^n(\mathbb{R}^n)}(t) \\
&\quad \leq C(T_*, B)\Bigg[||(\Phi,\Phi')||_{H^{n+1}(\mathbb{R}^n)\times H^n(\mathbb{R}^n)} + \int_0^T ||(\phi,\partial_t\phi)||_{H^{n+1}(\mathbb{R}^n)\times H^n(\mathbb{R}^n)}(t)\, dt\Bigg].
\end{aligned}$$

By applying a Grönwall inequality (Lemma 3.2), we arrive at

$$\sup_{t\in[0,T]} ||(\phi,\partial_t\phi)||_{H^{n+1}(\mathbb{R}^n)\times H^n(\mathbb{R}^n)}(t) \leq C(T_*, B)\Bigg[||(\Phi,\Phi')||_{H^{n+1}(\mathbb{R}^n)\times H^n(\mathbb{R}^n)}\Bigg].$$

We proceed to estimate the $H^{n+2} \times H^{n+1}$ norm. The idea is to use the above bound for the $H^{n+1} \times H^n$ norm in combination with a more careful analysis of the quadratic terms above. We suppose first that n is odd to estimate:

$$\begin{aligned}
\sum_{|I_1|+|I_2|\leq n+1} ||\partial_x^{I_1}\partial\phi\cdot\partial_x^{I_2}\partial\phi||_{L^2(\mathbb{R}^n)} \leq &\sum_{|I_1|\leq\frac{n-1}{2}} ||\partial_x^{I_1}\partial\phi||_{L^\infty(\mathbb{R}^n)} \sum_{|I_2|\leq n+1} ||\partial_x^{I_2}\partial\phi||_{L^2(\mathbb{R}^n)} \\
&+ \sum_{|I_1|=\frac{n+1}{2}} ||\partial_x^{I_1}\partial\phi||_{L^\infty(\mathbb{R}^n)} \sum_{|I_2|=\frac{n+1}{2}} ||\partial_x^{I_2}\partial\phi||_{L^2(\mathbb{R}^n)}.
\end{aligned}$$

[25] Exercise: Show this.

The first term on the right-hand side can be estimated by $B||(\phi,\partial_t\phi)||_{H^{n+2}(\mathbb{R}^n)\times H^{n+1}(\mathbb{R}^n)}(t)$. To estimate the second term, we first applying a Sobolev embedding theorem (Theorem A.2):

$$\sum_{|I_1|=\frac{n+1}{2}} ||\partial_x^{I_1}\partial\phi||_{L^\infty(\mathbb{R}^n)} \leq C||(\phi,\partial_t\phi)||_{H^{s+1}(\mathbb{R}^n)\times H^s(\mathbb{R}^n)}(t)$$

for $s > \frac{n+1}{2} + \frac{n}{2} = n + \frac{1}{2}$ to conclude that we can estimate:

$$\sum_{|I_1|=\frac{n+1}{2}} ||\partial_x^{I_1}\partial\phi||_{L^\infty(\mathbb{R}^n)} \sum_{|I_2|=\frac{n+1}{2}} ||\partial_x^{I_2}\partial\phi||_{L^2(\mathbb{R}^n)}(t) \leq C||(\phi,\partial_t\phi)||_{H^{n+2}(\mathbb{R}^n)\times H^{n+1}(\mathbb{R}^n)}(t)\cdot \sum_{|I_2|=\frac{n+1}{2}} ||\partial_x^{I_2}\partial\phi||_{L^2(\mathbb{R}^n)}(t).$$

Now, note that by the fact that $\frac{n+1}{2} \leq n$, the second factor on the right-hand side above is bounded by $||(\phi,\partial_t\phi)||_{H^{n+1}(\mathbb{R}^n)\times H^n(\mathbb{R}^n)}(t)$, which we have already estimated.

Suppose now that n is even. Then we can immediately estimate

$$\begin{aligned}\sum_{|I_1|+|I_2|\leq n+1} ||\partial_x^{I_1}\partial\phi\cdot\partial_x^{I_2}\partial\phi||_{L^2(\mathbb{R}^n)} &\leq \sum_{|I_1|\leq\frac{n}{2}} ||\partial_x^{I_1}\partial\phi||_{L^\infty(\mathbb{R}^n)} \sum_{|I_2|\leq n+1} ||\partial_x^{I_2}\partial\phi||_{L^2(\mathbb{R}^n)} \\ &\leq B||(\phi,\partial_t\phi)||_{H^{n+2}(\mathbb{R}^n)\times H^{n+1}(\mathbb{R}^n)}(t).\end{aligned}$$

We then obtain

$$\begin{aligned}&\sup_{t\in[0,T]} ||(\phi,\partial_t\phi)||_{H^{n+2}(\mathbb{R}^n)\times H^{n+1}(\mathbb{R}^n)}(t) \\ &\leq C(T_*,B,||(\Phi,\Phi')||_{H^{n+1}(\mathbb{R}^n)\times H^n(\mathbb{R}^n)})\left[1+\int_0^T ||(\phi,\partial_t\phi)||_{H^{n+2}(\mathbb{R}^n)\times H^{n+1}(\mathbb{R}^n)}(t)\,dt\right].\end{aligned}$$

and conclude boundedness of the $H^{n+2}\times H^{n+1}$ norm after applying Grönwall.

In order to prove (iii), we need to make use of the following lemma:

Lemma 4.5. *Let $n, m \in \mathbb{N}$.*

(i) *Let $F : \mathbb{R}^m \to \mathbb{R}$ be a smooth function that satisfies $F(0) = 0$. Let $s \in \mathbb{N}_0$ and $g \in H^s(\mathbb{R}^n;\mathbb{R}^m) \cap L^\infty(\mathbb{R}^n;\mathbb{R}^m)$. Then there exists a constant $C = C(F, ||g||_{L^\infty(\mathbb{R}^n;\mathbb{R}^m)}, n, m, s) > 0$ such that*

$$||F(g)||_{H^s(\mathbb{R}^n)} \leq C\cdot||g||_{H^s(\mathbb{R}^n;\mathbb{R}^m)}.$$

(ii) *Let $s > 0$ and $f, g \in H^s(\mathbb{R}^n)\cap L^\infty(\mathbb{R}^n)$. Then, there exists a constant $C = C(n,s) > 0$ such that:*

$$||f\cdot g||_{H^s(\mathbb{R}^n)} \leq C\left(||f||_{H^s(\mathbb{R}^n)}||g||_{L^\infty(\mathbb{R}^n)} + ||g||_{H^s(\mathbb{R}^n)}||f||_{L^\infty}\right).$$

The proof of (iii), using Lemma 4.5, is done in **Problem 4.4**. □

4.2 Problems

Problem 4.1. *Let $k \in \mathbb{N}_0$ and let (f_i) be a sequence in $C^0([0,T];H^{k+1}(\mathbb{R}^n))\cap C^1([0,T];H^k(\mathbb{R}^n))$ that is*

(i) *uniformly bounded in $C^0([0,T];H^{k+1}(\mathbb{R}^n))\cap C^1([0,T];H^k(\mathbb{R}^n))$,*

(ii) *Cauchy in the larger space $C^0([0,T];H^1(\mathbb{R}^n))\cap C^1([0,T];L^2(\mathbb{R}^n))$.*

a) Show that the sequence $(f_i,\partial_t f_i)(t)$ has a subsequence with a weak limit *in $H^{k+1}(\mathbb{R}^n)\times H^k(\mathbb{R}^n)$ for all $t\in[0,T]$ by applying Theorem B.2 (the Banach–Alaoglu Theorem)., i.e. show that there exists an $(f(t),f'(t))\in H^{k+1}(\mathbb{R}^n)\times H^k(\mathbb{R}^n)$ such that*

$$\lim_{j\to\infty}\langle(f_{i_j}(t),\partial_t f_{i_j}(t)),(g,g')\rangle_{H^{k+1}(\mathbb{R}^n)\times H^k(\mathbb{R}^n)} = \langle(f(t),f'(t)),(g,g')\rangle_{H^{k+1}(\mathbb{R}^n)\times H^k(\mathbb{R}^n)},$$

for all $(g,g')\in H^{k+1}(\mathbb{R}^n)\times H^k(\mathbb{R}^n)$ and for all $t\in[0,T]$, with (f_{i_j}) a subsequence of (f_i).

b) Show that for all $t \in [0,T)$, $(f_i, \partial_t f_i)(t)$ converges with respect to the $H^1(\mathbb{R}^n) \times L^2(\mathbb{R}^n)$-norm and explain why the limit must lie in the smaller space $H^{k+1}(\mathbb{R}^n) \times H^k(\mathbb{R}^n)$ for all $t \in [0,T]$.

c) Conclude that $(f, \partial_t f) \in L^\infty([0,T]; H^{k+1}(\mathbb{R}^n)) \times L^\infty([0,T]; H^k(\mathbb{R}^n))$ and also:

$$\lim_{t \to t_0} \langle f(t,\cdot), v \rangle_{H^{k+1}(\mathbb{R}^n)} = \langle f(t_0,\cdot), v \rangle_{H^{k+1}(\mathbb{R}^n)},$$
$$\lim_{t \to t_0} \langle \partial_t f(t,\cdot), v \rangle_{H^{k}(\mathbb{R}^n)} = \langle \partial_t f(t_0,\cdot), w \rangle_{H^{k}(\mathbb{R}^n)}$$

for all $v \in H^k(\mathbb{R}^n)$ and $w \in H^{k+1}(\mathbb{R}^n)$.

Problem 4.2. *Let $f \in C_c^\infty(\mathbb{R}^n)$. Define the Fourier transform $\mathcal{F} : C_c^\infty(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ as follows $\mathcal{F}(f) = \hat{f}$, with:*

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} e^{-2\pi i x \cdot \xi} f(x)\, dx.$$

You may assume that $\mathcal{F}$ is well-defined and the following estimate holds: there exists a $C > 0$, such that

$$||f||^2_{H^s(\mathbb{R}^n)} \leq \int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\hat{f}|^2(\xi)\, d\xi \leq C||f||^2_{H^s(\mathbb{R}^n)}. \tag{4.10}$$

Then there exists a unique extension $\mathcal{F} : H^s(\mathbb{R}^n)) \to (1+|\xi|)^{-\frac{s}{2}} L^2(\mathbb{R}^n)$, with

$$(1+|x|)^{-\frac{s}{2}} L^2(\mathbb{R}^n) = \{ f \in L^2(\mathbb{R}^n) \,|\, (1+|x|)^{\frac{s}{2}} f \in L^2(\mathbb{R}^n) \}.$$

Prove Lemma 4.2 by employing the above Fourier transform and its corresponding inequalities. [Hint: use the Hölder inequality $||fg||_{L^1(\mathbb{R}^n)} \leq ||f||_{L^p(\mathbb{R}^n)} ||f||_{L^q(\mathbb{R}^n)}$, with $\frac{1}{p} + \frac{1}{q} = 1$.]

Remark 4.2. *The equivalence of norms in (4.10) allow us to identify $H^s(\mathbb{R}^n)$ with the space of functions $f \in L^2(\mathbb{R}^n)$ with Fourier transforms satisfying $(1+|\xi|)^{\frac{s}{2}} \hat{f} \in L^2(\mathbb{R}^n)$. In fact, since $(1+|\xi|)^{-\frac{s}{2}} L^2(\mathbb{R}^n)$ is also well-defined for $s \notin \mathbb{Z}$ it can be used to extend the notion of L^2-based Sobolev spaces to non-integer powers s.*

Problem 4.3. *a) Explain how the statement and proof of Theorem 4.1 need to be modified to obtain local wellposedness when we additionally allow for smooth dependence of $a(\phi, D\phi)$ and $F(\phi, D\phi)$ on $(t, \mathbf{x})$, i.e. we consider $F(t, \mathbf{x}, \phi, D\phi)$ and $a(t, \mathbf{x}, \phi, D\phi)$. You may additionally assume that*

(i) *The functions $a(\cdot,\cdot,v,w)$ and $F(\cdot,\cdot,v,w)$ are bounded on $\mathbb{R}^{n+1}$ for each fixed $v, w \in \mathbb{R} \times \mathbb{R}^{n+1}$ and so are their derivatives, up to arbitrarily high order.*

(ii) *$F(t, \mathbf{x}, 0, 0) = 0$,*

(iii) *There exists an $0 < \epsilon_0 < 1$ such that for all $(t, \mathbf{x})$, $(v, w) \in \mathbb{R} \times \mathbb{R}^{n+1}$:*

$$|a^{00} - m^{00}|(t, \mathbf{x}, v, w) \leq \epsilon_0,$$
$$\sqrt{\sum_{i,j=1}^n |a^{ij} - m^{ij}|^2(t, \mathbf{x}, v, w)} \leq \epsilon_0,$$

b) What kind of additional boundedness/regularity assumption is required for F if we drop the assumption $F(t, \mathbf{x}, 0, 0) = 0$ in part a)?

Problem 4.4. *Prove* (iii) *of Theorem 4.4 by applying Lemma 4.5.*

Problem 4.5. ** The goal of this problem is to show that $(\phi, \partial_t \phi) \in C^0([0,T]; H^{m+1}(\mathbb{R}^n)) \times C^0([0,T]; H^m(\mathbb{R}^n))$. Let $m \geq n+1$ and $(\phi, \partial_t \phi) \in L^\infty([0,T]; H^{m+1}(\mathbb{R}^n)) \times L^\infty([0,T]; H^m(\mathbb{R}^n))$ be defined as in Theorem 4.1. Let $t \in [0,T]$. Define the bilinear map $B_{m,t} : (H^{m+1}(\mathbb{R}^n) \times H^m(\mathbb{R}^n))^2 \to \mathbb{R}$ as follows:*

$$B_{m,t}[(f,f'),(g,g')] := \sum_{|I| \leq m} \int_{\mathbb{R}^n} -a^{00}(\phi(t,\mathbf{x}), D\phi(t,\mathbf{x})) \partial_x^I f' \cdot \partial_x^I g' + a^{ij}(\phi(t,\mathbf{x}), D\phi(t,\mathbf{x})) \partial_x^I \partial_i f \cdot \partial_x^I \partial_j g \, d\mathbf{x}.$$

a) By revisiting the energy estimates in Theorem 3.1 and Corollary 3.3 and appealing to the proof of (i)(A) of Theorem 4.1, show that there exists a $C = C(m, n, F, a) > 0$ such that for all $t_0, t' \in [0, T]$:

$$B_{m,t'}[(\phi(t',\cdot),\partial_t\phi(t',\cdot)),(\phi(t',\cdot),\partial_t\phi(t',\cdot))] \leq \Big[B_{m,t_0}((\phi(t_0,\cdot),\partial_t\phi(t_0,\cdot)),(\phi(t_0,\cdot),\partial_t\phi(t_0,\cdot))) + C|t'-t_0| \sup_{t\in[0,T]} ||(\phi,\partial_t\phi)||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t)\Big] \times e^{C|t'-t_0|\sup_{t\in[0,T]}||(\phi,\partial_t\phi)||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t)}.$$

b) Show that for all $t' \in [0, T]$

$$||(\phi(t',\cdot)-\phi(t_0,\cdot),\partial_t\phi(t',\cdot)-\partial_t\phi(t_0,\cdot))||_{H^m(\mathbb{R}^n)\times H^{m-1}(\mathbb{R}^n)} \leq C(t'-t_0)\sup_{t\in[0,T]}||(\phi,\partial_t\phi)||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t) \times e^{C|t'-t_0|\sup_{t\in[0,T]}||(\phi,\partial_t\phi)||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)}(t)}.$$

c) Show that $\lim_{t\to t_0}||(\phi(t,\cdot)-\phi(t_0,\cdot),\partial_t\phi(t,\cdot)-\partial_t\phi(t_0,\cdot))||_{H^{m+1}(\mathbb{R}^n)\times H^m(\mathbb{R}^n)} = 0$ for all $t_0 \in [0, T]$ and conclude that $(\phi, \partial_t\phi) \in C^0([0, T]; H^{m+1}(\mathbb{R}^n)) \times C^0([0, T]; H^m(\mathbb{R}^n))$. [Hint: Expand

$$B_{m,t_0}[(\phi(t,\cdot)-\phi(t_0,\cdot),\partial_t\phi(t,\cdot)-\partial_t\phi(t_0,\cdot)),(\phi(t,\cdot)-\phi(t_0,\cdot),\partial_t\phi(t,\cdot)-\partial_t\phi(t_0,\cdot))]+\langle\phi(t)-\phi(t_0),\phi(t)-\phi(t_0)\rangle_{L^2(\mathbb{R}^n)},$$

consider the limit as $t \to t_0$ and apply (a), (b) and the results of **Problem 4.1**(c).]

5 Global well-posedness: power nonlinearities

In this section, we prove a global well-posedness result for semilinear wave equations with power nonlinearities. We consider the following initial value problems:

$$\begin{aligned} \Box\phi &= \pm|\phi|^{p-1}\phi \quad \text{in } [0,T]\times\mathbb{R}^n, \\ (\phi|_{t=0},\partial_t\phi|_{t=0}) &= (\Phi,\Phi') \in H^{n+2}(\mathbb{R}^n)\times H^{n+1}(\mathbb{R}^n), \end{aligned} \tag{5.1}$$

with $p > 1$.

For sufficiently regular ϕ, the following energy is conserved in time t (provided ϕ is sufficiently regular and decaying in $|\mathbf{x}|$); see **Problem 5.2**):

$$E_{NLW}[\phi](t) := \int_{\mathbb{R}^n} \frac{1}{2}(\partial_t\phi)^2 + \frac{1}{2}\sum_{i=1}^n(\partial_i\phi)^2 \pm \frac{1}{p+1}|\phi|^{p+1}\,d\mathbf{x}.$$

Note that in the case $\pm \to +$, the above integrand is non-negative and therefore the energy is coercive. We refer to this case as the *defocusing* case. We refer to $\pm \to -$ as the *focusing* case.

We will require the following inequality:

Lemma 5.1 (Gagliardo–Nirenberg inequality)**.** *Let $f \in H^1(\mathbb{R}^n)$. Then there exists a $C = C(n) > 0$, such that for $1 \leq p \leq n$ and $\frac{1}{q} = \frac{1}{p} - \frac{1}{n}$:*

$$||f||_{L^q(\mathbb{R}^n)} \leq C||Df||_{L^p(\mathbb{R}^n)}.$$

Proof. We will omit the proof, but you can find it, for example, in §5.6 of:

Evans, L.C., 2010. *Partial Differential Equations* (Vol. 19). AMS . □

Note that for $n = 3$ and $p = 2$, the above inequality implies that:

$$||\phi||_{L^6(\mathbb{R}^3)}(t) \leq C\sqrt{E_{NLW}[\phi](t)}.$$

It will moreover be useful to use that for $f, g, h \in L^6(\mathbb{R}^n)$:

$$||f\cdot g\cdot h||_{L^2(\mathbb{R}^n)} \leq ||f||_{L^6(\mathbb{R}^n)}||g||_{L^6(\mathbb{R}^n)}||h||_{L^6(\mathbb{R}^n)}, \tag{5.2}$$

Exercise: Show that (5.2) holds by applying Hölder inequalities.

Theorem 5.2. *There exists a unique solution $\phi \in C^0([0,\infty); H^5(\mathbb{R}^3)) \cap C^1([0,\infty); H^4(\mathbb{R}^3))$ to (5.1) with $n = 3$ and $p = 3$. If (Φ, Φ') are smooth, then ϕ is also smooth in $[0,\infty) \times \mathbb{R}^3$.*

Proof. Suppose the maximal time of existence satisfies $T_* < \infty$. By Theorem 4.4 (ii),[26] it is sufficient to show that there exists a $B > 0$ such that

$$\sum_{|I|\leq 2} ||\partial_x^I \phi||_{L^\infty(\mathbb{R}^n)}(t) + \sum_{|I|\leq 1} ||\partial_x^I \partial_t \phi||_{L^\infty(\mathbb{R}^n)}(t) \leq B,$$

for all $0 \leq t \leq T_*$ in order to reach a contradiction. By the Sobolev embedding theorem, this follows from boundedness of

$$||\phi||_{H^4(\mathbb{R}^n)}(t) + ||\partial_t \phi||_{H^3(\mathbb{R}^n)}(t).$$

Smoothness of ϕ given smooth initial data, then follows from Theorem 4.3.

We first consider the $H^2 \times H^1$ norm. By (3.10), we have that for all $T < T_*$:

$$\sup_{t\in[0,T]} ||(\phi, \partial_t \phi)||_{H^2(\mathbb{R}^n)\times H^1(\mathbb{R}^n)}(t) \leq C(1+T)\left[||(\Phi, \Phi')||_{H^2(\mathbb{R}^n)\times H^1(\mathbb{R}^n)} + \int_0^T ||F||_{H^1(\mathbb{R}^n)}(t)\, dt\right],$$

with $F = \phi^3$.

We further estimate:

$$\begin{aligned}
||F||_{H^1(\mathbb{R}^3)}(t) &\leq C||\phi^3||_{L^2(\mathbb{R}^3)}(t) + C||\phi \cdot \phi \cdot \partial_x \phi||_{L^2(\mathbb{R}^3)}(t) \\
&\overset{(5.2)}{\leq} C||\phi||^3_{L^6(\mathbb{R}^3)}(t) + C\sum_{i=1}^n ||\phi||^2_{L^6(\mathbb{R}^3)}(t)||\partial_{x^i}\phi||_{L^6(\mathbb{R}^3)}(t) \\
&\overset{\text{Gagliardo–Nirenberg}}{\leq} C||\partial_x \phi||^3_{L^2(\mathbb{R}^3)}(t) + C||\partial_x \phi||^2_{L^2(\mathbb{R}^3)}(t)||\phi||_{H^2(\mathbb{R}^3)}(t) \\
&\leq CE_{NLW}[\phi](t)(E_{NLW}[\phi]^{\frac{1}{2}}(t) + ||\phi||_{H^2(\mathbb{R}^3)}(t)) \\
&\overset{\text{energy conservation}}{=} CE_{NLW}[\phi](0)(E_{NLW}[\phi]^{\frac{1}{2}}(0) + ||\phi||_{H^2(\mathbb{R}^3)}(t)).
\end{aligned}$$

After applying Grönwall, we can therefore find a constant $A > 0$ such that:

$$\sup_{t\in[0,T_*)} ||(\phi, \partial_t \phi)||_{H^2(\mathbb{R}^3)\times H^1(\mathbb{R}^3)}(t) \leq C(T_*, E_{NLW}[\phi](0))\left[||(\Phi, \Phi')||_{H^2(\mathbb{R}^3)\times H^1(\mathbb{R}^3)} + 1\right] \leq A$$

and also:

$$||F||_{H^1(\mathbb{R}^n)}(t) \leq CA.$$

We can now estimate the H^3 norm of F. We have that

$$\begin{aligned}
||F||_{H^3(\mathbb{R}^3)}(t) &\leq ||F||_{H^1(\mathbb{R}^3)}(t) + C\sum_{2\leq |I|+|J|+|K|\leq 3} ||\partial_x^I \phi \cdot \partial_x^J \phi \cdot \partial_x^K \phi||_{L^2(\mathbb{R}^3)}(t) \\
&\overset{(5.2)}{\leq} ||F||_{H^1(\mathbb{R}^3)}(t) + C\left(\sum_{|I|\leq 1} ||\partial_x^I \phi||_{L^6(\mathbb{R}^3)}\right)^2 \sum_{|I|\leq 2} ||\partial_x^I \phi||_{L^6(\mathbb{R}^3)} \\
&\quad + C||\phi||^2_{L^\infty(\mathbb{R}^n)} \sum_{|I|=3} ||\partial_x^I \phi||_{L^2(\mathbb{R}^3)} \\
&\overset{\text{Sobolev } H^2 \hookrightarrow L^\infty \text{ \& Gagliardo–Nirenberg}}{\leq} ||F||_{H^1(\mathbb{R}^3)}(t) + C||\phi||^2_{H^2(\mathbb{R}^3)}||\phi||_{H^3(\mathbb{R}^3)}.
\end{aligned}$$

[26] We can also apply Theorem 4.4 (iii), which simplifies the proof a little (Exercise). However, Theorem 4.4 (ii) is sufficient for our purposes.

By (3.10), Grönwall and already established bound for $||\phi||_{H^2(\mathbb{R}^3)}(t)+||F||_{H^3(\mathbb{R}^3)}(t)$, we therefore obtain

$$\sup_{t\in[0,T_*)} ||(\phi,\partial_t\phi)||_{H^4(\mathbb{R}^3)\times H^3(\mathbb{R}^3)}(t)$$
$$\leq C(T_*, E_{NLW}[\phi](0))\left[||(\Phi,\Phi')||_{H^4(\mathbb{R}^n)\times H^3(\mathbb{R}^n)} + A^2\right] =: B.$$

To obtain smoothness of ϕ given smooth, compactly supported data, we can apply Theorem 4.3. Finally, in the case of smooth data that are not necessarily compactly supported, we apply the domain of dependence property: $\phi(t,\mathbf{x})$ depends only on initial data in $B_{(0,\mathbf{x})}(t)$; see **Problem 5.1**, so the smoothness of ϕ at $(t,\mathbf{x})$ holds also when the data are not compactly supported. □

5.1 Scaling and criticality

In this section, we will introduce the notion of *criticality* by looking at the behaviour of conservation laws as we "zoom in" to smaller length scales.

Let ϕ denote a solution to (5.1). We introduce the following rescaled quantity for $\lambda\in\mathbb{R}_+$ and $\alpha\in\mathbb{R}$:

$$\phi_\lambda(t,\mathbf{x}) := \lambda^\alpha\phi(\lambda t,\lambda\mathbf{x}).$$

If we choose $\alpha=\frac{2}{p-1}$, then ϕ_λ solve the nonlinear wave equation (5.1) for any $\lambda\in\mathbb{R}_+$ (see **Problem 5.2**). The energy rescales in the following way:

$$E_{NLW}[\phi_\lambda](t) = \lambda^{2\frac{p+1}{p-1}-n}E_{NLW}[\phi](\lambda t);$$

see **Problem 5.2**. In the defocusing case, $E_{NLW}[\phi][t]$ provides controls over the L^2 norm of the derivatives of ϕ at time t, and by conservation of the energy in time, the value of $E_{NLW}[\phi][t]$ is fixed at $t=0$. If we restrict to a bounded spacetime region around $(t,\mathbf{x})=0$, then the consideration of ϕ_λ with $\lambda<1$ can be interpreted as zooming in at $(t,\mathbf{x})=0$. The energy of the rescaled quantity $E_{NLW}[\phi_\lambda](0)$ either grows, shrinks or stays the same as we zoom in. We say (5.1) is:

- **energy subcritical** if $E_{NLW}[\phi_\lambda](0)\to\infty$ as $\lambda\downarrow 0$,
- **energy supercritical** if $E_{NLW}[\phi_\lambda](0)\downarrow 0$ as $\lambda\downarrow 0$,
- **energy critical** if $E_{NLW}[\phi_\lambda](0) = E_{NLW}[0]$.

In view of the above scaling property of the energy, energy subcriticality in the $n\geq 2$ case corresponds to $1<p<\frac{n+2}{n-2}$, energy supercriticality corresponds to $p>\frac{n+2}{n-2}$ and criticality corresponds to $p=\frac{n+2}{n-2}$. The assumptions in Theorem 5.2 therefore correspond to the case of energy subcriticality. In fact, when $n=3$, global wellposedness for the defocusing (5.1) can be shown to hold for $p<5$ using more advanced methods ("Strichartz estimates").

This illustrates the validity of the following heuristic principle: "for nonlinear evolutionary PDE which are subcritical with respect to a coercive, monotone quantity, global wellposedness in time for large initial data holds because the monotone quantity provides better control at smaller scales". For equations that are supercritical, global wellposedness for large data might not hold. And indeed, little is known about the large data global wellposedness properties for the energy supercritical, defocusing (5.1). Critical equations are "borderline". In the case of energy critical defocusing (5.1), the "borderline case", global wellposedness has been shown, but requires additional ideas.

The notion of criticality extends more generally to evolutionary PDE, including the Navier–Stokes equations, which may be thought of as being supercritical in three dimensions with respect to known monotone quantities. This fact is related to the phenomenon of *turbulence* in fluid dynamics, which can be interpreted as the transfer of energy to smaller and smaller length scales. In contrast, the Navier–Stokes equations are critical in two dimensions and it has been shown that global wellposedness holds in that setting.

The notion of criticality is linked to Sobolev norms relating to monotone quantities. A supercritical PDE can become (sub)critical if new monotone quantities are discovered. This is what happened in Perelman's proof of the Poincaré conjecture: a key element of the proof was the discovery of new, coercive, monotone quantities for the Ricci flow equations, with respect to which the equations become critical.

5.2 Problems

Problem 5.1. *Show that Corollary 2.3 also holds for general semilinear wave equations:*

$$\Box\phi = F(\phi, D\phi) \quad \text{in } \mathbb{R}\times\mathbb{R}^n,$$

with $F : \mathbb{R}\times\mathbb{R}^{n+1} \to \mathbb{R}$ *smooth and* $F(0,0) = 0$.

Problem 5.2. *Let* ϕ *denote a solution to* (5.1). *Define* ϕ_λ *as follows:*

$$\phi_\lambda(t,\mathbf{x}) := \lambda^\alpha \phi(\lambda t, \lambda\mathbf{x}).$$

a) Show that $E_{NLW}[\phi](t)$ *is independent of* t *for* ϕ *sufficiently regular and decaying in* $|\mathbf{x}|$.

b) Show that the function ϕ_λ *is a solution to* (5.1) *for all* $\lambda \in \mathbb{R}_+$ *if* $\alpha = \frac{2}{p-1}$.

c) Show that $E_{NLW}[\phi_\lambda](t) = \lambda^{2\frac{p+1}{p-1}-n} E_{NLW}[\phi](\lambda t)$.

d) Define the homogeneous $H^s(\mathbb{R}^n)$ *norm*[27] *on* $C_c^\infty(\mathbb{R}^n)$ *as follows:*

$$||f||_{\dot{H}^s(\mathbb{R}^n)} := \sqrt{\sum_{|I|=s} \int_{\mathbb{R}^n} |\partial^I f|^2 \, d\mathbf{x}}.$$

Given an $s \in \mathbb{N}_0$ *with* $s < 2n$, *for which value(s) of* $p \in (1,\infty)$ *is the* $\dot{H}^s(\mathbb{R}^n)$ *norm of* $\phi_\lambda(t,\cdot)$ *independent of* λ?[28]

Problem 5.3. *a) Let* $n \in \mathbb{N}$ *and* $p > 1$ *and find the value of* $c_p \in \mathbb{R}$ *so that* $\phi_p(t,\mathbf{x}) = c_p(t-t_0)^{-\frac{2}{p-1}}$ *is a solution to the focusing* (5.1).

b) Construct smooth, compactly supported initial data for the focusing (5.1) *that give rise to a solution* ϕ *that blows up in finite time.*

You may use the following domain of dependence property *(as well as the result of* ***Problem 5.1****): let* ϕ_1 *and* ϕ_2 *be two solutions to*

$$\Box\phi_j = F(\phi_j, D\phi_j) \quad \text{in } \mathbb{R}\times\mathbb{R}^n,$$

with $j = 1,2$, $F : \mathbb{R}\times\mathbb{R}^{n+1} \to \mathbb{R}$ *smooth and* $F(0,0) = 0$, *such that* $(\phi_1 - \phi_2, \partial_t\phi_1 - \partial_t\phi_2)|_{t=0} = (0,0)$ *when restricted to the ball* $B_R(\mathbf{x}) \subset \mathbb{R}^3$. *Then* $\phi_1 \equiv \phi_2$ *in* $I^-_{(R,\mathbf{x})} \cap I^+_{(-R,\mathbf{x})}$.

[Hint: Modify the initial data that give rise to ϕ_p in a).]

Remark 5.1. *In the special case where* $(\phi_j, \partial_t\phi_j) = (0,0)$ *in* $B_{\mathbf{x}}(R)$ *for* $j = 1,2$, *this domain of dependence property follows directly from* ***Problem 5.1****. The more general case follows by revisiting the proof of Theorem 4.1(ii) in the spacetime region* $I^-_{(R,\mathbf{x})} \cap I^+_{(-R,\mathbf{x})}$ *and using that appropriate energy estimates still apply in this restricted region.*

6 Global linear theory

In this section, we establish global decay-in-time estimates (or "dispersive estimates") for the standard linear wave equation with inhomogeneity

$$\begin{aligned} \Box\phi &= F \quad \text{in } [0,\infty)\times\mathbb{R}^3, \\ (\phi, \partial_t\phi)|_{t=0} &= (\Phi,\Phi') \in C_c^\infty(\mathbb{R}^3)\times C_c^\infty(\mathbb{R}^3), \end{aligned} \tag{6.1}$$

which will allow us to study the global properties of nonlinear wave equations with more complicated nonlinearities with an appropriate smallness assumption on the initial data.

[27] Optional: Why is $||\cdot||_{\dot{H}^s(\mathbb{R}^n)}$ a well-defined norm?

[28] One may also consider $s \in \mathbb{R}_+ \setminus \mathbb{N}_0$ here by defining Sobolev spaces via Fourier transforms; see Remark 4.2.

Recall from (2.2) that the above equation takes the form in spherical coordinates (t, r, θ, φ):

$$-\partial_t^2\phi + r^{-2}\partial_r(r^2\partial_r\phi) + r^{-2}\not{\Delta}_{\mathbb{S}^2}\phi = F,$$
$$\not{\Delta}_{\mathbb{S}^2}(\cdot) = \frac{1}{\sin\theta}\partial_\theta(\sin\theta\partial_\theta(\cdot)) + \frac{1}{\sin^2\theta}\partial_\varphi^2(\cdot).$$

6.1 Energy boundedness

We revisit the energy boundedness statement from § 2.2. Rather than considering energies with respect to level sets of the time function t, we will introduce another time function τ that will be better suited for capturing the property of "energy radiating towards infinity".

Let $u = \frac{1}{2}(t-r)$ and $v = \frac{1}{2}(t+r)$ and $R > 2$. We define with respect to the radial function r:

$$\tau = t \quad \text{for } r \leq R,$$
$$\tau = t - r + R = 2u + R \quad \text{for } r > R.$$

We now introduce the following hypersurfaces:

$$\Sigma_{\tau'} = \{\tau = \tau'\},$$
$$N_{\tau'} = \{\tau = \tau'\} \cap \{r \geq R\},$$
$$I_{v'} = \{v = v', u \in \mathbb{R}\},$$

the 2-spheres:

$$S^2_{u',v'} = \{u = u\} \cap \{v = v'\}$$

and the spacetime regions

$$\mathcal{R}^{\tau_2}_{\tau_1} := \{\tau_1 < \tau < \tau_2\} \quad \text{for } 0 \leq \tau_1 < \tau_2,$$
$$\mathcal{D}^{\tau_2}_{\tau_1} := \mathcal{R}^{\tau_2}_{\tau_1} \cap \{r \geq R\}.$$

We can express ∂_u and ∂_v in (u, v, θ, φ) coordinates as:

$$\partial_u = \partial_t - \partial_r,$$
$$\partial_v = \partial_t + \partial_r.$$

Furthermore, we will apply shorthand notation for the following limiting integrals:

$$\int_{\mathcal{I}^+\cap\{\tau_1<\tau<\tau_2\}} (\ldots)\, d\sigma d\tau := 2\lim_{v\to\infty}\int_{I_v\cap\mathcal{R}^{\tau_2}_{\tau_1}} (\ldots)\, d\sigma du,$$

where

$$d\sigma = \sin\theta d\theta d\varphi.$$

Now we define the following energy quantity:

$$E[\phi](\tau) := \frac{1}{2}\int_{\Sigma_\tau\cap\{r\leq R\}} r^2(\partial_t\phi)^2 + r^2(\partial_r\phi)^2 + |\nabla\!\!\!/_{\mathbb{S}^2}\phi|^2\, d\sigma dr + \frac{1}{2}\int_{N_\tau} r^2(\partial_v\phi)^2 + |\nabla\!\!\!/_{\mathbb{S}^2}\phi|^2\, d\sigma dr,$$

with

$$|\nabla\!\!\!/_{\mathbb{S}^2}(\cdot)|^2 := (\partial_\theta(\cdot))^2 + \frac{1}{\sin^2\theta}(\partial_\varphi(\cdot))^2.$$

The following Leibniz rule holds on $\mathbb{S}^2$:

$$(f\cdot\not{\Delta}_{\mathbb{S}^2}g)\sin\theta = -\left[\partial_\theta f\cdot\partial_\theta g + \frac{1}{\sin^2\theta}\partial_\varphi f\cdot\partial_\varphi g\right]\sin\theta + \partial_\theta(\sin\theta f\cdot\partial_\theta g) + \partial_\varphi\left(\frac{1}{\sin\theta^2}f\cdot\partial_\varphi g\right).$$

Note that the total derivative terms on the right-hand side vanish after integration over θ, φ.[29]

[29] For those with a background in Riemannian geometry: the total derivative terms can be written more geometrically as follows: $\sin\theta\mathrm{div}_{\mathbb{S}^2}(\sin\theta f\nabla\!\!\!/_{\mathbb{S}^2}g)$, where $\nabla\!\!\!/_{\mathbb{S}^2}$ is the Levi–Civita connection and $\mathrm{div}_{\mathbb{S}^2}$ the corresponding divergence operator.

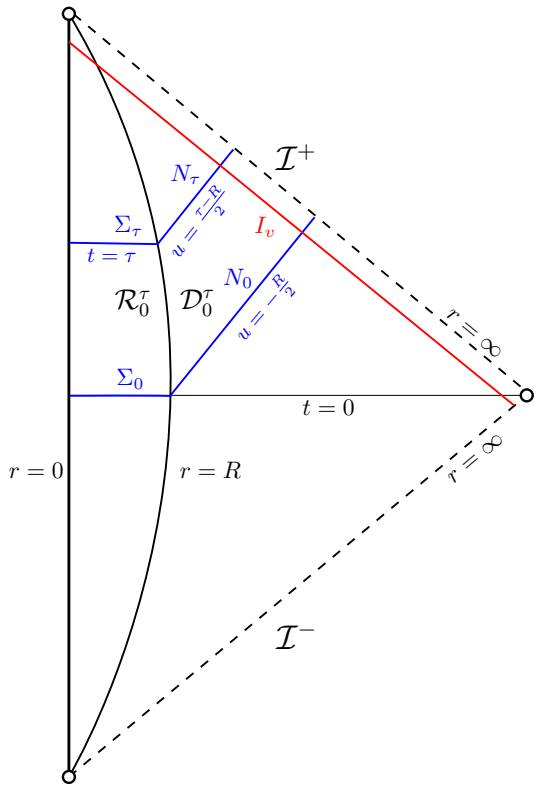


Figure 3: A pictorial representation of the relevant spacetime regions.

Proposition 6.1. *Let ϕ be the solution to (6.1) with F suitably regular. Then we can estimate*

$$E[\phi](\tau_2) - E[\phi](\tau_1) + \frac{1}{4}\int_{\mathcal{I}^+\cap\{\tau_1<\tau<\tau_2\}} r^2(\partial_u\phi)^2 + |\nabla\!\!\!/_{\mathbb{S}^2}\phi|^2\, d\sigma d\tau = -\int_{\tau_1}^{\tau_2}\int_{\Sigma_\tau} \partial_t\phi \cdot F\, r^2 d\sigma dr d\tau. \tag{6.2}$$

Proof. We can apply the Leibniz rule to obtain

$$\begin{aligned} -r^2\partial_t\phi\cdot F =& -r^2\partial_t\phi\cdot\left(-\partial_t^2\phi + r^{-2}\partial_r(r^2\partial_r\phi) + r^{-2}\Delta\!\!\!/_{\mathbb{S}^2}\phi\right)\\ =& \frac{1}{2}\partial_t(r^2(\partial_t\phi)^2) - \partial_r(r^2\partial_r\phi\partial_t\phi) + \frac{1}{2}\partial_t((\partial_r\phi)^2) + \frac{1}{2}\partial_t(|\nabla\!\!\!/_{\mathbb{S}^2}\phi|^2) + \mathrm{div}_{\mathbb{S}^2}(\ldots), \end{aligned}$$

with $\mathrm{div}_{\mathbb{S}^2}(\ldots)$ schematic notation for angular divergence terms, which vanish after integration over $\mathbb{S}^2$.

Note that $v = u + r \leq \frac{1}{2}\tau_2 + R$ along $\Sigma_{\tau_i}\cap\{r\leq R\}$, $i = 1, 2$. Integrating in $\bigcup_{\tau_1\leq\tau\leq\tau_2}\Sigma_\tau\cap\{v\leq V\}$, with $V \geq \frac{1}{2}\tau_2 + R$ and using that $dr = -du = -\frac{1}{2}d\tau$ along I_V, we obtain:

$$\begin{aligned} \int_{\tau_1}^{\tau_2}\int_{\Sigma_\tau\cap\{v\leq V\}} -\partial_t\phi\cdot F\, r^2 d\sigma dr d\tau =& \frac{1}{2}\int_{\Sigma_{\tau_2}\cap\{r\leq R\}} r^2(\partial_t\phi)^2 + r^2(\partial_r\phi)^2 + |\nabla\!\!\!/_{\mathbb{S}^2}\phi|^2\, d\sigma dr\\ &- \frac{1}{2}\int_{\Sigma_{\tau_1}\cap\{r\leq R\}} r^2(\partial_t\phi)^2 + r^2(\partial_r\phi)^2 + |\nabla\!\!\!/_{\mathbb{S}^2}\phi|^2\, d\sigma dr\\ &+ \frac{1}{2}\int_{N_{\tau_2}\cap\{v\leq V\}} r^2(\partial_t\phi)^2 + r^2(\partial_r\phi)^2 + 2r^2\partial_r\phi\partial_t\phi + |\nabla\!\!\!/_{\mathbb{S}^2}\phi|^2\, d\sigma dr\\ &- \frac{1}{2}\int_{N_{\tau_1}\cap\{v\leq V\}} r^2(\partial_t\phi)^2 + r^2(\partial_r\phi)^2 + 2r^2\partial_r\phi\partial_t\phi + |\nabla\!\!\!/_{\mathbb{S}^2}\phi|^2\, d\sigma dr\\ &+ \frac{1}{2}\int_{I_V\cap\{\tau_1<\tau<\tau_2\}} r^2(\partial_t\phi)^2 + r^2(\partial_r\phi)^2 - 2r^2\partial_r\phi\partial_t\phi + |\nabla\!\!\!/_{\mathbb{S}^2}\phi|^2\, d\sigma du. \end{aligned}$$

We can furthermore express:

$$r^2(\partial_t\phi)^2 + r^2(\partial_r\phi)^2 + 2r^2\partial_r\phi\partial_t\phi = r^2(\partial_v\phi)^2,$$
$$r^2(\partial_t\phi)^2 + r^2(\partial_r\phi)^2 - 2r^2\partial_r\phi\partial_t\phi = r^2(\partial_u\phi)^2$$

and take the limit $V \to \infty$ to obtain (6.2). □

Remark 6.1. *Contrast Proposition 6.1, which implies for $F \equiv 0$ that $E(\tau_1) < E(\tau_2)$ if the integral along $\mathcal{I}^+$ is strictly possible (positive energy flux through infinity), with Corollary 2.4, which implies $E(t_1) = E(t_2)$ for energies defined with respect to t-level sets rather than τ-level set. Since we eventually want to establish* decay *in time, and not just boundedness, it is therefore more natural to consider τ-level sets.*

6.2 Integrated local energy estimates

From § 6.1, it is possible to show that solutions to the standard wave equation can be bounded *uniformly in time*, i.e. we can estimate $||\phi||_{L^\infty(\Sigma_\tau)}$ in terms of initial energies with constants $C > 0$ that are independent of time.[30] However, to understand the global properties of the nonlinear equations in later sections, we will need more than boundedness: we will need decay.

A "weak" type of energy decay statement would be:

$$\int_{\tau_1}^{\tau_2} E[\phi](\tau)\, d\tau \leq CE[\phi](\tau_1). \tag{6.3}$$

for all $0 \leq \tau_1 < \tau_2 \leq \infty$. Indeed, by the mean-value theorem, boundedness of the left-hand side would already tell us that there exists a sequence of times $\{\tau_i\}$, with $\tau_i \to \infty$ along which $E[\psi](\tau)$ needs to decay. In fact, from **Problem 6.1** it follows that (6.3) together with (6.1) (with $F \equiv 0$) lead to exponential decay in τ for $E[\phi](\tau)$.

The estimate (6.3) will turn out to be a little too strong. Let us introduce a *local energy*: let $R_0 > 0$, then

$$E[\phi](\tau; R_0) := \frac{1}{2}\int_{\Sigma_\tau \cap \{r \leq \max\{R, R_0\}\}} r^2(\partial_t\phi)^2 + r^2(\partial_r\phi)^2 + |\nabla\!\!\!/_{\mathbb{S}^2}\phi|^2\, d\sigma dr + \frac{1}{2}\int_{N_\tau \cap \{r \leq R_0\}} r^2(\partial_v\phi)^2 + |\nabla\!\!\!/_{\mathbb{S}^2}\phi|^2\, d\sigma dr.$$

The aim of this section is to show instead the *integrated local energy estimate*:

$$\int_{\tau_1}^{\tau_2} E[\phi](\tau; R_0)\, d\tau \leq CE[\phi](\tau_1),$$

where $C > 0$ can depend on $R_0 > 0$. This type of estimates is also known as a *Morawetz estimate*.

To obtain an integrated local energy estimate, we will apply once more a vector field multiplier argument. However, rather than considering $X\phi\Box\phi$, for a suitably choice of vector field X, it is more convenient to rewrite (6.1) as an equation for $\psi = r\phi$ (recall the derivation of the Huygens principles in § 2.1):

$$-\partial_u\partial_v\psi + r^{-2}\Delta\!\!\!/_{\mathbb{S}^2}\psi = rF, \tag{6.4}$$

and then multiply it with $X\psi$.

Proposition 6.2. *Let ϕ be the solution to (6.1) with F suitably regular. Then:*

(i) *Let $\delta > 0$. Then there exists a constant $C = C(R, \delta) > 0$, such that for all $0 \leq \tau_1 < \tau_2 \leq \infty$:*

$$\begin{aligned}\int_{\tau_1}^{\tau_1}\int_{\Sigma_\tau} r^2(1+r)^{-1-\delta}\left[(\partial_u\phi)^2 + (\partial_v\phi)^2 + r^{-2}\phi^2\right] + r^{-1}|\nabla\!\!\!/_{\mathbb{S}^2}\phi|^2 d\sigma dr d\tau \leq CE[\phi](\tau_1)\\ + C\int_{\tau_1}^{\tau_1}\int_{\Sigma_\tau} r^2(1+r)^{1+\delta}|F|^2\, d\sigma dr d\tau.\end{aligned} \tag{6.5}$$

[30] Obtaining estimates for L^∞ norms via energy norms along Σ_τ is done in § 6.5.

(ii) *There exists a constant $C = C(R, R_0) > 0$, such that for all $0 \leq \tau_1 < \tau_2 \leq \infty$:*

$$\int_{\tau_1}^{\tau_2} E[\phi](\tau; R_0)\, d\tau + \int_{\tau_1}^{\tau_1} \int_{\Sigma_\tau \cap \{r \leq R_0\}} \phi^2 \, d\sigma dr d\tau \leq C E[\phi](\tau_1) + C \int_{\tau_1}^{\tau_1} \int_{\Sigma_\tau} r^2 (1+r)^{1+\delta} |F|^2 \, d\sigma dr d\tau. \tag{6.6}$$

Proof. We consider the vector field $X = f(r)(\partial_v - \partial_u) = 2f(r)\partial_r$, where $f : \mathbb{R}_+ \to \mathbb{R}$ will be chosen later. Then we apply the Leibniz rule, together with $\partial_v r = -\partial_u r = 1$ to obtain:

$$\begin{aligned} X\psi \cdot rF &= f(r)(\partial_v \psi - \partial_u \psi)\left(-\partial_u \partial_v \psi + r^{-2} \not{\Delta}_{\mathbb{S}^2} \psi\right) \\ &= -\frac{1}{2} f(r) \partial_u \left((\partial_v \psi)^2\right) + \frac{1}{2} f(r) \partial_v \left((\partial_u \psi)^2\right) - \frac{1}{2} r^{-2} f(r) (\partial_v - \partial_u) \left(|\not{\nabla}_{\mathbb{S}^2} \psi|^2\right) + \mathrm{div}_{\mathbb{S}^2}(\ldots) \\ &= -\frac{1}{2} \partial_u \left[f(r)(\partial_v \psi)^2 - r^{-2} f(r) |\not{\nabla}_{\mathbb{S}^2} \psi|^2\right] + \frac{1}{2} \partial_v \left[f(r)(\partial_u \psi)^2 - r^{-2} f(r) |\not{\nabla}_{\mathbb{S}^2} \psi|^2\right] \\ &\quad - \frac{1}{2} \frac{df}{dr} \left[(\partial_v \psi)^2 + (\partial_u \psi)^2\right] + \frac{d(r^{-2} f)}{dr} |\not{\nabla}_{\mathbb{S}^2} \psi|^2 + \mathrm{div}_{\mathbb{S}^2}(\ldots). \end{aligned}$$

Step 1: $f \equiv -1$ We first take $f \equiv -1$ to obtain:

$$X\psi \cdot rF = \frac{1}{2} \partial_u \left[(\partial_v \psi)^2 - r^{-2} |\not{\nabla}_{\mathbb{S}^2} \psi|^2\right] - \frac{1}{2} \partial_v \left[(\partial_u \psi)^2 - r^{-2} |\not{\nabla}_{\mathbb{S}^2} \psi|^2\right] + r^{-3} |\not{\nabla}_{\mathbb{S}^2} \psi|^2 + \mathrm{div}_{\mathbb{S}^2}(\ldots).$$

Hence, after integration over $\bigcup_{\tau \in [\tau_1 \tau_2]} \Sigma_\tau$, we obtain a spacetime ("bulk") term with a good sign:

$$\int_{\mathcal{R}_{\tau_1}^{\tau_2}} 2r^{-3} |\not{\nabla}_{\mathbb{S}^2} \psi|^2 d\sigma du dv.$$

We moreover obtain the boundary terms:

$$\begin{aligned} &\frac{1}{2} \int_{\Sigma_{\tau_2} \cap \{r \leq R\}} (\partial_v \psi)^2 - (\partial_u \psi)^2 \, d\sigma dr - \frac{1}{2} \int_{\Sigma_{\tau_1} \cap \{r \leq R\}} (\partial_v \psi)^2 - (\partial_u \psi)^2 \, d\sigma dr \\ &+ \frac{1}{2} \lim_{\epsilon \downarrow 0} \int_{r=\epsilon} (\partial_v \psi)^2 + (\partial_u \psi)^2 - 2r^{-2} |\not{\nabla}_{\mathbb{S}^2} \psi|^2 \, d\sigma dv - \frac{1}{2} \int_{\mathcal{I}^+ \cap \{\tau_1 < \tau < \tau_2\}} (\partial_u \psi)^2 - r^{-2} |\not{\nabla}_{\mathbb{S}^2} \psi|^2 \, \frac{1}{2} d\sigma d\tau \\ &+ \frac{1}{2} \int_{N_{\tau_2}} (\partial_v \psi)^2 - r^{-2} |\not{\nabla}_{\mathbb{S}^2} \psi|^2 \, d\sigma dr - \frac{1}{2} \int_{N_{\tau_1}} (\partial_v \psi)^2 - r^{-2} |\not{\nabla}_{\mathbb{S}^2} \psi|^2 \, d\sigma dr. \end{aligned}$$

We observe that at $r = \epsilon$, using that $\Delta = r^{-2} \partial_r (r^2 \partial_r (\cdot)) + r^{-2} \not{\Delta}_{\mathbb{S}^2}(\cdot)$:

$$\begin{aligned} \int_{r=\epsilon} |\not{\nabla}_{\mathbb{S}^2} \phi|^2 \, d\sigma dv &= -\int_{r=\epsilon} \phi \not{\Delta}_{\mathbb{S}^2} \phi \, d\sigma dv \\ &= -\int_{r=\epsilon} r^2 \phi \left(\Delta \phi - \partial_r (r^2 \partial_r \phi)\right) \, d\sigma dv \\ &\to 0 \quad \text{as } \epsilon \downarrow 0, \end{aligned}$$

by the assumption that ϕ is sufficiently regular at $r = 0$. Note moreover that $\partial_v \psi|_{r=0} = -\partial_u \psi|_{r=0} = \phi|_{r=0}$. Hence, the $r = 0$ boundary term has a good sign.

We can estimate the remaining boundary terms via (6.2). First, we note that:

$$\begin{aligned} (\partial_u \psi)^2 &\leq 2r^2 (\partial_u \phi)^2 + 2\phi^2, \\ (\partial_v \psi)^2 &\leq 2r^2 (\partial_v \phi)^2 + 2\phi^2. \end{aligned}$$

To estimate the integral of ϕ^2 along Σ_τ, we will apply a *Hardy inequality*.

Lemma 6.3 (Hardy inequalities)**.** *Let $a, b \in \mathbb{R}$ with $a < b$ and let $f \in C^1([a, b])$. Then for $q \in \mathbb{R} \setminus \{-1\}$:*

$$\int_a^b x^q f^2(x) \, dx \leq \frac{2}{q+1} \left[b^{q+1} f^2(b) - a^{q+1} f^2(a)\right] + \frac{4}{(q+1)^2} \int_a^b x^{q+2} \left(\frac{df}{dx}\right)^2 (x) \, dx. \tag{6.7}$$

Proof. See **Problem 6.2**. □

By taking $q = 0$, we can estimate:

$$\int_{\Sigma_\tau} \phi^2 \, d\sigma dr \leq 4 \int_{\Sigma_\tau \cap \{r \leq R\}} r^2 (\partial_r \phi)^2 \, d\sigma dr + 4 \int_{N_\tau} r^2 (\partial_v \phi)^2 \, d\sigma dr + 2 \limsup_{v \to \infty} \int_{S^2_{\frac{\tau - R}{2}, v}} r \phi^2 \, d\sigma.$$

Using **Problem 6.3**, we can moreover estimate:

$$\limsup_{v \to \infty} \int_{S^2_{\frac{\tau - R}{2}, v}} r \phi^2 \, d\sigma \leq CE[\phi](\tau) + C \int_{\mathcal{R}^\tau_{\tau_1}} r^2 |F| |\partial_t \phi| \, d\sigma du dv,$$

so that for all $\tau_1 \leq \tau \leq \tau_2$:

$$\int_{\Sigma_\tau} \phi^2 \, d\sigma dr \leq CE[\phi](\tau_1) + C \int_{\mathcal{R}^{\tau_2}_{\tau_1}} r^2 |F| |\partial_t \phi| \, d\sigma du dv. \tag{6.8}$$

We will leave it as an Exercise to conclude that moreover:

$$\int_{\mathcal{I}^+ \cap \{\tau_1 < \tau < \tau_2\}} (\partial_u \psi)^2 \, d\sigma du \leq CE[\phi](\tau_1) + C \int_{\mathcal{R}^{\tau_2}_{\tau_1}} r^2 |F| |\partial_t \phi| \, d\sigma du dv.$$

Hence, we conclude **Step 1**, with the following estimate:

$$\int_{\mathcal{R}^{\tau_2}_{\tau_1}} r^{-3} |\nabla\!\!\!/_{\mathbb{S}^2} \psi|^2 d\sigma du dv + \int_{\{r=0\} \cap \{\tau_1 < \tau < \tau_2\}} \phi^2 \, d\sigma dv \leq CE[\phi](\tau_1) + C \int_{\mathcal{R}^{\tau_2}_{\tau_1}} r |F| (|\partial_r \psi| + |\partial_t \psi|) \, d\sigma du dv.$$

Step 2: $f(r) = (1+r)^{-\delta}$ Now that we have control over the angular derivatives as well as a $r = 0$ boundary terms, via **Step 1**. We consider X with $f(r) = (1+r)^{-\delta}$, with $0 < \delta \leq 1$, to control the remaining derivatives. The boundary terms can be estimated as above and we now see the following bulk terms:

$$\int_{\mathcal{R}^{\tau_2}_{\tau_1}} \frac{\delta}{2} (1+r)^{-1-\delta} \left[(\partial_u \psi)^2 + (\partial_v \psi)^2 \right] - \frac{1}{2} r^{-3} (1+r)^{-1-\delta} (2 + (2+\delta) r) |\nabla\!\!\!/_{\mathbb{S}^2} \psi|^2 d\sigma du dv$$

Note that the angular derivative term has a bad sign, but we can already control in with our previous multiplier estimate ($f \equiv -1$). We therefore obtain

$$\begin{aligned} \int_{\mathcal{R}^{\tau_2}_{\tau_1}} (1+r)^{-1-\delta} \left[(\partial_u \psi)^2 + (\partial_v \psi)^2 \right] + r^{-3} |\nabla\!\!\!/_{\mathbb{S}^2} \psi|^2 d\sigma du dv + \int_{\{r=0\} \cap \{\tau_1 < \tau < \tau_2\}} \phi^2 \, d\sigma dv \leq CE[\phi](\tau_1) \\ + C \int_{\mathcal{R}^{\tau_2}_{\tau_1}} r |F| (|\partial_r \psi| + |\partial_t \psi|) \, d\sigma du dv. \end{aligned} \tag{6.9}$$

Note that we can split via Young's inequality:

$$\begin{aligned} Cr|F|(|\partial_r \psi| + |\partial_t \psi|) &\leq \frac{1}{2} (1+r)^{-1-\delta} ((\partial_r \psi)^2 + (\partial_t \psi)^2) + C^2 (1+r)^{1+\delta} r^2 |F|^2 \\ &\leq \frac{1}{4} (1+r)^{-1-\delta} ((\partial_u \psi)^2 + (\partial_v \psi)^2) + C^2 (1+r)^{1+\delta} r^2 |F|^2 \end{aligned}$$

and absorb the $(\partial_u \psi)^2$ and $(\partial_v \psi)^2$ terms in the corresponding integral into the left-hand side of (6.9).

Step 3: spacetime integral ϕ^2 In the final step, we establish control over a spacetime integral of ϕ^2. Two applications of Lemma 6.3 (with $q = -2$ and $q = -3 + \delta$, respectively) allow use to conclude that:

$$\int_{\Sigma_\tau \cap \{r \leq R\}} r^{-2} \psi^2 \, d\sigma dr + \int_{\Sigma_\tau \cap \{r = R\}} r^{-1} \psi^2 \, d\sigma \leq C \int_{\Sigma_\tau \cap \{r \leq R\}} (\partial_r \psi)^2 \, d\sigma dr,$$

$$\int_{N_\tau} r^{-3-\delta} \psi^2 \, d\sigma dr \leq C \int_{N_\tau} r^{-1-\delta} (\partial_v \psi)^2 \, d\sigma dr + C \int_{\Sigma_\tau \cap \{r = R\}} r^{-2-\delta} \psi^2 \, d\sigma.$$

Hence, using (6.9) together with the above two Hardy inequalities, we can conclude that (6.5) holds, from which (6.6) immediately follows. □

6.3 A hierarchy of r-weighted energy estimates

In this section, we develop the tools necessary to go from a *local* integrated energy estimate to a global energy estimate. We will in particular show that for $F \equiv 0$:

$$\int_{\tau_1}^{\tau_2} E[\phi](\tau)\, d\tau \leq CE[\phi](\tau_1) + C\int_{N_{\tau_1}} r\cdot(\partial_v\psi)^2\, d\sigma dr. \tag{6.10}$$

In contrast with (6.3), there is an extra factor r in front of the 2nd term on the RHS, which will prevent us from concluding exponential decay in τ! The tool that we will use are the *Dafermos–Rodnianski r-weighted energy estimates*.

Since it only remains to improve the integrated energy estimate from § 6.2 for large r, we will restrict to the domain $\mathcal{D}^{\tau_2}_{\tau_1} = \mathcal{R}^{\tau_2}_{\tau_1}\cap\{r\geq R\}$.

Proposition 6.4 (Dafermos–Rodnianski r-weighted energy estimates)**.** *Let ϕ be the solution to (6.1) with F suitably regular. Let $0<p\leq 2$, then there exists a constant $C=C(R,p)>0$, such that:*

$$\begin{aligned}&\int_{N_{\tau_2}} r^p(\partial_v\psi)^2\, d\sigma dr + \frac{1}{2}\int_{\mathcal{I}^+\cap\{\tau_1<\tau<\tau_2\}} r^{p-2}|\nabla\!\!\!/_{\mathbb{S}^2}\psi|^2\, d\sigma du + \frac{1}{2}\int_{\tau_1}^{\tau_2}\int_{N_\tau} pr^{p-1}(\partial_v\psi)^2 + (2-p)r^{p-3}|\nabla\!\!\!/_{\mathbb{S}^2}\psi|^2\, d\sigma dr d\tau\\ &\quad\leq \int_{N_{\tau_1}} r^p(\partial_v\psi)^2\, d\sigma dr + CE[\psi](\tau_1) + C\int_{\mathcal{R}^{\tau_2}_{\tau_1}} r^{p+3}|F|^2 + r^2(1+r)^{1+\delta}|F|^2\, d\sigma dr d\tau\end{aligned} \tag{6.11}$$

Proof. Let $\chi:\mathbb{R}_+\to\mathbb{R}$ denote a smooth cut-off function satisfying $\chi(r)=1$ for $r\geq R+1$ and $\chi(r)=0$ for $r\leq R$. We then multiply the equation (6.4) with $-r^p\chi\partial_v\psi$, with p to be determined, and we apply the Leibniz rule to obtain:

$$\begin{aligned}-r^p\chi\partial_v\psi\cdot rF =& -r^p\chi\partial_v\psi\left[-\partial_u\partial_v\psi + r^{-2}\Delta\!\!\!/_{\mathbb{S}^2}\psi\right]\\ =& \frac{1}{2}\chi r^p\partial_u((\partial_v\psi)^2) + \frac{1}{2}\chi r^{p-2}\partial_v(|\nabla\!\!\!/_{\mathbb{S}^2}\psi|^2) + \mathrm{div}_{\mathbb{S}^2}(\ldots)\\ =& \partial_u\left[\frac{1}{2}\chi r^p(\partial_v\psi)^2\right] + \partial_v\left[\frac{1}{2}\chi r^{p-2}|\nabla\!\!\!/_{\mathbb{S}^2}\psi|^2\right]\\ &+\frac{1}{2}pr^{p-1}\chi(\partial_v\psi)^2 + \frac{1}{2}(2-p)\chi r^{p-3}|\nabla\!\!\!/_{\mathbb{S}^2}\psi|^2 + \frac{1}{2}\frac{d\chi}{dr}\left[r^p(\partial_v\psi)^2 - r^{p-2}|\nabla\!\!\!/_{\mathbb{S}^2}\psi|^2\right]\end{aligned}$$

Integrating the above equation over $\mathcal{D}^{\tau_2}_{\tau_1}$, we can therefore estimate:

$$\begin{aligned}&\int_{N_{\tau_2}}\chi r^p(\partial_v\psi)^2\, d\sigma dr + \int_{\mathcal{I}^+\cap\{\tau_1\leq\tau\leq\tau_2\}} r^{p-2}|\nabla\!\!\!/_{\mathbb{S}^2}\psi|^2\, d\sigma du\\ &\quad+\int_{\mathcal{D}^{\tau_2}_{\tau_1}}\chi pr^{p-1}(\partial_v\psi)^2 + (2-p)\chi r^{p-3}|\nabla\!\!\!/_{\mathbb{S}^2}\psi|^2\, d\sigma dr du\\ &\leq \int_{N_{\tau_1}}\chi r^p(\partial_v\psi)^2\, d\sigma dr + C\int_{\mathcal{D}^{\tau_2}_{\tau_1}\cap\{R\leq r\leq R+1\}} r^{p-2}|\nabla\!\!\!/_{\mathbb{S}^2}\psi|^2\, d\sigma du dv + 2\int_{\mathcal{D}^{\tau_2}_{\tau_1}}\chi r^{p+1}|F||\partial_v\psi|\, d\sigma du dv.\end{aligned} \tag{6.12}$$

Now note that by applying (6.6) we can not only estimate the second integral on the RHS of (6.12), but also extend the integration range on the LHS from $r\geq R+1$ to $r\geq R$.

Now, note that

$$2r^{p+1}\chi|F||\partial_v\psi| \leq \frac{p}{2}\chi r^{p-1}(\partial_v\psi)^2 + \frac{2}{p}\chi r^{p+3}F^2$$

so we can absorb the $r^{p-1}\chi(\partial_v\psi)^2$ term into the left-hand side of (6.12).

Furthermore, the spacetime integral in $\{R\leq r\leq R+1$ on the RHS of (6.12) can be estimated via (6.6).

Finally, we can get rid of the χ factors, which are not equal to 1 when $R\leq r\leq R+1$ in (6.12) as follows: 1) in the spacetime integrals, we apply (6.6), 2) we can estimate the integral of $r^P($
$partial_v\psi)^2$ over $N_{\tau_i}\cap\{R\leq r\leq R+1\}$ by first splitting

$$(\partial_v\psi)^2 \leq 2r^2(\partial_v\phi)^2 + 2\phi^2,$$

and then applying (6.8) in combination with (6.2) □

In particular, applying (6.11) with $p = 1$ gives the global integrated energy estimate (6.10). The estimates (6.11) are known as the *Dafermos–Rodnianski r-weighted energy estimates*.

6.4 Energy decay

In this section, we will show that $E[\psi](\tau)$ in fact decays (at least) polynomially in time, in the homogeneous setting ((6.1) with $F \equiv 0$).

The pidgeonhole principle We will first introduce a useful tool for converting integrated energy estimates to decay-in-time estimates. We start by defining the notion of *dyadic sequences*:

Definition 6.1. *We say a strictly monotonically increasing sequence $\{a_i\}_{i\in\mathbb{N}_0}$ in $\mathbb{R}$ is dyadic if there exists constants $c, C > 0$, such that*

$$a_i \leq a_{i+1} \leq Ca_i,$$
$$ca_i \leq a_{i+1} - a_i \leq Ca_i.$$

Below we will apply the mean-value theorem along dyadic sequences. This is also known as the *pigeonhole principle*. Let $f, g : [0, \infty) \to \mathbb{R}$ be non-negative functions. If there exist constants $C, B > 0$, such that for all $0 \leq a < b \leq \infty$

$$\int_a^b f(x)\,dx \leq Cg(a),$$
$$\int_a^b g(x)\,dx \leq B$$

then given a dyadic sequence $\{x_i\}$ in $[0, \infty)$ (for example $x_i = 2^i$), we can apply the mean value theorem in the intervals $[x_i, x_{i+1}]$ to conclude that there exists a sequence $\{x_i'\}$, with $x_i \leq x_i' \leq x_{i+1}$, satisfying:

$$g(x_i') = \frac{1}{(x_i - x_{i+1})}\int_{x_i}^{x_{i+1}} g(x)\,dx \leq \frac{B}{(x_{i+1} - x_i)} \leq CB(x_i')^{-1}.$$

Now, define $y_i = x'_{2i+1}$, then it follows that $\{y_i\}$ is also dyadic.[31] Hence, we can repeat the above argument along $\{y_i\}$ to obtain the existence of a sequence $\{y_i'\}$, $y_i \leq y_i' \leq y_{i+1}$ with:

$$\begin{aligned} f(y_i') =& \frac{1}{(y_i - y_{i+1})}\int_{y_i}^{y_{i+1}} f(x)\,dx \leq C(y_{i+1} - y_i)^{-1}g(y_i) \leq C(y_i')^{-1}g(y_i) \\ &\leq CB(y_i')^{-1}y_i^{-1} \leq C(y_i')^{-2}. \end{aligned}$$

The above estimates illustrate how we can obtain x^{-2}-decay of f along a sequence, given a hierarchy of integrated estimates of length 2. If $B = Ch(a)$ and we have an integrated estimate for $h(x)$, then we would be able to apply the above arguments to obtain x^{-3}-decay along a sequence. The decay rate therefore corresponds to the length of the hierarchy of estimates.

Theorem 6.5. *Let ϕ be the solution to (6.1) with $F \equiv 0$. There exists a $C = C(R) > 0$, such that for all $\tau \geq 0$:*

$$E(\tau) \leq C(1+\tau)^{-2}\left[E[\phi](0) + \int_{N_0} r^2(\partial_v\psi)^2\,d\sigma dr\right]. \tag{6.13}$$

Furthermore,

$$\int_{N_\tau} r^2(\partial_v\psi)^2\,d\sigma dr \leq C\left[E[\phi](0) + \int_{N_0} r^2(\partial_v\psi)^2\,d\sigma dr\right]. \tag{6.14}$$

Proof. The strategy is to apply the above pigeonhole principle together with the integrated energy estimates that follow from Proposition 6.4 with $p = 1, 2$, in combination with the integrated local energy estimate (6.6).

[31] Exercise: Convince yourself of this.

Step 1: τ^{-1}-decay By applying (6.11) with $p = 1$ in combination with (6.6), we obtain: there exists a constant $C > 0$, such that for all $\tau_1 \leq \tau \leq \tau_2$

$$\int_{\tau_1}^{\tau_2} E[\phi](\tau)\, d\tau \leq CE[\phi](\tau_1) + C\int_{N_{\tau_1}} r(\partial_v\psi)^2\, d\sigma dr.$$

We can replace the RHS above with τ_1 replaced by 0, using (6.2) in combination with (6.11) with $p = 1$ and conclude via the pigeonhole principle that there exists a dyadic sequence $\{\tau_i\}$ such that:

$$E[\phi](\tau_i) \leq C\tau_i^{-1}\left[E[\phi](0) + \int_{N_0} r(\partial_v\psi)^2\, d\sigma dr\right].$$

To extend the above estimate to all $\tau_i \leq \tau \leq \tau_{i+1}$, we apply (6.2) again, together with the dyadicity of $\{\tau_i\}$ and conclude:

$$\begin{aligned} E[\phi](\tau) &\leq CE[\phi](\tau_i) \leq C(1+\tau_i)^{-1}\left[E[\phi](0) + \int_{N_0} r(\partial_v\psi)^2\, d\sigma dr\right] \\ &\leq C(1+\tau_{i+1})^{-1}\left[E[\phi](0) + \int_{N_0} r(\partial_v\psi)^2\, d\sigma dr\right] \\ &\leq C(1+\tau)^{-1}\left[E[\phi](0) + \int_{N_0} r(\partial_v\psi)^2\, d\sigma dr\right]. \end{aligned} \tag{6.15}$$

Step 2: τ^{-2}-decay Now we apply (6.11) with $p = 1$ *and* $p = 2$ in combination with (6.6), and in addition (6.15) to obtain the hierarchy: for all $0 \leq \tau_1 \leq \tau_2$

$$\int_{\tau_1}^{\tau_2} E[\phi](\tau)\, d\tau \leq C\int_{N_{\tau_1}} r(\partial_v\psi)^2\, d\sigma dr + C(1+\tau_1)^{-1}\left[E[\phi](0) + \int_{N_0} r(\partial_v\psi)^2\, d\sigma dr\right],$$

$$\int_{N_\tau} r^2(\partial_v\psi)^2\, d\sigma dr + \int_{\tau_1}^{\tau_2}\int_{N_\tau} r(\partial_v\psi)^2\, d\tau \leq C\int_{N_0} r^2(\partial_v\psi)^2\, d\sigma dr + CE[\phi](0).$$

The estimate (6.14) follows immediately from the second inequality above. Applying the pigeonhole principle once, gives moreover (as in **Step 1** above),

$$\int_{N_{\tau_i}} r(\partial_v\psi)^2\, d\tau \leq C\tau_i^{-1}\left[\int_{N_0} r^2(\partial_v\psi)^2\, d\sigma dr + E[\phi](0)\right]$$

along a dyadic sequence $\{\tau_i\}$. Plugging this estimate above, we therefore obtain the existence of another dyadic sequence $\{\tau_i'\}$ such that

$$E[\phi](\tau_i') \leq C\tau_i'^{-2}\left[E[\phi](0) + \int_{N_0} r^2(\partial_v\psi)^2\, d\sigma dr\right].$$

Finally, this can be extended to all τ by applying (6.2) again, as in **Step 1** to conclude the proof. □

6.5 Pointwise estimates

In order to go from energy decay estimates to pointwise decay estimates, we will make use of another symmetry, *the rotational invariance* of the standard wave equation. Define the angular momentum operators Ω_i, $i = 1, 2, 3$ as follows:

$$\begin{aligned} \Omega_1 &= -\sin\varphi\partial_\theta - \cot\theta\cos\varphi\partial_\varphi, \\ \Omega_2 &= \cos\varphi\partial_\theta - \cot\theta\sin\varphi\partial_\varphi, \\ \Omega_3 &= \partial_\varphi. \end{aligned}$$

It is straightforward to show that $\Omega_1^2 + \Omega_2^2 + \Omega_3^2 = \not{\Delta}_{\mathbb{S}^2}$. Furthermore, $[\not{\Delta}_{\mathbb{S}^2}, \Omega_i] = [\Box, \Omega_i] = 0$.

Let $I = (I_1, I_2, I_3) \in \mathbb{N}_0^3$. We introduce the notation $\Omega^I = \Omega_1^{I_1}\Omega_2^{I_2}\Omega_3^{I_3}$.

We can obtain pointwise estimates from L^2 estimates for Ω^I-derivatives on $\mathbb{S}^2$. Indeed, we have the following Sobolev inequality on $\mathbb{S}^2$: there exists a constant $C > 0$, such that

$$||f||_{L^\infty(\mathbb{S}^2)} \leq C\sqrt{\sum_{|I|\leq 2} ||\Omega^I f||^2_{L^2(\mathbb{S}^2)}},$$

which is proved in Lemma A.4.

Proposition 6.6. *Then there exists a constant $C = C(R) > 0$, such that for all $\tau \geq 0$:*

$$||r^{\frac{1}{2}}\phi||_{L^\infty(\Sigma_\tau)} \leq C\sum_{|I|\leq 2}\sqrt{E[\Omega^I\phi](\tau)}, \tag{6.16}$$

$$||r\phi||_{L^\infty(\Sigma_\tau)} \leq C\sum_{|I|\leq 2}\left(E[\Omega^I\phi](\tau)\right)^{\frac{1}{4}}\left(E[\Omega^I\phi](\tau) + \int_{N_\tau} r^2(\partial_v\Omega^I\psi)^2\, d\sigma dr\right)^{\frac{1}{4}}, \tag{6.17}$$

$$||\phi||_{L^\infty(\Sigma_\tau\cap\{r<1\})} \leq C\sum_{k\leq 1}\sqrt{E[\partial_t^k\phi](\tau)}. \tag{6.18}$$

Furthermore, if ϕ the solution to (6.1) with $F \equiv 0$, then:

$$||r^{\frac{1}{2}}\phi||_{L^\infty(\Sigma_\tau)} \leq C(1+\tau)^{-1}\sum_{|I|\leq 2}\sqrt{E[\Omega^I\phi](0) + \int_{N_0} r^2(\partial_v\Omega^I\psi)^2\, d\sigma dr}, \tag{6.19}$$

$$||r\phi||_{L^\infty(\Sigma_\tau)} \leq C(1+\tau)^{-\frac{1}{2}}\sum_{|I|\leq 2}\sqrt{E[\Omega^I\phi](0) + \int_{N_0} r^2(\partial_v\Omega^I\psi)^2\, d\sigma dr}, \tag{6.20}$$

$$||\phi||_{L^\infty(\Sigma_\tau\cap\{r\leq 1\})} \leq C(1+\tau)^{-1}\sum_{k\leq 1}\sqrt{E[\partial_t^k\phi](0) + \int_{N_0} r^2(\partial_v\partial_t^k\psi)^2\, d\sigma dr}. \tag{6.21}$$

We can combine (6.19)–(6.22) and write:

$$||(r+1)^{\frac{1}{2}}(v+1)^{\frac{1}{2}}(\tau+1)^{\frac{1}{2}}\phi||_{L^\infty(\Sigma_\tau)} \leq C\sum_{k+|I|\leq 2}\sqrt{E[\partial_t^k\Omega^I\phi](0) + \int_{N_0} r^2(\partial_v\partial_t^k\Omega^I\psi)^2\, d\sigma dr}. \tag{6.22}$$

Proof. Note that (6.19)–(6.22), follow directly from (6.16)–(6.18) by plugging in the energy time decay estimates of Theorem 6.5.

To prove (6.16), we integrate $\partial_v\phi$ along N_τ and $\partial_r\phi$ along $\Sigma_\tau \cap \{r \leq R\}$ to obtain: if $r \geq R$,

$$\begin{aligned}\phi^2(\tau, r, \theta, \varphi) &\leq \lim_{r\to\infty}\phi^2(\tau, r, \theta, \varphi) + \left(\int_r^\infty |\partial_v\phi|(\tau, r, \theta, \varphi)\, dr'\right)^2 \\ &\leq \int_r^\infty r'^{-2}\, dr' \cdot \int_r^\infty r^2(\partial_v\phi)^2(\tau, r, \theta, \varphi)\, dr',\end{aligned}$$

where we used that $\lim_{r\to\infty}\phi^2(\tau, r, \theta, \varphi) = 0$ (see **Problem 6.3**) and we applied Cauchy–Schwarz. Similarly, for $r < R$

$$\phi^2(\tau, r, \theta, \varphi) \leq \phi^2(\tau, R, \theta, \varphi) + \int_r^\infty r'^{-2}\, dr' \cdot \int_r^\infty r^2(\partial_r\phi)^2(\tau, r, \theta, \varphi)\, dr',$$

so we can conclude that

$$||r^{\frac{1}{2}}\phi||^2_{L^2(\mathbb{S}^2)}(\tau, r) \leq E[\phi](\tau). \tag{6.23}$$

We conclude (6.16) by applying Lemma A.4.

We prove (6.17) by integrating $\partial_v(\psi^2)$ along N_τ:

$$\begin{aligned}\psi^2(\tau,r,\theta,\varphi) &\leq \psi^2(\tau,R,\theta,\varphi) + \int_R^r \partial_v(\psi^2)(\tau,r,\theta,\varphi)\,dr' \\ &\leq \psi^2(\tau,R,\theta,\varphi) + 2\int_R^r |\psi||\partial_v\psi|(\tau,r,\theta,\varphi)\,dr' \\ &\leq \psi^2(\tau,R,\theta,\varphi) + \sqrt{\int_R^r r^2(\partial_v\psi)^2(\tau,r',\theta,\varphi)\,dr'}\cdot\sqrt{\int_R^r \phi^2(\tau,r',\theta,\varphi)\,dr'}.\end{aligned}$$

We obtain

$$||r\phi||^4_{L^2(\mathbb{S}^2)}(\tau,r) \leq E[\phi](\tau)^2 + E[\phi](\tau)\cdot\int_{N_\tau} r^2(\partial_v\psi)^2\,d\sigma dr$$

by applying Lemma 6.3 to estimate:

$$\int_R^r \phi^2(\tau,r',\theta,\varphi)\,dr' \leq 4\int_R^r r^2(\partial_v\phi)^2(\tau,r',\theta,\varphi)\,dr' + 2r\phi^2(\tau,R,\theta,\varphi)$$

and using that the right-hand side can be bounded by $E[\phi](\tau)$ after integration along $\mathbb{S}^2$ and application of (6.23). We conclude (6.17) by applying Lemma A.4.

To prove (6.22), we first apply a Sobolev inequality restricted to $\Sigma_\tau\cap\{r<1\}$ (see Theorem A.2 and Remark A.1) to estimate

$$||\phi||_{L^\infty(\Sigma_\tau\cap\{r<1\})} \leq ||\phi||_{H^2(\Sigma_\tau\cap\{r<1\})} \leq ||(1-\Delta)\phi||_{L^2(\Sigma_\tau\cap\{r<1\})}$$

Then, we multiply ϕ with a smooth cut-off function χ, with $\chi\equiv 1$ in $r\leq 1$ and $\chi\equiv 0$ in $r\geq 2$ and we split:

$$(I-\Delta)(\chi\phi) = (I-\Delta)(\chi)\phi - 2\sum_i \partial_i(\chi)\partial_i\phi - \chi\partial_t^2\phi$$

so that we can apply Theorem B.1 to conclude that

$$\begin{aligned}||\phi||_{H^2(\Sigma_\tau\cap\{r<1\})} \leq ||\chi\phi||_{H^2(\Sigma_\tau\cap\{r<2\})} \leq ||(I-\Delta)(\chi\phi)||_{L^2(\Sigma_\tau\cap\{r<2\})} &\leq CE[\phi](\tau) + C\int_{\Sigma_\tau\cap\{r\leq 2\}} \phi^2 + (\partial_t^2\phi)^2\,d\sigma dr \\ &\leq C(E[\phi](\tau) + E[\partial_t\phi](\tau)),\end{aligned}$$

where we used Lemma 6.3 again to estimate

$$\int_{\Sigma_\tau\cap\{r\leq 2\}} \phi^2\,d\sigma dr \leq 4\int_{\Sigma_\tau\cap\{r\leq 2\}} r^2(\partial_r\phi)^2\,d\sigma dr + 2\int_{\Sigma_\tau\cap\{r=2\}} r\phi^2\,d\sigma,$$

where the right-hand side can be bounded by $E[\phi](\tau)$ after applying (6.23). □

Remark 6.2. *We note that in the context of wave equations on Minkowski, there are other vector field methods for arriving at pointwise decay-in-time estimates, like the classical method of* Klainerman–Sobolev inequalities*, which rely more on the symmetry properties of the Minkowski spacetime. We refer to Sogge for further details.*

6.6 Additional higher-order estimates

In order to derive global properties for the nonlinear wave equations in the next section, we need to appeal to versions of energy boundedness (Proposition 6.1), integrated local energy decay (Proposition 6.2) and r-weighted energy estimates (Proposition 6.4) that include appropriate higher-order derivatives. For this purpose, we introduce the following notation. We denote with Z a vector field belonging to $\{T,\Omega_1,\Omega_2,\Omega_3,(r+1)\partial_v\}$, with $T=\partial_t$ in (t,r) coordinates. A nd we denote for $I\in\mathbb{N}_0^5$: $Z^I = (\Omega_1)^{I_2}(\Omega_2)^{I_2}(\Omega_3)^{I_3}T^{I_4}((r+1)\partial_v)^{I_5}$.

We will make use of the notation $E_{\leq V}[\phi](\tau)$ to indicate a restriction to the region $\Sigma_\tau\cap\{v\leq V\}$ in the integrals defining the energy $E[\phi](\tau)$.

Proposition 6.7. *Let ϕ be the solution to (6.1) with F suitably regular. Let $N \in \mathbb{N}_0$, $\delta > 0$ and $0 < p \leq 2$. Then there exists a constant $C = C(R, N, \delta, p) > 0$, such that for all $0 \leq \tau_1 < \tau_2$:*

$$\begin{aligned}
&\sum_{|I|\leq N} \sup_{\tau_1\leq\tau\leq\tau_2} E_{\leq V}[Z^I\phi](\tau) \qquad (6.24)\\
&\quad + \int_{\tau_1}^{\tau_2}\int_{\Sigma_\tau\cap\{v\leq V\}} r^2(r+1)^{-1-\delta}\left[(\partial_u Z^I\phi)^2 + (\partial_v Z^I\phi)^2 + r^{-2}|\nabla\!\!\!/_{\mathbb{S}^2} Z^I\phi|^2 + r^{-2}(Z^I\phi)^2\right] d\sigma dr d\tau\\
&\leq C\sum_{|I|\leq N} E_{\leq V}[Z^I\phi](\tau_1) + C\int_{\mathcal{R}^{\tau_2}_{\tau_1}\cap\{v\leq V\}} r^2(r+1)^{1+\delta}|Z^I F|^2\, d\sigma dr d\tau,\\
&\sum_{|I|\leq N} \sup_{\tau_1\leq\tau\leq\tau_2} \int_{N_\tau\cap\{v\leq V\}} r^p(\partial_v Z^I\psi)^2\, d\sigma dr + \int_{I_V\cap\{\tau_1\leq\tau\leq\tau_2\}} r^{p-2}|\nabla\!\!\!/_{\mathbb{S}^2} Z^I\psi|^2\, d\sigma du \qquad (6.25)\\
&\quad + \int_{\tau_1}^{\tau_2}\int_{N_\tau\cap\{v\leq V\}} pr^{p-1}(\partial_v Z^I\psi)^2 + (2-p)r^{p-3}|\nabla\!\!\!/_{\mathbb{S}^2} Z^I\psi|^2\, d\sigma dr d\tau\\
&\leq C\sum_{|I|\leq N}\int_{N_{\tau_1}\cap\{v\leq V\}} r^p(\partial_v Z^I\psi)^2\, d\sigma dr + CE_{\leq V}[Z^I\psi](\tau_1)\\
&\quad + \int_{\mathcal{R}^{\tau_2}_{\tau_1}\cap\{v\leq V\}} r^{p+3}|Z^I F|^2 + r^2(r+1)^{1+\delta}|Z^I F|^2\, d\sigma dr d\tau.
\end{aligned}$$

Proof. We will only provide a sketch of the proof. First of all, one may immediately apply the estimates from Propositions 6.1, 6.2 and 6.4 to ϕ replaced by $T^k\Omega^I\phi$, for all $k \in \mathbb{N}_0$, $I \in \mathbb{N}_0^3$, since $[\Box, T] = [\Box, \Omega_i] = 0$, so $T^k\Omega^I\phi$ satisfies the same equation as ϕ, but with F replaced by $T^k\Omega^I F$. It therefore only remains to control derivatives in $(r+1)\partial_v$. Via elliptic estimates, we immediately obtain control of all higher-order derivatives in the region $r \leq R$ from control of higher-order T derivatives. Indeed, we multiply ϕ with a smooth cut-off function χ, with $\chi \equiv 1$ in $r \leq R$ and $\chi \equiv 0$ in $r \geq R+1$ and we split:

$$(I - \sum_i \partial_i^2)(\chi\phi) = (I - \sum_i \partial_i^2)(\chi)\phi - 2\sum_i \partial_i(\chi)\partial_i\phi - \chi\partial_t^2\phi - \chi F =: f$$

so that we can apply Theorem B.1.

To conclude that (6.24) holds, we can use that when considering $(r\partial_v)^k\psi$ with $k \geq 1$ instead of ψ, Proposition 6.4 applies in fact also with $p = 0$ (this is not true when $k = 0$), thereby providing control of general Z^I derivatives in the region $r \geq R$ with the desired powers in r.

Finally, (6.25) follows similarly from the fact that Proposition 6.4 remains valid for $0 < p \leq 2$ when replacing ψ with $(r\partial_v)^k\psi$, $k \geq 0$.

We leave it as an Exercise for the interested reader to fill in the details of the proof. □

Corollary 6.8. *Let ϕ be the solution to (6.1) with F suitably regular. Let $N \in \mathbb{N}_0$, $\delta > 0$ and $0 < p \leq 2$. Then there exists a constant $C = C(R, N, \delta, p) > 0$, such that for all $z\tau \geq 0$:*

$$\sum_{|I|\leq N-2} ||(r+1)^{\frac{1}{2}}(v+1)^{\frac{1}{2}}(\tau+1)^{\frac{1}{2}} Z^I\phi||_{L^\infty(\Sigma_\tau)} \leq C\sum_{|I|\leq N}\sqrt{E[Z^I\phi](0) + \int_{N_0} r^2(\partial_v Z^I\psi)^2\, d\sigma dr}, \qquad (6.26)$$

$$\sum_{|I|\leq N-3}\sum_{i=1}^{3} ||(\tau+1)(r^{-1}\Omega_i)Z^I\phi||_{L^\infty(\Sigma_\tau\cap\{r\leq 1\})} \leq C\sum_{|I|\leq N}\sqrt{E[Z^I\phi](0) + \int_{N_0} r^2(\partial_v Z^I\psi)^2\, d\sigma dr}. \qquad (6.27)$$

Proof. The estimate (6.26) is the higher-order analogue of (6.22) and follows by repeating the arguments in the proof of Theorem 6.5 and Proposition 6.6, but making use of the higher-order energy estimates from Proposition 6.7.

To conclude (6.27), we first observe that for any suitably regular function $f : \Sigma_\tau \to \mathbb{R}$ and for $r \leq 1$:

$$\sum_{i=1}^{3} ||r^{-1}\Omega_i f||_{L^\infty(S^2_{\tau',r})} \overset{(A.3)}{\leq} \sum_{i=1}^{3} ||r^{-1}\Omega_i f||_{H^2(B_2(0))}(\tau') \leq C||r^{-1}\nabla\!\!\!/_{\mathbb{S}^2} f||_{H^2(B_2(0))}(\tau) \leq C||f||_{H^3(B_2(0))}(\tau),$$

where we used that $||\nabla\!\!\!/_{\mathbb{S}^2} h||_{L^2(\mathbb{S}^2)} = \sum_{i=1^3} ||\Omega_i h||_{L^2(\mathbb{S}^2)}$ to arrive at the second inequality above,[32] and the third inequality simply follows by writing down the H^3-norm in spherical coordinates.

Taking $f = Z^I\phi$ and applying an elliptic estimate as in the proof of Proposition 6.7, we can further estimate:

$$||Z^I\phi||_{H^3(B_2(0))}(\tau) \leq C \sum_{k\leq 2+|I|} E[T^k\phi](\tau) + C \sum_{k\leq 1+|I|} ||T^k F||_{L^2(B_3(0))}(\tau).$$

For $F \equiv 0$, (6.27) the follows from the energy decay properties of $Z^I\phi$. □

6.7 Problems

Problem 6.1. *Let $f : [0,\infty) \to \mathbb{R}_+$ be a continuous function. Assume that there exist $B > 0$ such that for all $\tau \geq \tau' \geq 0$:*

$$f(\tau) + \int_{\tau'}^{\tau} f(s)\, ds \leq B f(\tau').$$

a) Let $A > 0$. Show that there exists a sequence $\{\tau_i\}$, with $\tau_0 = 0$ and $\frac{A}{2} \leq \tau_{i+1} - \tau_i \leq A$, such that

$$f(\tau_i) \leq \left(\frac{4B^2}{A}\right)^i f(0).$$

b) Conclude that there exist constants $C = C(B) > 0$ and $c = c(B) > 0$ such that for all $\tau \in [0,\infty)$:

$$f(\tau) \leq C f(0) e^{-c\tau}.$$

Problem 6.2. *Prove Lemma 6.3.* [Hint: integrate the identity $\frac{d}{dx}(x^{q+1}f^2) = (q+1)x^q f^2 + 2x^{q+1} f\frac{df}{dx}$ and apply a weighted Young's inequality: $2A \cdot B \leq wA^2 + \frac{1}{w}B^2$, for any $w \in \mathbb{R}_+$ and $A, B \in \mathbb{R}$.]

Problem 6.3. *Let ϕ be the solution to (6.1) and assume that F is smooth.*

a) Let $\tau_0 > 0$. Show that for all $V > 0$

$$\frac{1}{2}\int_{I_V \cap \{\tau \leq \tau_0\}} r^2(\partial_u\phi)^2 + |\nabla\!\!\!/_{\mathbb{S}^2}\phi|^2\, d\sigma du \leq E[\phi](0) + \int_{\{\tau\leq\tau_0\}\cap\{v\leq V\}} |F||\partial_t\phi| r^2\, d\sigma du dv.$$

b) Let $0 \leq \tau_1 < \tau_2$. Show that for all (u,v) with $\frac{\tau_1 - R}{2} \leq u \leq \frac{\tau_2 - R}{2}$ and $V \geq \frac{\tau_2 + R}{2}$:

$$\int_{S^2_{u,V}} \phi^2\, d\sigma \leq 2\max\left\{\left(u + \frac{R}{2}\right) r^{-2}(u,V), r^{-1}(u,V)\right\}\left[E[\phi](\tau_1) + \int_{\mathcal{R}^{\tau_2}_{\tau_1}\cap\{v\leq V\}} |F||\partial_t\phi| r^2\, d\sigma du dv\right]$$

and conclude that $\lim_{V\to\infty} r^q\phi(u,V) \to 0$ for all $q < 1$ when the spacetime integral involving F is finite in the limit where $V \to \infty$. [Hint: Apply the fundamental theorem of calculus along I_V.]

Problem 6.4. *a) Let $f : \Sigma_\tau \to \mathbb{R}$ be a continuous function. Assume that for some $m_1, m_2 \in \mathbb{R}$ with $m_1 < m_2$, there exist $q \in \mathbb{R}$, $D_1, D_1 > 0$ and $V > 0$, such that:*

$$\int_{\Sigma_\tau\cap\{v\leq V\}} r^{m_1} f^2\, d\sigma dr \leq (1+\tau)^{-q} D_1, \tag{6.28}$$

$$\int_{\Sigma_\tau\cap\{v\leq V\}} r^{m_2} f^2\, d\sigma dr \leq (1+\tau)^{-q+(m_2-m_1)} D_2. \tag{6.29}$$

Show that for all $m_1 \leq m \leq m_2$ the following interpolation inequality holds:

$$\int_{\Sigma_\tau\cap\{v\leq V\}} r^m f^2\, d\sigma dr \leq (1+\tau)^{-q+m-m_1}(D_1 + D_2).$$

[Hint: Split $\Sigma_\tau \cap \{v \leq V\} = (\Sigma_\tau \cap \{v \leq V\} \cap \{r \leq (1+\tau)\}) \cup (\Sigma_\tau \cap \{v \leq V\} \cap \{r > (1+\tau)\})$.]

[32] Exercise: Convince yourself that this holds.

b) Using the interpolation inequality from a), revisit the proof of Proposition 6.5 and derive the following alternative energy decay statement: for $0 \leq \delta \leq \frac{1}{2}$:

$$E(\tau) \leq C(1+\tau)^{-2+2\delta}\left[E[\phi](0) + \int_{N_0} r^{2-2\delta}(\partial_v \psi)^2 \, d\sigma dr\right].$$

7 Global well-posedness: more general nonlinearities

In this chapter, we will investigate nonlinear wave equations with quadratic (or higher-order) nonlinearities involving *derivatives* of ϕ. For example,

$$\Box\phi = (\partial_t \phi)^2, \quad \Box\phi = \partial_u\phi\partial_v\phi.$$

These types of nonlinearities show up when considering, for example, Einstein's equations of general relativity and the compressible Euler's equations. We will see that the *structure* of the nonlinearity is very important for determining global well-posedness of solutions. In particular, we will see that in the case of the nonlinearity $(\partial_t\phi)^2$, there is no well-posedness, whereas there *is* well-posedness when considering the nonlinearity $\partial_u\phi\partial_v\phi$.

7.1 A counter-example to global well-posedness

We consider the initial value problem

$$\begin{aligned} \Box\phi &= (\partial_t\phi)^2 \quad \text{in } [0,T)\times\mathbb{R}^3, \\ (\phi, \partial_t\phi)|_{t=0} &= (\Phi,\Phi') \in C_c^\infty(\mathbb{R}^3)\times C_c^\infty(\mathbb{R}^3). \end{aligned} \tag{7.1}$$

By Theorem 4.1, (7.1) admits a unique solution in $[0,T)\times\mathbb{R}^3$, provided $T>0$ is chosen suitably small, depending on (Φ,Φ'). In the theorem below, we show that for any non-trivial data, we cannot take $T\to\infty$.

Theorem 7.1. *Let ϕ be a solution to the initial value problem (7.1) with $(\Phi,\Phi') \neq (0,0)$. Then the maximal time of existence T_* (as defined in Theorem 4.3) satisfies $T_* < \infty$ and*

$$\limsup_{t\uparrow T_*} \sum_{|I|\leq 1} ||\partial_x^I \phi||_{L^\infty(\mathbb{R}^n)}(t) + ||\partial_t\phi||_{L^\infty(\mathbb{R}^n)}(t) = \infty. \tag{7.2}$$

Proof. Note first of all that (7.2) follows from the property $T_* < \infty$ by Theorem 4.4. We will argue by contradiction that $T_* < \infty$. The strategy of the proof is to assume $T_* = \infty$ and then show that this implies that $\phi \equiv 0$, which is in contradiction with $(\Phi,\Phi') \neq (0,0)$.

To avoid confusion when passing between different coordinate choices, we denote $\mathbf{T} = \partial_t$ in (t,r) coordinates. The wave equation in (7.1) is equivalent to the following equation for $\psi = r\phi$ in (u,v) coordinates:

$$\partial_u\partial_v\psi - r^{-2}\not{\Delta}_{\mathbb{S}^2}\psi = -r^{-1}(\mathbf{T}\psi)^2$$

Denoting with f_0 the spherical mean of a function f in $[0,\infty)\times\mathbb{R}^3$ along the 2-spheres of constant (u,v), we obtain

$$\partial_u\partial_v\psi_0 = -r^{-1}((\mathbf{T}\psi)^2)_0 \leq -r^{-1}(\mathbf{T}\psi_0)^2,$$

where we used that $(f^2)_0 \geq f_0^2$, which follows from Cauchy–Schwarz along the spheres.

We will take $R>0$ suitably large so that $\mathrm{supp}((\Phi,\Phi')) \subset B_R(0)$. Since (7.1) is semilinear, the arguments in the proof of Corollary 2.3 still apply to conclude that $\phi\equiv 0$ in the domain of dependence of $\mathbb{R}^3\setminus B_R(0)$, see **Problem 5.1**. In other words, the weak Huygens principle still holds.

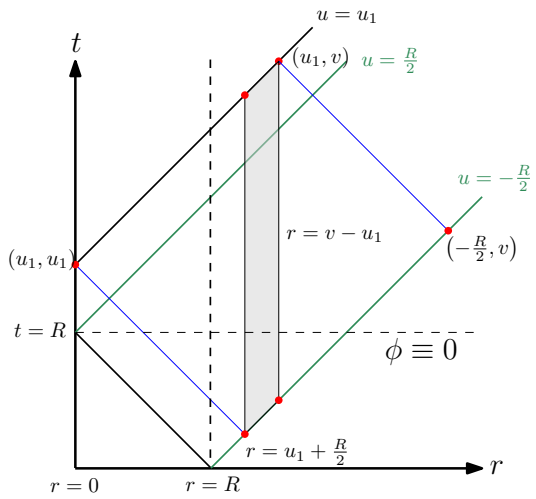


Figure 4: The spacetime integration regions.

Let $u_1 > \frac{R}{2}$. We will investigate the behaviour of ψ_0 along $u = u_1$ by integrating first in the v-direction, starting from $r = 0$ and then in the u-direction, starting from $u = -\frac{R}{2}$:

$$\begin{aligned}
\psi_0(u_1, v) &= \int_{u_1}^{v} \partial_v \psi_0(u_1, v')\, dv' \\
&= \int_{u_1}^{v} \int_{-\frac{R}{2}}^{u_1} \partial_v \partial_u \psi_0(u', v')\, du' dv' \\
&= \int_{u_1}^{v} \int_{-\frac{R}{2}}^{u_1} -r^{-1}((\mathbf{T}\psi)^2)_0(u', v')\, du' dv' \\
&\leq \int_{u_1}^{v} \int_{-\frac{R}{2}}^{u_1} -r^{-1}(\mathbf{T}\psi_0)^2(u', v')\, du' dv'.
\end{aligned} \tag{7.3}$$

Since the integrand is non-positive definite, we can restrict the integration region further after switching to (u, r) coordinates and assuming $v > 2u_1 + \frac{R}{2}$:

$$\begin{aligned}
\int_{u_1}^{v} \int_{-\frac{R}{2}}^{u_1} -r^{-1}(\mathbf{T}\psi_0)^2(u', v')\, du' dv' &= \int_{-\frac{R}{2}}^{u_1} \int_{u_1 - u}^{v-u} -r'^{-1}(\mathbf{T}\psi_0)^2(u', v')\, dr' du' \\
&\leq -\int_{u_1 + \frac{R}{2}}^{v-u_1} r'^{-1} \int_{-\frac{R}{2}}^{u_1} (\mathbf{T}\psi_0)^2(u', r')\, du' dr'.
\end{aligned}$$

See Figure 4 for a depiction of the relevant integration regions.

By the fundamental theorem of calculus together with Cauchy–Schwarz, using that $\mathbf{T} = \partial_u$ in (u, r) coordinates, we can write:

$$\begin{aligned}
\psi_0^2(u_1, r') &= \left(0 + \int_{-\frac{R}{2}}^{u_1} \mathbf{T}\psi_0(u', r')\, du'\right)^2 \\
&\leq \left(u_1 + \frac{R}{2}\right) \int_{-\frac{R}{2}}^{u_1} (\mathbf{T}\psi_0)^2(u', r')\, du'.
\end{aligned}$$

We combine the above inequalities to obtain for $v > 2u_1 + \frac{R}{2}$:

$$\psi_0(u_1, v) \leq -\left(u_1 + \frac{R}{2}\right)^{-1} \int_{u_1 + \frac{R}{2}}^{v-u_1} (r')^{-1} \psi_0^2(u_1, r')\, dr' =: -\Psi(v).$$

Hence,

$$\Psi'(v) = \left(u_1 + \frac{R}{2}\right)^{-1} r^{-1}\psi_0^2(u_1, v) \geq \left(u_1 + \frac{R}{2}\right)^{-1} (v - u_1)^{-1}\Psi(v)^2,$$

which we can rewrite as follows:

$$\left(\Psi^{-1}\right)'(v) \leq -\left(u_1 + \frac{R}{2}\right)^{-1} (v - u_1)^{-1}.$$

Now suppose there exists a $v_* > 2u_1 + \frac{R}{2}$ such that $\Psi(v_*) \neq 0$. Since Ψ is non-negative definite, this means that $\Psi(v_*) > 0$. Then for $v > v_*$, we have that

$$\Psi^{-1}(v) \leq \Psi^{-1}(v_*) - \left(u_1 + \frac{R}{2}\right)^{-1} \log\left(\frac{v - u_1}{v_* - u_1}\right).$$

Hence, there must exist a $V > v_*$ such that $\Psi^{-1}(V) = 0$. But then $\Psi(V)$ is not well-defined, which is a contradiction with ϕ being globally well-defined.

We conclude that $\Psi(v) = 0$ for all $v > 2u_1 + \frac{R}{2}$ and hence $\psi_0(u_1, v) = 0$ for all $v \geq 2u_1 + \frac{R}{2}$.

By the first inequality in (7.3) with $v \geq 2u_1 + \frac{R}{2}$, we can therefore conclude that $((\mathbf{T}\psi)^2)_0 \equiv 0$ and hence $\mathbf{T}\psi \equiv 0$ in $[-\frac{R}{2}, u_1]_u \times [u_1, \infty)_v \times \mathbb{S}^2$. Taking $u_1 \downarrow \frac{R}{2}$, we therefore have, in particular, that $T\psi \equiv 0$ in the region $\{t \geq R\}$. Now, applying the wave equation, it follows that $\Delta_{\mathbb{R}^3}\phi(t, \cdot) = 0$ for all $t \geq R$. Since the support of ϕ on constant t slices must be compact, by the weak Huygens principle, we can integrate by parts $\phi\Delta_{\mathbb{R}^3}\phi = 0$ in $\mathbb{R}^3$ to finally conclude that $\phi \equiv 0$ in $\{t \geq R\}$.

Finally, by uniqueness of solutions (evolving in the backwards time direction), we can conclude that $\phi \equiv 0$ in $\{t < R\}$, so $\phi \equiv 0$ in $\mathbb{R} \times \mathbb{R}^3$, which contradicts $(\phi, \partial_t\phi)|_{t=0} \neq (0, 0)$. □

Remark 7.1. *The proof of Theorem 7.1 works with minor modifications when $\Box\phi \geq (\partial_t\phi)^2$. Replacing ϕ with $-\phi$, the proof also applies to $\Box\phi \leq -(\partial_t\phi)^2$.*

7.2 The null condition

Theorem 7.2 illustrates that we cannot hope to prove global well-posedness for solutions with general quadratic derivative nonlinearities, even when considering initial data arbitrarily with arbitrarily small norms. It turns out that global wellposedness *does* hold for small initial data norm if we make *structural assumptions* on the derivatives.

Definition 7.1. *Let $\phi : \mathbb{R}^{1+3} \supseteq U \to \mathbb{R}^m$ with U open, and consider the system of wave equations*

$$\sum_{B=1}^{m} a_{AB}^{\alpha\beta}(\phi, D\phi)\partial_\alpha\partial_\beta\phi_B = F_A(\phi, D\phi) \quad \text{in } \mathbb{R}^{1+3}, \text{ for } A = 1, \ldots, m, \tag{7.4}$$

with $a_{AB}^{\alpha\beta} = a_{BA}^{\alpha\beta}$.[33]

Assume that a and F take the following form:

$$F_A(v, w) = Q_{ABC}^{\alpha\beta} w_{B\alpha} w_{C\beta} + O((|v| + |w|)^3),$$
$$a_{AB}^{\alpha\beta}(v, w) = m^{\alpha\beta}\delta_{AB} + L_{ABC}^{\alpha\beta\kappa} w_{C\kappa} + O((|v| + |w|)^2),$$

with δ_{AB} the Kronecker delta, $Q_{ABC}^{\alpha\beta} = Q_{ABC}^{\beta\alpha} = Q_{ACB}^{\alpha\beta}$ and $L_{ABC}^{\alpha\beta\kappa} = L_{ABC}^{\beta\alpha\kappa} = L_{BAC}^{\beta\alpha\kappa}$.

This is equivalent to: 1) $\phi \equiv 0$ is a solution to (7.4), 2) the linearization of (7.4) around the zero solution is $\Box\phi_A = 0$ and 3) all quadratic terms in F and linear terms in a involve only $D\phi$.

We say (7.4) satisfies the ***null condition*** *if:*

$$Q_{ABC}^{\alpha\beta}\xi_\alpha\xi_\beta = 0 \quad \forall A, B, C \in \{1, \ldots m\},$$

[33] This symmetry condition is present to ensure that analogues of the (local) energy estimates still apply when considering a system of wave equations instead of a single wave equation.

$$L^{\alpha\beta\kappa}_{ABC}\xi_\alpha\xi_\beta\xi_\kappa = 0 \quad \forall A, B, C \in \{1, \dots m\},$$

for all $\xi \in \mathbb{R}^{1+3}$ *such that* $m^{\alpha\beta}\xi_\alpha\xi_\beta = 0$.[34]

Consider the coordinates (u, v, θ, φ) The null condition guarantees that we do not see terms of the form:

$$\begin{aligned} &\partial_u\phi_B\partial_u\phi_C,\ \partial_v\phi_B\partial_v\phi_C \quad \text{in } F, \\ &\partial_u\phi_B\partial_u^2\phi_C,\ \partial_v\phi_B\partial_v^2\phi_C \quad \text{in } a^{\alpha\beta}_{AB}(\phi, D\phi)\partial_\alpha\partial_\beta\phi_B. \end{aligned}$$

It is therefore easy to verify that (7.1) does not satisfy the null condition as the nonlinearity $(\partial_t\phi)^2$ features $(\partial_u\phi)^2$ and $(\partial_v\phi)^2$, when considering (u, v) coordinates. Similarly, the nonlinearity $2\partial_t\phi\partial_t^2\phi$ in (7.10) (**Problem 7.1**) does not satisfy the null condition, as it involves $\partial_u\phi\partial_u^2\phi$ and $\partial_v\phi\partial_v^2\phi$ when written out in (u, v) coordinates.

As we will see in § 7.3, when considering solutions to (7.4) for $t \geq 0$, the terms $\partial_u\phi_B\partial_u\phi_C$ are problematic for establishing future global existence of solutions, whereas in the $t \leq 0$ case the terms $\partial_v\phi_B\partial_v\phi_C$ are problematic for establishing past global existence. This has to do with the fact that when considering the future time-decay behaviour of solutions to $\Box\phi = 0$, $\phi \sim r^{-1}$ along constant-u level set and:

$$\begin{aligned} &\partial_u\phi \sim r^{-1}, \text{ but} \\ &\partial_v\phi \sim r^{-2}. \end{aligned}$$

It is the extra power in r^{-1} that makes $\partial_v\phi$ factors in quadratic terms "good", but $\partial_u\phi$ factors "bad" (with the roles of u and v when considering constant-v level sets). Cubic and higher-order nonlinear terms are not relevant, because each factor of ϕ or $\partial\phi$ introduces (at least) and additional factor r^{-1} in the r-decay behaviour along null cones.

7.3 Small-data global well-posedness with null condition

Consider the following semilinear wave equation:

$$\begin{aligned} \Box\phi = A(\phi)m^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi = A(\phi)(-\partial_u\phi\partial_v\phi + r^{-2}|\not\nabla_{\mathbb{S}^2}\phi|^2) \quad &\text{in } [0, \infty) \times \mathbb{R}^3, \\ (\phi, \partial_t\phi)|_{t=0} = (\Phi, \Phi') \quad &\text{on } \mathbb{R}^3, \end{aligned} \tag{7.5}$$

where $A : \mathbb{R} \to \mathbb{R}$ is a smooth function. This wave equation satisfies the null condition. Note that global well-posedness for small data can be obtained via a shortcut in the special case $A \equiv 1$ (see **Problem 7.3**).

Theorem 7.2. *Let* $R > 0$ *and* $(\Phi, \Phi') \in C_c^\infty(B_R(0)) \times C_c^\infty(B_R(0))$ *and assume that there exist* $\epsilon_0 > 0$ *such that:*

$$\sum_{|I|\leq 6} E[Z^I\phi](0) + \int_{N_0} r^2(\partial_v Z^I\psi)^2\, d\sigma dr \leq \epsilon_0^2.$$

Let $\Delta > 0$ *be arbitrarily small. Then, for* $\epsilon_0 = \epsilon_0(A, R, \delta) > 0$ *suitably small, there exists a unique solution* ϕ *to* (7.5) *in* $[0, \infty) \times \mathbb{R}^3$ *that satisfies: there exists* $C = C(R, \delta) > 0$*, such that*

$$\sup_{\tau\in[0,\infty)} \sum_{|I|\leq 6} (1+\tau)^{2-\delta} E[Z^I\phi](\tau) + \int_{N_\tau} r^{2-\delta}(\partial_v Z^I\psi)^2\, d\sigma dr \leq C\epsilon_0^2, \tag{7.6}$$

$$\sup_{\tau\in[0,\infty)} \sum_{|I|\leq 4} ||(1+v)^{\frac{1}{2}}(1+r)^{\frac{1}{2}}(u+R)^{\frac{1-\delta}{2}} Z^I\phi||_{L^\infty(\Sigma_\tau)} \leq C\epsilon_0, \tag{7.7}$$

$$\sup_{\tau\in[0,\infty)} \sum_{|I|\leq 3} \sum_{i=1}^{3} ||(u+R)^{1-\frac{\delta}{2}}(r^{-1}\Omega_i)Z^I\phi||_{L^\infty(\Sigma_\tau\cap\{r\leq 1\})} \leq C\epsilon_0. \tag{7.8}$$

Proof. First of all, by domain of dependence argument (weak Huygens principle), we have that ϕ and its derivatives vanish on N_0.

[34] Vectors ξ satisfying this condition are called *null vectors*.

Strategy: The proof will proceed via a continuity argument. We restrict to the finite spacetime region $\tau \in [0, T)$ and $v \leq V$.

We will make the following bootstrap assumption in $\{\tau \leq T, v \leq V\}$:

$$\mathcal{L}_N(T) := \sup_{\tau\in[0,T)} \sum_{|I|\leq \frac{N+1}{2}} ||(v+1)^{\frac{1}{2}}(r+1)^{\frac{1}{2}}(u+R)^{\frac{1-\delta}{2}} Z^I \phi||_{L^\infty(\Sigma_\tau \cap \{v\leq V\})}$$
$$+ \sup_{\tau\in[0,T)} \sum_{|I|\leq \frac{N-1}{2}} \sum_{i=1}^{3} ||(u+R)^{1-\frac{\delta}{2}}(r^{-1}\Omega_i) Z^I \phi||_{L^\infty(\Sigma_\tau\cap\{r\leq 1\}\cap\{v\leq V\})} \leq \epsilon_0 \Delta_0, \quad (7.9)$$

for $N \in \mathbb{N}$ and $\Delta_0 \in \mathbb{R}_+$ that will be chosen suitably large later on (we will see that $N = 6$ suffices) and $\epsilon_0 \leq 1$.

Indeed, by the local-in-time energy estimates from Theorem 4.1, it follows that there exist $T > 0$ and $V > \frac{R}{2}$ such that there exists a unique solution ϕ in $\{\tau \leq T, v \leq V\}$ and (7.9) holds for a $\Delta_0 = C$, with $C = C(R, \delta) > 0$.

Our goal is to *improve* the bootstrap assumption, by proving that for a suitably choice of Δ_0 depending only on D_0, R, δ, the estimate in (7.9) holds with Δ_0 replaced by $\frac{1}{2}\Delta_0$. Then one can appeal again to local energy estimates in $\{\tau \leq T + \nu\}$ to show that a unique solution ϕ in fact exists in a larger region $\{\tau \leq T + \nu, v \leq V + \nu\}$, such that (7.9) holds, with $\nu = \nu(\epsilon, R, \delta) > 0$; see Figure 5 for a pictorial version of the argument. From this, it follows in fact that the solution ϕ in the full region $\mathcal{R}_0^\infty$ and (7.9) holds for all $T = \infty$. Along the way, we will establish (7.6).

Improving the bootstrap assumption: We will improve (7.7) by first establishing energy decay estimates as in Theorem 6.5. We will do this by showing that the terms involving $Z^I F$ on the RHS of the estimates in Proposition 6.7 can be *absorbed* into LHS, by making use of the bootstrap assumption (7.9).

We write $F = A(\phi)(-\partial_u\phi\partial_v\phi + r^{-2}|\nabla\!\!\!/_{\mathbb{S}^2}\phi|^2)$.

By applying the chain rule when Z^I acts on $A(\phi)$ and using that the argument of A is bounded, as a result of $||\phi||_{L^\infty(\{\tau\leq T, v\leq V\})} \leq \epsilon_0\Delta_0$ we can write:

$$\sum_{|I|\leq N} |Z^I F| \leq C_{A,\Delta_0}\Bigg[\sum_{|I_1|+|I_2|\leq N} |\partial_v Z^{I_1}\phi||\partial_u Z^{I_2}\phi| + r^{-2}|\nabla\!\!\!/_{\mathbb{S}^2} Z^{I_1}\phi||\nabla\!\!\!/_{\mathbb{S}^2} Z^{I_2}\phi| + \dots$$
$$+ \sum_{|I_1|+|I_2|+\dots+|I_N|\leq N} (|\partial_v Z^{I_1}\phi||\partial_u Z^{I_2}\phi| + r^{-2}|\nabla\!\!\!/_{\mathbb{S}^2} Z^{I_1}\phi||\nabla\!\!\!/_{\mathbb{S}^2} Z^{I_2}\phi|)|Z^{I_3}\phi|\cdot\dots\cdot|Z^{I_N}\phi|\Bigg],$$

with $C_{A,\Delta_0} > 0$ depending on the choice of A, on Δ_0 and on N. In the remainder of the proof, we will only treat the quadratic terms in derivatives of ϕ, as the remaining terms are strictly easier to deal with.

Now, we will estimate the terms appearing on the RHS of Proposition 6.7. For the sake of notational convenience, we will omit the restriction "$\leq V$" in the notation below, but it is implied in all the integrals. We have that for all $0 \leq \tau_1 \leq \tau_2 \leq T$:

$$\sum_{|I|\leq N} \int_{\mathcal{R}_{\tau_1}^{\tau_2}} r^2(r+1)^{1+\delta}|Z^I F|^2 \, d\sigma dr d\tau$$
$$\leq C_{A,\Delta_0}^2 \sum_{|I_1|+|I_2|\leq N} \int_{\mathcal{R}_{\tau_1}^{\tau_2}} r^2(r+1)^{1+\delta}\left[|\partial_v Z^{I_1}\phi|^2|\partial_u Z^{I_2}\phi|^2 + r^{-4}|\nabla\!\!\!/_{\mathbb{S}^2} Z^{I_1}\phi|^2|\nabla\!\!\!/_{\mathbb{S}^2} Z^{I_2}\phi|^2\right] d\sigma dr d\tau$$
$$\leq C_{A,\Delta_0}^2 \mathcal{L}_N(T)^2 \sum_{|I|\leq N} \int_{\mathcal{R}_{\tau_1}^{\tau_2}} \frac{(r+1)^\delta}{(v+1)(u+R)^{1-\delta}}\left[r^2|\partial_v Z^I\phi|^2 + \left(\frac{r}{r+1}\right)^2|\partial_u Z^{I_2}\phi|^2 + (r+1)^{-2}|\nabla\!\!\!/_{\mathbb{S}^2} Z^I\phi|^2\right] d\sigma dr d\tau$$
$$\leq C_{A,\Delta_0}^2 \epsilon_0^2\Delta_0^2 \sum_{|I|\leq N} \int_{\tau_1}^{\tau_2} (\tau+1)^{-2+2\delta} \int_{\Sigma_\tau} r^2|\partial_v Z^I\phi|^2 \, d\sigma dr d\tau$$
$$+ \int_{\mathcal{R}_{\tau_1}^{\tau_2}} \frac{r^2(r+1)^{-3+\delta}}{(u+R)^{1-\delta}}\left[|\partial_u Z^I\phi|^2 + r^{-2}|\nabla\!\!\!/_{\mathbb{S}^2} Z^I\phi|^2\right] d\sigma dr d\tau.$$

By taking ϵ_0 suitably small compared depending on Δ_0, A and N, we can absorb the first term on the very RHS above into $\sum_{|I|\leq N}\sup_{\tau\in[0,T]}E[Z^I\phi](\tau)$ and the second term into the spacetime integral on the LHS of (6.24) (for $\delta<\frac{1}{2}$).

We moreover have that:

$$\begin{aligned}
&\int_{\mathcal{D}^{\tau_2}_{\tau_1}} r^{p+3}|Z^I F|^2\,d\sigma dr d\tau\\
&\leq C^2_{A,\Delta_0}\sum_{|I_1|+|I_2|\leq N}\int_{\mathcal{D}^{\tau_2}_{\tau_1}} r^{p+3}\left[|\partial_v Z^{I_1}\phi|^2|\partial_u Z^{I_2}\phi|^2+r^{-4}|\nabla\!\!\!/_{\mathbb{S}^2}Z^{I_1}\phi|^2|\nabla\!\!\!/_{\mathbb{S}^2}Z^{I_2}\phi|^2\right]\,d\sigma dr d\tau\\
&\leq C^2_{A,\Delta_0}\mathcal{L}_N(T)^2\sum_{|I|\leq N}\int_{\mathcal{D}^{\tau_2}_{\tau_1}}\frac{r^p}{(v+1)(u+R)^{1-\delta}}\left[|\partial_v Z^I\psi|^2+r^{-2}(Z^I\psi)^2+|\partial_u Z^I\phi|^2+r^{-2}|\nabla\!\!\!/_{\mathbb{S}^2}Z^I\phi|^2\right]\,d\sigma dr d\tau\\
&\leq C^2_{A,\Delta_0}\epsilon_0^2\Delta_0^2\sum_{|I|\leq N}\int_{\mathcal{D}^{\tau_2}_{\tau_1}}\frac{r^{p-1}}{(u+R)^{1-\delta}}|\partial_v Z^I\psi|^2+\frac{r^{p-3}}{(u+R)^{1-\delta}}|Z^I\psi|^2\\
&\quad+\frac{r^{p-1}}{(u+R)^{1-\delta}}\left[|\partial_u Z^I\phi|^2+r^{-2}|\nabla\!\!\!/_{\mathbb{S}^2}Z^I\phi|^2\right]\,d\sigma dr d\tau
\end{aligned}$$

For ϵ_0 suitably small, the first term on the very RHS above can be absorbed into the LHS of (6.25) for $p>0$ and the second term can then be absorbed similarly after application of Lemma 6.3 with $q=-2$ and (6.24) to control the resulting boundary term at $r=R$. The third and fourth terms be absorbed into the LHS of (6.24), provided we take $p\leq 2-\delta$.

We therefore conclude that (6.24) and (6.25) hold also without F-terms on the RHS and with $0<p\leq 2-2\delta$, if we take ϵ_0 suitably small compared depending on Δ_0, A, N.

Now we can conclude that:

$$\sup_{\tau\in[0,T)}\sum_{|I|\leq N}(1+\tau)^{2-\delta}E[Z^I\phi](\tau)+\int_{N_\tau} r^{2-\delta}(\partial_v Z^I\psi)^2\,d\sigma dr\leq C\epsilon_0^2$$

by applying the proof of Theorem 6.5, modified with an additional interpolation argument (**Problem 6.4**) that deals with the fact that we are taking $p\leq 2-\delta$ instead of $p\leq 2$.

Then, by applying the pointwise estimates from Corollary 6.8 (including the modification from taking $p\leq 2-\delta$ instead of $p\leq 2$),[35] we conclude that there exists a $C=C(R,\delta)>0$ such that:

$$\mathcal{L}_N(T)\leq C\epsilon_0.$$

By taking $N\geq 6$, we have that $N-2\geq\frac{N}{2}+1$ and by defining $\Delta_0=2C$, we obtain an improvement of the bootstrap assumption (7.9). □

7.4 Problems

Problem 7.1. *In this problem, we will apply the mechanism developed in the proof of Theorem 7.1 to show that global C^2 solutions to the quasilinear wave equation*

$$\Box\phi=2\partial_t\phi\partial_t^2\phi \tag{7.10}$$

must be equal to the zero solution if we assume a certain non-negativity condition on the initial data at $t=0$.

a) Show that (7.10) satisfies the domain of dependence property of Corollary 2.3.

[35] Note that in the elliptic estimates from the proof of Corollary 6.8 feature also derivatives of F, but it can be shown that these are also controlled by $E[Z^I\phi]$. EXERCISE: Convince yourself of this.

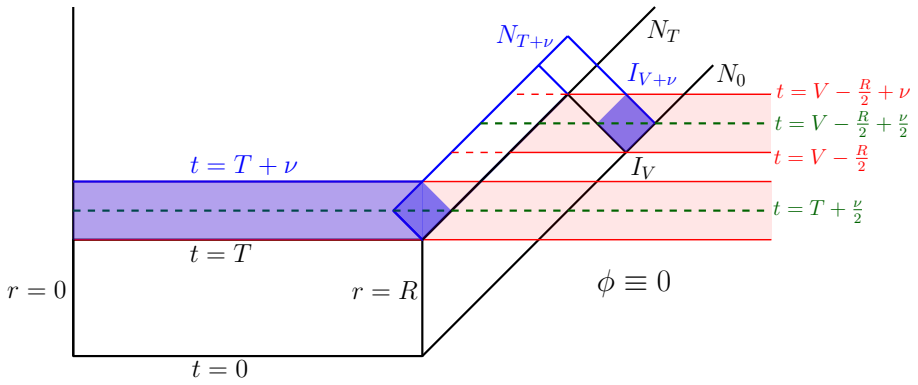


Figure 5: A pictorial representation of the following argument for extending the domain of existence to $\tau \leq T+\nu$ and $v \leq v+\nu$ in the proof of Theorem 7.2: 1) By local-in-time energy estimates in $T \leq t \leq T+\nu$, the bootstrap assumption is extended and then improved in the dark shaded (blue) region on the left. 2) Repeat 1), but now starting from $t = T + \frac{\nu}{2}$ and ending at $t = T + \frac{3}{2}\nu$ and continue this procedure until the bootstrap assumption is derived and improved in $\{0 \leq \tau \leq T+\nu,\, v \leq V\}$. 3) By local-in-time energy estimates in $T - \frac{R}{2} \leq t \leq T - \frac{R}{2} + \nu$, the bootstrap assumption is extended to the dark shaded (blue) diamond on the right. 4) Repeat 3) until the bootstrap assumption is improved in $\{\tau \leq T+\nu,\, v \leq V+\nu\}$.

b) Define the time integral $\widetilde{\phi}(t,\mathbf{x}) = \int_0^t \phi(s,\mathbf{x})\, ds$ *and denote* $\widetilde{\psi} = r\widetilde{\phi}$*. By considering the equation satisfied by* $\widetilde{\phi}$*, show that global* C^2 *solutions to* (7.10) *must be equal to the zero solution, if we make the following additional assumption on the initial data* (Φ, Φ') *corresponding to* ϕ *at* $t = 0$*:*

$$\int_{\mathbb{R}^3} (\Phi')^2 - \Phi'\, d\mathbf{x} \geq 0.$$

[Hint: Try applying the proof of Theorem 7.1 to the equation satisfied by $\widetilde{\phi}$ and show/use that $\int_{-\frac{R}{2}}^{u_1} \partial_u \widetilde{\psi}_0(u,r)\, du = \int_{-\frac{R}{2}}^{u_1} (u_1 - u)\partial_u^2 \widetilde{\psi}_0(u,r)\, du$.]

Problem 7.2. *Show that* (1.7) *from* ***Example 1.4*** *does not satisfy the null condition.*

Problem 7.3. *Consider the initial value problem:*

$$\begin{aligned} \Box\phi &= m^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi \quad \text{in } [0,\infty)\times\mathbb{R}^3, \\ (\phi, \partial_t\phi)|_{t=0} &= (\Phi,\Phi') \in C_c^\infty(\mathbb{R}^3). \end{aligned} \tag{7.11}$$

a) Define $\Phi = 1 - e^{-\phi}$ *and show that* $\Box\Phi = 0$*.*

b) What conditions need to be applied on the initial data (Φ,Φ') *to obtain global well-posedness for* (7.11) *from global well-posedness for the standard linear wave equation via the consideration of* Φ *(i.e. circumventing the use of Theorem 7.2)?*

A Sobolev spaces

A.1 $L^2(\mathbb{R}^n)$: a "pedestrian" definition

Consider the space $C_c^0(\mathbb{R}^n)$ of continuous, compactly supported (real- or complex-valued) functions on $\mathbb{R}^n$. We can associate to $C_c^0(\mathbb{R}^n)$ the L^2-norm, defined as follows:

$$||f||_{L^2(\mathbb{R}^n)} := \left(\int_{\mathbb{R}^n} |f|^2\, dx\right)^{\frac{1}{2}}.$$

In order to apply techniques of functional analysis, we would like $(C_c^0(\mathbb{R}^n), ||\cdot||_{L^2(\mathbb{R}^n)})$ to be a Hilbert space, i.e. a complete, normed vector space where the norm arises from an inner product, in this case:

$$\langle f, g\rangle_{L^2(\mathbb{R}^n)} := \int_{\mathbb{R}^n} f\overline{g}\, dx.$$

However: $(C_c^0(\mathbb{R}^n), ||\cdot||_{L^2(\mathbb{R}^n)})$ fails to be complete (see **Problem** A.1). Nevertheless, we can show obtain a natural larger function space in which $(C_c^0(\mathbb{R}^n), ||\cdot||_{L^2(\mathbb{R}^n)})$ embeds by applying the following theorem:

Theorem A.1. *Let $(X, \langle\cdot,\cdot\rangle)$ be an inner product space. Then there exist a Hilbert space $(H, \langle\langle\cdot,\cdot,\rangle\rangle)$, a dense subspace $Y < H$ and an isometric isomorphism $A : X \to Y$. The above Hilbert space H is unique up to isometric isomorphisms.*

Applying the above theorem with $X = C_c^0(\mathbb{R}^n)$ and $\langle\cdot,\cdot\rangle = \langle\cdot,\cdot\rangle_{L^2(\mathbb{R}^n)}$, we can obtain a Hilbert space $(H, \langle\langle\cdot,\cdot\rangle\rangle)$ (that is unique up to isometric isomorphism), that we will denote with $L^2(\mathbb{R}^n) = H$. We will abuse notation and also denote $\langle\cdot,\cdot\rangle_{L^2(\mathbb{R}^n)} = \langle\langle\cdot,\cdot\rangle\rangle$, such that $C_c^0(\mathbb{R}^n) \subset L^2(\mathbb{R}^n)$ is a dense subspace. More concretely, we can identify elements in $L^2(\mathbb{R}^n)$ with equivalence classes of Cauchy sequences in $(C_c^0(\mathbb{R}^n), ||\cdot||_{L^2(\mathbb{R}^n)})$,[36] just like how we can think of elements in $\mathbb{R}$ as equivalence classes of Cauchy sequences in $\mathbb{Q}$ (i.e. $\mathbb{R}$ is the completion of $\mathbb{Q}$ with respect to the standard Euclidean inner product).

Similarly, we can define Banach spaces $L^p(\mathbb{R}^n)$ for $1 \le p < \infty$ as completions of $C_c^0(\mathbb{R}^n)$ with respect to the norm $\left(\int_{\mathbb{R}^n} |f|^p\, dx\right)^{\frac{1}{p}}$, using a Banach space version of the theorem above (where the word "Hilbert" is replaced by "Banach" and "$\langle\cdot,\cdot\rangle$" by "$||\cdot||$").[37]

A.2 $L^2(\mathbb{R}^n)$: a measure-theoretic definition

It is possible to give a more concrete definition of $L^2(\mathbb{R}^n)$ where elements are identified with equivalence classes of *functions* rather than with equivalence classes of *Cauchy sequences* of functions. Let $f : \mathbb{R}^n \to \mathbb{C}$ be Lebesgue measurable. Then we denote:

$$[f] = \{g : \mathbb{R}^n \to \mathbb{C} \mid g \text{ Lebesgue measurable} \mid \operatorname{supp}(f - g) \text{ has zero measure}\}.$$

Note that $[f]$ are equivalence classes.

Now we define

$$L^2(\mathbb{R}^n) = \{[f] \mid f : \mathbb{R}^n \to \mathbb{C} \text{ Lebesgue measurable} \mid |f|^2 \text{ Lebesgue integrable}\}.$$

Then

$$\langle f, g\rangle_{L^2(\mathbb{R}^n)} := \int_{\mathbb{R}^n} f\overline{g}\, dx,$$
$$||f||_{L^2(\mathbb{R}^n)} := \int_{\mathbb{R}^n} |f|^2\, dx$$

are well-defined in terms of Lebesgue integration and finite-valued.[38] We can similarly define spaces $L^p(\mathbb{R}^n)$ for $1 \le p < \infty$.

Applying the measure theory framework, we can moreover define $L^\infty(\mathbb{R}^n)$ as follows:

$$L^\infty(\mathbb{R}^n) = \{[f] \mid f : \mathbb{R}^n \to \mathbb{C} \text{ Lebesgue measurable and bounded almost everywhere}\}.$$

Then the norm is given by the *essential supremum*:

$$||f||_{L^\infty(\mathbb{R}^n)} := \inf\{C > 0 \mid |f(x)| \le C \text{ for almost every } x\}.$$

[36] Two Cauchy sequences $\{f_n\}$ and $\{g_n\}$ are in the same equivalence class if $\{f_n - g_n\}$ converges to 0. We can think of elements $f \in C_c^0(\mathbb{R}^n) \subset L^2(\mathbb{R}^n)$ as the equivalence class containing the constant Cauchy sequence $\{f, f, \ldots\}$.

[37] In fact, this completion theorem holds even more generally for metric spaces (X, d).

[38] Exercise: Show that the inner product is a well-defined Lebesgue integral, given the assumptions on f and g.

A.3 Sobolev spaces: a "pedestrian" definition

On the space $C_c^\infty(\mathbb{R}^n)$, we can define for any $s \in \mathbb{N}_0$:

$$\langle f, g\rangle_{H^s(\mathbb{R}^n)} := \sum_{|I|\le s} \int_{\mathbb{R}^n} \partial^I f \partial^I \overline{g}\, dx,$$

$$||f||_{H^s(\mathbb{R}^n)} := \left(\sum_{|I|\le s} \int_{\mathbb{R}^n} |\partial^I f|^2\, dx\right)^{\frac{1}{2}},$$

with $I = (I_1, I_2, \ldots, I_n) \in \mathbb{N}_0^n$ denoting a multiindex, $\partial^I = \partial_{x^1}^{I_1}\partial_{x^2}^{I_2}\ldots\partial_{x^n}^{I_n}$ and $|I| = \sum_{i=1}^n I_n$.

By applying Theorem A.1 to $X = C_c^\infty(\mathbb{R}^n)$ with inner product $\langle f, g\rangle_{H^s(\mathbb{R}^n)}$, we obtain the completion $H^s(\mathbb{R}^n)$ with elements equivalence classes of Cauchy sequences of functions $\{f_n\}$ with respect to the norm $||\cdot||_{H^s(\mathbb{R}^n)}$. Since the partial derivative operators $\partial^I : C_c^\infty(\mathbb{R}^n) \to C_c^\infty(\mathbb{R}^n)$ satisfy $||\partial^I f||_{L^2(\mathbb{R}^n)} \le ||f||_{H^s(\mathbb{R}^n)}$, they are clearly bounded and therefore[39] they extend uniquely to bounded operators $\partial^I : H^s(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ for all $|I| \le s$.

Similarly, we can define

$$||f||_{W^{s,p}(\mathbb{R}^n)} = \left(\sum_{|I|\le s} \int_{\mathbb{R}^n} |\partial^I f|^p\, dx\right)^{\frac{1}{p}}$$

and the corresponding completion we denote by $W^{s,p}(\mathbb{R}^n)$.

A.4 Sobolev spaces : a measure-theoretic definition

As in the case of the spaces $L^2(\mathbb{R}^n)$, it is possible to give a more concrete definition of $H^s(\mathbb{R}^n)$ and $W^{s,p}(\mathbb{R}^n)$ via Lebesgue integration.

Definition A.1 (Local integrability)**.** *We say a function $f : \mathbb{R}^n \to \mathbb{C}$ is in $L^1_{\text{loc}}(\mathbb{R}^n)$ if for all $\chi \in C_c^\infty(\mathbb{R}^n)$, $\chi \cdot f \in L^1(\mathbb{R}^n)$.*

Definition A.2 (Weak derivatives)**.** *Let I be a multiindex and let $f \in L^1_{\text{loc}}(\mathbb{R}^n)$. Then $g \in L^1_{\text{loc}}(\mathbb{R}^n)$ is the* I-th weak derivative *of f if the following identity holds (with respect to Lebesgue integration):*

$$\int_{\mathbb{R}^n} f\partial^I \phi\, dx = (-1)^{|I|} \int_{\mathbb{R}^n} g\phi\, dx \quad \forall \phi \in C_c^\infty(\mathbb{R}^n).$$

We then denote $\partial^I f = g$.

Exercise: Show that the I-th weak derivative of f coincides with the regular I-th derivative $\partial^I f = \partial_{x^1}^{I_1}\ldots\partial_{x^n}^{I_n} f$ when $f \in C^{|I|}(\mathbb{R}^n)$.

Now we can define the Sobolev space $H^s(\mathbb{R}^n)$, with $s \in \mathbb{N}_0$ as follows:

$$H^s(\mathbb{R}^n) = \{f \in L^2(\mathbb{R}^n) \,|\, \partial^I f \in L^2(\mathbb{R}^n) \text{ if } |I| \le s\},$$

where $\partial^I f$ above is to be interpreted as a weak derivative.

We can equip the above space with the inner product and norms:

$$\langle f, g\rangle_{H^s(\mathbb{R}^n)} := \sum_{|I|\le s} \int_{\mathbb{R}^n} \partial^I f \partial^I \overline{g}\, dx,$$

$$||f||_{H^s(\mathbb{R}^n)} := \left(\sum_{|I|\le s} \int_{\mathbb{R}^n} |\partial^I f|^2\, dx\right)^{\frac{1}{2}}.$$

[39] Exercise: Convince yourself of this.

By replacing $L^2(\mathbb{R}^n)$ with $L^p(\mathbb{R}^n)$ for $1 \leq p \leq \infty$ in the Sobolev space definition, and by considering the norm:

$$||f||_{W^{s,p}(\mathbb{R}^n)} = \left(\sum_{|I| \leq s} \int_{\mathbb{R}^n} |\partial^I f|^p \, dx \right)^{\frac{1}{p}},$$

we can define the more general Sobolev spaces $W^{s,p}(\mathbb{R}^n)$, which are Banach spaces.

A.5 A Sobolev embedding theorem

It turns out that for sufficiently high s, elements in $H^s(\mathbb{R}^n)$ can be considered as bounded, continuous, or k times continuously differentiable functions (see Theorem A.2 below). We introduce therefore the function spaces:

$$C_b^k(\mathbb{R}^n) = \{f \in C^k(\mathbb{R}^n) \,|\, \sup_{x \in \mathbb{R}^n} |\partial^I f|(x) < \infty \quad \forall |I| \leq k\}.$$

If we equip $C_b^k(\mathbb{R}^n)$ with the norm defined as follows:

$$||f||_{C^k(\mathbb{R}^n)} := \max_{I \in \{0,\ldots,k\}^n} \sup_{x \in \mathbb{R}^n} |\partial^I f|(x),$$

we obtain a Banach space $(C_b^k(\mathbb{R}^n), ||\cdot||_{C^k(\mathbb{R}^n)})$.[40]

Theorem A.2 (A Sobolev embedding theorem)**.** *Let $k \in \mathbb{N}_0$, then $H^s(\mathbb{R}^n) \subset C_b^k(\mathbb{R}^n)$ for $s > k + \frac{n}{2}$ and there exists a constant $C > 0$ such that*

$$||f||_{C^k(\mathbb{R}^n)} \leq C||f||_{H^s(\mathbb{R}^n)}.$$

Before we prove this theorem, we establish first the following lemma:

Lemma A.3. *Let $u \in C^\infty(B_R(0))$ and $m \in \mathbb{N}_0$. Then there exists a $C = C(m,n) > 0$, such that for all $R > 0$*

$$|u(0)| \leq C \left[\sum_{|I| \leq m-1} R^{-n} \int_{B_R(0)} |\partial^I u||y|^{|I|} \, dy + \sum_{|I|=m} \int_{B_R(0)} |\partial^I u||y|^{m-n} \, dy \right]. \tag{A.1}$$

Proof. Define $f : \mathbb{R}_+ \to \mathbb{R}$ as follows: $f(r) = u(rz)$, for some $z \in \mathbb{R}^n$ with $||z|| = 1$. By Taylor's theorem, we can write:

$$f(0) = \sum_{j \leq m-1} \frac{1}{j!} f^{(j)}(r)(0-r)^j + \frac{1}{(m-1)!} \int_r^0 f^{(j)}(s)(0-s)^{m-1} \, ds$$

Integrating both sides over $(0,R)$, with respect to $r^{n-1}dr$, we obtain:

$$\frac{1}{n-1}|f(0)|R^n \leq \sum_{j \leq m-1} \frac{1}{j!} \int_0^R |f^{(j)}|(r) r^{n-1+j} \, dr + \frac{1}{(m-1)!} \int_0^R \int_0^r |f^{(m)}|(s) s^{m-1} r^{n-1} ds dr.$$

We estimate the second integral further, by switching the order of integration:

$$\int_0^R \int_0^r |f^{(m)}|(s) s^{m-1} r^{n-1} ds dr = \int_0^R s^m |f^{(m)}|(s) \int_s^R r^{n-1} dr ds = \frac{1}{n} \int_0^R |f^{(m)}|(s) s^{m-1} (R^n - s^n) ds.$$

Hence, there exists a constant $C = C(n,m) > 0$ such that we can estimate

$$|f(0)| \leq CR^{-n} \sum_{j \leq m-1} \int_0^R |f^{(j)}|(r) r^{j+n-1} \, dr + \int_0^R |f^{(m)}|(r) r^{m-1} \, dr.$$

[40] Since this space is complete, there is no need to extend it further via an abstract completion, as was the case for $L^p(\mathbb{R}^n)$ with $1 \leq p < \infty$. Nevertheless, it is sometimes useful to consider the space $L^\infty(\mathbb{R}^n)$, defined above using measure theory, which is analogous to $C_b^0(\mathbb{R}^n)$, but it contains also functions that are not continuous.

Now we apply the chain rule to write:

$$f^{(j)}(r) = \sum_{|I|=j} \frac{j!}{I_1!I_2!\dots I_n!} \partial^I u(rz) \cdot z_1^{I_1} z_2^{I_2} \dots z_n^{I_n}$$

and obtain:

$$|u(0)| \leq CR^{-n} \sum_{|I|\leq m-1} \int_0^R |\partial^I u|(rz) r^{j+n-1}\, dr + \sum_{|I|=m} \int_0^R |\partial^I u|(rz) r^{m-1}\, dr.$$

Then, (A.1) follows after integrating both sides of the equation above over theunit round $n-1$-sphere $\mathbb{S}^{n-1}$ and using $dy = c_n r^{n-1} d\sigma_{n-1} dr$, with $d\sigma_{n-1}$ the standard volume form on $\mathbb{S}^{n-1}$. □

Proof of Theorem A.2. Assume first that $u \in C_c^\infty(\mathbb{R}^n)$. By applying (A.1), together with Cauchy–Schwarz, we obtain:

$$\begin{aligned} |u(0)| \leq C \sum_{|I|\leq m-1} R^{-n} \left(\int_0^R \int_{\mathbb{S}^{n-1}} r^{2|I|+n-1}\, d\sigma_{n-1} dr \right)^{\frac{1}{2}} \cdot ||\partial^I u||_{L^2(B_R(0))} \\ + C \sum_{|I|=m} \left(\int_0^R \int_{\mathbb{S}^{n-1}} r^{2(m-n)+n-1}\, d\sigma_{n-1} dr \right)^{\frac{1}{2}} \cdot ||\partial^I u||_{L^2(B_R(0))}. \end{aligned}$$

The integral factors above are finite provided $2(m-n)+n > 0$, or equivalently, $m > \frac{n}{2}$. Since the above estimate is still valid when we translate the origin, we can conclude that

$$||u||_{C^0(\mathbb{R}^n)} \leq C||u||_{H^s(\mathbb{R}^n)}, \tag{A.2}$$

if we take $s > \frac{n}{2}$.

Suppose now that $u \in H^s(\mathbb{R}^n)$. Then u is the limit of a sequence $\{u_i\}$, with $u_i \in C_c^\infty(\mathbb{R}^n)$ which satisfy (A.2). Hence, $\{u_i\}$ is also a Cauchy sequence in $C_b^0(\mathbb{R}^n)$ and by completeness of $C_b^0(\mathbb{R}^n)$, it converges in $C_b^0(\mathbb{R}^n)$, so we can conclude that $u \in C_b^0(\mathbb{R}^n)$.

Finally, by repeating the above arguments with u replaced by $\partial^I u$, with $|I| \leq k$, we conclude the statement in the theorem. □

Remark A.1. *This proof works also for functions defined on a smaller domain, e.g. we can replace $H^s(\mathbb{R}^n)$ and $C^k(\mathbb{R}^n)$ with $H^s(U)$ and $C^k(U)$, respectively, where $U \subset \mathbb{R}^n$ open.*

For more Sobolev embedding theorems, involving more general $W^{s,p}(U)$, with $U \subset \mathbb{R}^n$ an appropriate subset, see for example:

Adams, R. A., & Fournier, J. J. F. (2003). *Sobolev spaces.* Elsevier.

Lemma A.4 (A Sobolev inequality on $\mathbb{S}^2$)**.** *Let $k \in \mathbb{N}$, $\ell \in \mathbb{N}_0$ and $\Omega^I f \in L^2(\mathbb{S}^2)$ for all $I \in \mathbb{N}_0^3$ with $|I| \leq 2$. Then there exists a constant $C > 0$ such that*

$$||f||_{L^\infty(\mathbb{S}^2)} \leq C \sqrt{\sum_{|I|\leq 2} ||\Omega^I f||^2_{L^2(\mathbb{S}^2)}}. \tag{A.3}$$

Proof. Let $f \in C^\infty(\mathbb{S}^2)$ and let $F : \mathbb{R}^3 \to \mathbb{R}$ be defined as follows: $F(x) = f(x)$ if $x \in \mathbb{S}^2 \subset \mathbb{R}^3$ and $F(x) = 0$ if $x \notin \mathbb{S}^2$. Let $\widetilde{F}$ be defined as the convolution $\widetilde{F} = \phi * F$, where $\phi(x) = \chi(r)$, where $\chi : [0,2] \to \mathbb{R}_+$ is a smooth cut-off function with $\chi(r) = 1$ if $r \in [1/2, 3/2]$ and $\chi(r) = 0$ if $\in [0, \frac{1}{2}] \cup [2, \infty)$.

By Theorem A.2, we have that

$$||f||_{L^\infty(\mathbb{S}^2)} \leq ||\widetilde{F}||_{L^\infty(\mathbb{R}^3)} \leq C||\widetilde{F}||_{H^2(\mathbb{R}^3)}.$$

From the elliptic estimate in (B.2), it follows moreover that

$$||\widetilde{F}||_{H^2(\mathbb{R}^3)} \leq ||(1-\Delta)\widetilde{F}||_{L^2(\mathbb{R}^3)}.$$

In spherical coordinates (r, θ, φ), we can express $(1-\Delta)\widetilde{F} = \widetilde{F} - r^{-2}\partial_r(r^2\partial_r\widetilde{F}) - r^{-2}\not{\Delta}_{\mathbb{S}^2}\widetilde{F}$, so

$$||(1-\Delta)\widetilde{F}||_{L^2(\mathbb{R}^3)} \leq ||\phi * F - r^{-2}\partial_r(r^2\partial_r\phi) * F - r^{-2}\phi * \not{\Delta}_{\mathbb{S}^2}F||_{L^2(\mathbb{R}^n)} \leq C(||f||_{L^2(\mathbb{R}^n)} + ||\not{\Delta}_{\mathbb{S}^2}f||_{L^2(\mathbb{R}^n)}),$$

with $C > 0$ a constant depending on the choice of χ. Finally, we use that $\not{\Delta}_{\mathbb{S}^2} = \Omega_1^2 + \Omega_2^2 + \Omega_3^2$ to conclude the proof of (A.3) for $f \in C^\infty(\mathbb{S}^2)$. We conclude that (A.3) holds also for $f \in L^2(\mathbb{S}^2)$ satisfying $\Omega^I f \in L^2(\mathbb{S}^2)$, by approximating f with a sequence of smooth functions on $\mathbb{S}^2$ with respect to the norm $\sum_{|I|\leq 2} ||\Omega^I f||_{L^2(\mathbb{S}^2)}$. □

A.6 Duality between L^p spaces

Given a Banach space B, we can consider the corresponding *dual space*:

$$B^* = \{f : B \to \mathbb{R} \,|\, \text{bounded and linear}\}.$$

We equip B^* with the following norm:

$$||f||_{B^*} := \sup_{v \in B\setminus\{0\}} \frac{|f(v)|}{||v||}.$$

Furthermore, given a subset $E \subset \mathbb{R}^n$ which is open, compact, or an interval in the $n = 1$ case (see Footnote 41 for a more general characterization of E) and a Banach space $(B, ||\cdot||_B)$, we define the functions space $L^p(E; B)$ as in the sections above, but with $||\cdot||_{L^p(\mathbb{R}^n)}$ replaced by:

$$||f||_{L^p(E;B)} = \left(\int_E ||f||_B^p \, dx\right)^{\frac{1}{p}}, \quad f : E \to B.$$

Similarly, given a Hilbert space $(H, \langle\cdot,\cdot\rangle_H)$, we define $L^2(E; B)$ as in the sections above, but we replace $\langle\cdot,\cdot\rangle_{L^2(\mathbb{R}^n)}$ with:

$$\langle f, g\rangle_{L^2(E;H)} = \int_E \langle f, g\rangle_H \, dx, \quad f : E \to H.$$

Theorem A.5 (Duality of L^p-spaces)**.** *Let $1 \leq p \leq \infty$ and $1 < q \leq \infty$, such that $\frac{1}{q} + \frac{1}{p} = 1$, with the convention $\frac{1}{\infty} = 0$. Then for any subset $E \subset \mathbb{R}^n$, which is open, compact, or an interval in the $n = 1$ case:*[41]

(i)
$$L^p(E;\mathbb{R})^* = L^q(E;\mathbb{R}).$$
More precisely, for each $\phi \in L^p(E;\mathbb{R})^*$, there exists a unique $f \in L^q(E;\mathbb{R})$ so that
$$\phi(g) = \int_E f(x)g(x)\,dx \quad \forall g \in L^p(E;\mathbb{R}).$$

(ii) *For all $m \in \mathbb{N}$,*
$$L^p(E;\mathbb{R}^m)^* = L^q(E;\mathbb{R}^m).$$
More precisely, for each $\phi \in L^p(E;\mathbb{R}^m)^*$, there exists a unique $f \in L^q(E;\mathbb{R}^m)$ so that
$$\phi(g) = \sum_{j=1}^m \int_E f^j(x)g^j(x)\,dx \quad \forall g \in L^p(E;\mathbb{R}^m),$$
with $f^j(x), g^j(x)$ denoting the components of $f(x), g(x) \in \mathbb{R}^m$, respectively.

[41] More generally, we can take $E \subseteq \mathbb{R}^n$ to be any Lebesgue measure subset.

(iii) *Let $(H, \langle \cdot, \cdot \rangle_H)$ be a separable Hilbert space. Then*

$$L^p(E;H)^* = L^q(E;H^*),$$

up to isomorphism. More precisely, for each $\phi \in L^p(E;H)^*$, there exists a unique $f \in L^q(E;H^*)$ so that

$$\phi(g) = \int_E \langle f, g \rangle_H \, dx \quad \forall g \in L^p(E;H).$$

Proof. The proof of (i) lies outside the scope of this course, as it relies heavily on measure theory. One can, for example, find a full proof in §1.4 of

Stein, E.M. and Shakarchi, R., 2011. *Functional analysis.* Princeton University Press.

The proof of (ii) and (iii) is left as an exercise for those with a background in functional analysis:

EXERCISE: Use (i) to derive (ii).

EXERCISE: Use (ii) to derive (iii). [Hint: use that H admits a countable orthonormal basis, by the separability assumption.]

□

A.7 Problems

Problem A.1. *Show that $(C_c^0(\mathbb{R}^n), || \cdot ||_{L^2(\mathbb{R}^n)})$ is a normed vector space that is not complete.*

Problem A.2. **This is only for those familiar with the theory of Lebesgue integration.**

a) *Show that $L^p(\mathbb{R}^n)$, $1 \leq p \leq \infty$ as defined via measure theory are well-defined Banach spaces space (hint: apply Lebesgue integral convergence theorems for completeness and Minkowski inequality for vector space structure).*

b) *Show that the above spaces are isometrically isomorphic to the spaces "$L^p(\mathbb{R}^n)$ as completions of $C_c^0(\mathbb{R}^n)$", defined via Theorem A.1.*

B Elliptic regularity

In this section, we construct solutions to the equations $(1 - \Delta)u = f$ and show that u gains "two powers" in regularity compared to f. We will apply in particular Theorem A.2.

First, we introduce the following dual spaces to $H^m(\mathbb{R}^n)$:

$$H^{-m}(\mathbb{R}^n) := H^m(\mathbb{R}^n)^* = \{f : H^m(\mathbb{R}^n) \to \mathbb{R} \text{ bounded and linear}\},$$

with

$$||f||_{H^{-m}(\mathbb{R}^n)} = \sup_{v \in H^m(\mathbb{R}^n) \setminus \{0\}} \frac{|f(v)|}{||v||_{H^m(\mathbb{R}^n)}}.$$

Theorem B.1. *Let $f \in H^{-1}(\mathbb{R}^n)$. Then:*

(i) *There exists a unique $u \in H^1(\mathbb{R}^n)$ such that*

$$\langle u, v \rangle_{H^1(\mathbb{R}^n)} = f(v) \quad \forall v \in H^1(\mathbb{R}^n).$$

Moreover, for $f \in L^2(\mathbb{R}^n)$:

$$\int_{\mathbb{R}^n} u \cdot \overline{v} + \sum_{i=1}^n \partial_i u \cdot \partial_i \overline{v} \, dx = \int_{\mathbb{R}^n} f \cdot \overline{v} \, dx \quad \forall v \in H^1(\mathbb{R}^n). \tag{B.1}$$

(ii) *If $f \in H^m(\mathbb{R}^n)$ for $m \in \mathbb{N}_0$, then $u \in H^{m+2}(\mathbb{R}^n)$ with*

$$c||f||_{H^m(\mathbb{R}^n)} \leq ||u||_{H^{m+2}(\mathbb{R}^n)} \leq ||f||_{H^m(\mathbb{R}^n)}, \tag{B.2}$$

where $c = c(m) > 0$ is a numerical constant, and

$$(I - \Delta)u = f, \tag{B.3}$$

where the Laplacian Δ appearing in the left-hand side above can be thought of as either the extension of the regular Laplacian Δ to the completion $H^2(\mathbb{R}^n)$, or as a weak-derivative.

Furthermore, if $f \in \bigcap_{m\in\mathbb{N}} H^m(\mathbb{R}^n)$, then $u \in C^\infty(\mathbb{R}^n) \cap \bigcap_{m\in\mathbb{N}} H^m(\mathbb{R}^n)$.

Solutions $u \in H^1(\mathbb{R}^n)$ to (B.1) are known as weak solutions *to the equation (B.3).*

Proof. We will assume f is a real-valued function. The complex-valued case can be treated very similarly. The statement (i) follows directly from the Riesz representation theorem with respect to the Hilbert space $H^1(\mathbb{R}^n)$. In order to prove (ii) we consider first the case $m = 0$. We want to show that $u \in H^2(\mathbb{R}^n)$. We will do this by considering *difference quotients*.

Let $f \in L^2(\mathbb{R}^n)$ and denote with e_k the k-th basisvector in $\mathbb{R}^n$. Write $f^h(x) = f(x+he_k)$, with $h \in \mathbb{R}\backslash\{0\}$. Define the difference quotient with respect to e_k and h as follows:

$$D_k^h f(x) = h^{-1}[f^h(x) - f(x)].$$

Then the following properties hold (Exercise: Derive these properties.): for all $f, g \in L^2(\mathbb{R}^n)$

$$\begin{aligned}
D_k^h(fg) &= gD_k^h f + f^h D_k^h g,\\
[\partial_i, D_k^h] &= 0 \quad \forall 1 \leq i \leq n,\\
\int_{\mathbb{R}^n} gD_k^h f\, dx &= -\int_{\mathbb{R}^n} fD_k^{-h} g\, dx,\\
||D_k^h f||_{L^2(\mathbb{R}^n)} &\leq ||\partial_k f||_{L^2(\mathbb{R}^n)} \quad \text{if } \partial_k f \in L^2(\mathbb{R}^n).
\end{aligned}$$

Applying now (B.1) with $v = -D_k^{-h}(D_k^h u)$, we obtain:

$$\int_{\mathbb{R}^n} -u \cdot D_k^{-h}(D_k^h u) - \sum_{i=1}^n \partial_i u \cdot \partial_i D_k^{-h}(D_k^h u)\, dx = -\int_{\mathbb{R}^n} fD_k^{-h}(D_k^h u).$$

Hence, after applying the above properties of difference quotients and rearranging terms, we obtain:

$$\int_{\mathbb{R}^n} \sum_{i=1}^n |D_k^h \partial_i u|^2 + |D_k^h u|^2\, dx \leq \int_{\mathbb{R}^n} |f||D_k^{-h}(D_k^h u)|\, dx.$$

Hence, after applying Young's inequality to estimate $|f||D_k^{-h}(D_k^h u)| \leq \frac{1}{2}|D_k^{-h}(D_k^h u)|^2 + \frac{1}{2}|f|^2$, we obtain:

$$||D_k^h u||^2_{H^1(\mathbb{R}^n)} \leq \frac{1}{2}||D_k^{-h}(D_k^h u)||^2_{L^2(\mathbb{R}^n)} + \frac{1}{2}||f||^2_{L^2(\mathbb{R}^n)}$$

Since we can estimate $||D_k^{-h}(D_k^h u)||_{L^2(\mathbb{R}^n)}$ by $||D_k^h u||_{H^1(\mathbb{R}^n)}$ by the above properties, we can absorb this term into the LHS to obtain:

$$||D_k^h u||_{H^1(\mathbb{R}^n)} \leq ||f||_{L^2(\mathbb{R}^n)}.$$

In particular, given a sequence $\{h_j\}$ such that $h_j \to 0$, the sequence $\{D_k^{h_j}\partial_i u\}$ is therefore bounded in $L^2(\mathbb{R}^n)$ for all $1 \leq i \leq n$. This implies that the sequence has a subsequence $\{D_k^{h_j'}\partial_i u\}$ which converges to a *weakly limit* w_k by the following standard theorem from functional analysis:

Theorem B.2 (Banach–Alaoglu)**.** *Let $\{f_j\}_{j\in\mathbb{N}}$ be a bounded sequence in a Hilbert space H.*[42] *Then there exists a subsequence $\{f_{j_i}\}_{i\in\mathbb{N}} \subseteq \{f_j\}_{j\in\mathbb{N}}$ and a $f \in H$ such that:*

$$\lim_{i\to\infty} \langle f_{j_i}, g\rangle_H = \langle f, g\rangle_H \quad \forall g \in H.$$

We denote $f_{j_i} \rightharpoonup f$ and refer to g as the weak limit *of the sequence $\{f_{j_i}\}_{i\in\mathbb{N}}$.*

We will now show that w_k is a weak derivative of $\partial_i u$. Let $\phi \in C_c^\infty(\mathbb{R}^n)$, then:

$$\begin{aligned}\int_{\mathbb{R}^n} \partial_i u \partial_k \phi \, dx &= \lim_{j\to\infty} \int_{\mathbb{R}^n} \partial_i u D_k^{h'_j} \phi \, dx \quad \text{by the fact that } D_k^{h'_j}\phi \to \partial_k \phi \text{ in } L^2(\mathbb{R}^n) \\ &= -\lim_{j\to\infty} \int_{\mathbb{R}^n} D_k^{-h'_j} \partial_i u \phi \, dx \\ &= -\int_{\mathbb{R}^n} w_k \phi \quad \text{by the weak limit property.}\end{aligned}$$

Hence, in the sense of weak derivatives, $\partial_k \partial_i u = w_i \in L^2(\mathbb{R}^n)$. Since this is true for all $1 \le i, k \le n$, we can conclude that $u \in H^2(\mathbb{R}^n)$.

For the general m case, we observe that $[(I-\Delta), \partial_i] = 0$ for all $1 \le i \le n$. We can then apply an induction argument where we assume $u \in H^{m+2}(\mathbb{R}^n)$ and prove that $u \in H^{m+3}(\mathbb{R}^n)$ if $f \in H^{m+1}(\mathbb{R}^n)$ (Exercise: Complete the details of this induction argument.)

By using that $u \in H^2(\mathbb{R}^n)$ to be able to integrate by parts (B.1) and conclude that:

$$\int_{\mathbb{R}^n} [(I-\Delta)u - f] \cdot \phi \, dx = 0 \quad \forall \phi \in C_c^\infty(\mathbb{R}^n).$$

Hence $(I-\Delta)u = f$ in the sense of weak derivatives. This implies moreover after integrating by parts twice that

$$\begin{aligned}\int_{\mathbb{R}^n} f^2 \, dx &= \int_{\mathbb{R}^n} (u-\Delta u)\cdot(u-\Delta u)\, dx = \int_{\mathbb{R}^n} u^2 - 2u \sum_{1\le i\le n} \partial_i^2 u + \sum_{1\le i,j\le n} \partial_i^2 u \partial_j^2 u \, dx \\ &= \int_{\mathbb{R}^n} u^2 + 2\sum_{1\le i\le n} |\partial_i u|^2 + \sum_{1\le i,j\le n} |\partial_i \partial_j u|^2 \, dx.\end{aligned}$$

Hence,

$$\frac{1}{\sqrt{2}} ||f||_{L^2(\mathbb{R}^n)} \le ||u||_{H^2(\mathbb{R}^n)} \le ||f||_{L^2(\mathbb{R}^n)}.$$

By considering $(I-\Delta)\partial_i u = \partial_i f$ and repeating the above argument,

$$\frac{1}{\sqrt{3}} ||f||_{H^1(\mathbb{R}^n)} \le ||u||_{H^3(\mathbb{R}^n)} \le ||f||_{H^1(\mathbb{R}^n)}.$$

More generally, by considering $(I-\Delta)\partial^I u = \partial^I f$, with $|I| \le m$, we can find a numerical constant $c = c(m) > 0$ such that

$$c||f||_{H^m(\mathbb{R}^n)} \le ||u||_{H^{2+m}(\mathbb{R}^n)} \le ||f||_{H^m(\mathbb{R}^n)}.$$

Finally, if $f \in \bigcap_{m\in\mathbb{N}} H^m(\mathbb{R}^n)$, then by the above, $u \in \bigcap_{m\in\mathbb{N}} H^m(\mathbb{R}^n)$, so by Theorem A.2, $u \in C_b^k(\mathbb{R}^n)$ for all $k \in \mathbb{N}_0$, which implies that $u \in C^\infty(\mathbb{R}^n)$.

Remark B.1. *The strategy of the proof of* (ii) *of Theorem B.1 applies more generally to solutions of* elliptic equations *of the form:*

$$\sum_{1\le i,j\le n} \partial_i(a^{ij}(x)\partial_j u) + b^i(x)u + c(x)u = f,$$

with $a^{ij}(x)$ a positive-definite matrix.

□

[42] Or more generally, in a reflexive Banach space.